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## **Practical 1**

### **A. File System Analysis using The SleuthKit (Autopsy).**

#### **Literature Theory**

Sleuth Kit is a collection of command line tools and a C library that allows you to analyze disk images and recover files from them. It is used behind the scenes in Autopsy and many other opensource forensics tools.

There are multiple techniques that could be used to perform the analysis on the files, listed below are the techniques that could be used.

img\_stat: It displays the details associated with the image files. fls: It displays files and directory in a disk image

fsstat : It displays general details of a file system.

istat : It displays details of a meta-data structure (i.e. inode).

Autopsy : It is a digital forensics platform and graphical interface to the Sleuth Kit and other digital forensics tools



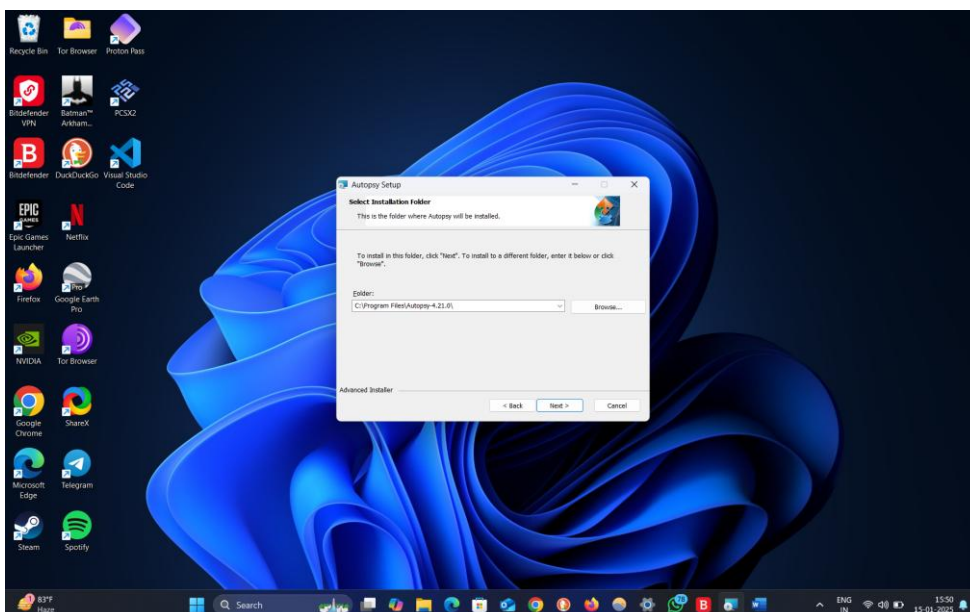
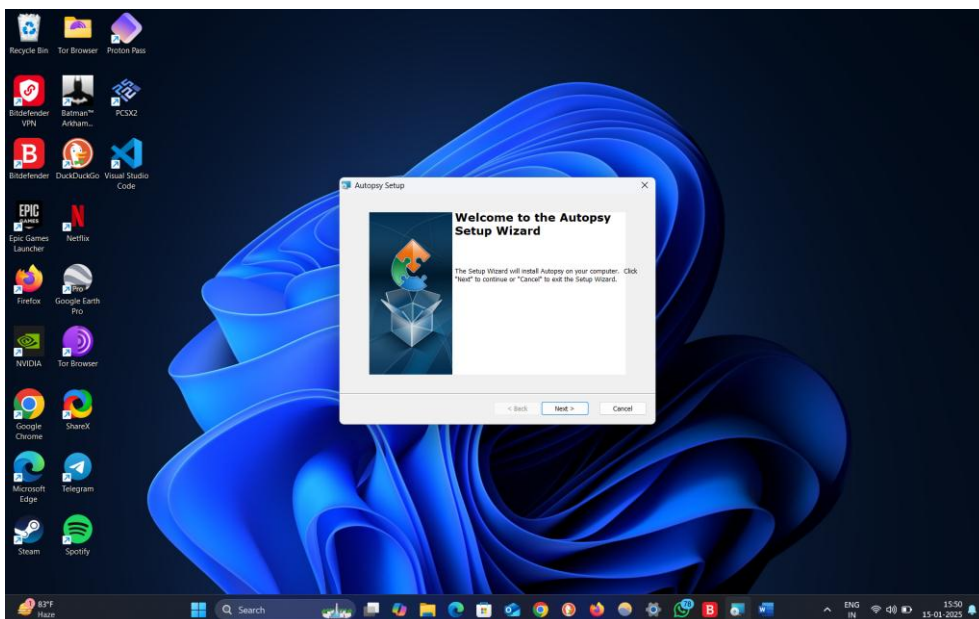
## Autopsy:

Autopsy is a digital forensics platform and graphical interface to The Sleuth Kit® and other digital forensics tools. It is used by law enforcement, military, and corporate examiners to investigate what happened on a computer. You can even use it to recover photos from your camera's memorycard.

Autopsy is an easy to use, GUI-based program that allows you to efficiently analyze hard drives and smart phones. It has a plug-in architecture that allows you to find add-on modules or develop custom modules in Java or Python.

### Easy to Use

Autopsy was designed to be intuitive out of the box. Installation is easy and wizards guide you through every step. All results are found in a single tree.



Autopsy was designed to be an end-to-end platform with modules that come with it out of the box and others that are available from third-parties. Some of the modules provide:

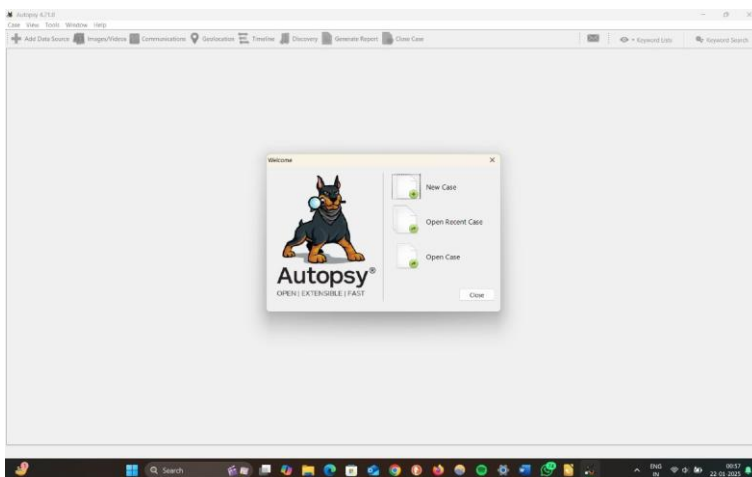
- Timeline Analysis - Advanced graphical event viewing interface (video tutorial included).
- Hash Filtering - Flag known bad files and ignore known good.
- Keyword Search - Indexed keyword search to find files that mention relevant terms.
- Web Artifacts - Extract history, bookmarks, and cookies from Firefox, Chrome, and IE.
- Data Carving - Recover deleted files from unallocated space using PhotoRec
- Multimedia - Extract EXIF from pictures and watch videos.
- Indicators of Compromise - Scan a computer using STIX.Fast

Everyone wants results yesterday. Autopsy runs background tasks in parallel using multiple cores and provides results to you as soon as they are found. It may take hours to fully search the drive, but you will know in minutes if your keywords were found in the user's home folder. See the fast results page for more details.

Cost Effective - Autopsy is free. As budgets are decreasing, cost effective digital forensics solutions are essential. Autopsy offers the same core features as other digital forensics tools and offers other essential features, such as web artifact analysis and registry analysis, that other commercial tools do not provide.

## Step 1 — Start a New Case

Click **New Case**. This will add a new case folder to the system and allow you to begin adding evidence. To begin, click **New Case**.



## Step 2 — Enter the Case Details

The screenshot shows the Autopsy 4.21.0 application window with the 'New Case Information' dialog box open. The dialog box has two tabs: 'Case Information' and 'Optional Information'. The 'Optional Information' tab is selected, showing fields for Case Number, Examiner Name, Phone, Email, Notes, and Organization. The 'Case Information' tab is also visible, showing a list of steps: 1. Case Information, 2. Optional Information, and 3. Generate Report. The 'Optional Information' tab is currently active, and the 'Case Number' field is empty. The 'Examiner Name' field contains the text 'deepM'. The 'Phone', 'Email', and 'Notes' fields are empty. The 'Organization' field is set to 'Not Specified' with a 'Manage Organizations' button next to it. The dialog box has 'Back', 'Next', 'Finish', 'Cancel', and 'Help' buttons at the bottom.

Autopsy 4.21.0

Case View Tools Window Help

+ Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

**New Case Information**

**Steps**

1. Case Information
2. Optional Information
3. Generate Report

**Optional Information**

Case Number:

Examiner Name:

Phone:

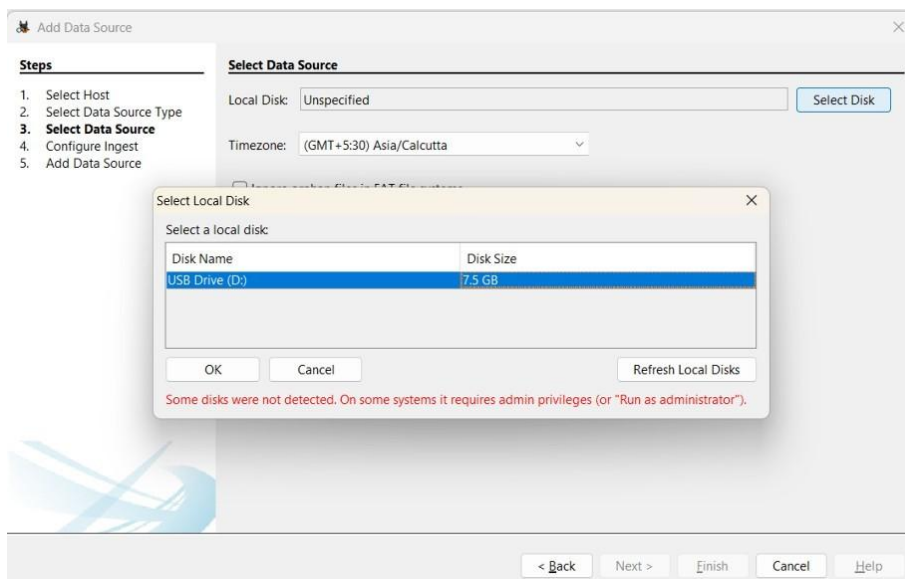
Email:

Notes:

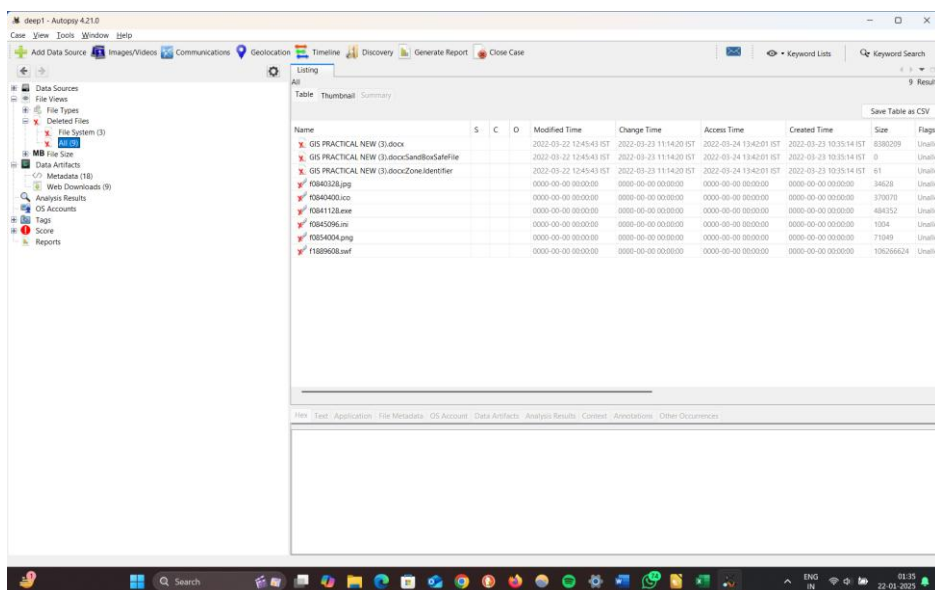
Organization analysis is being done for:  [Manage Organizations](#)

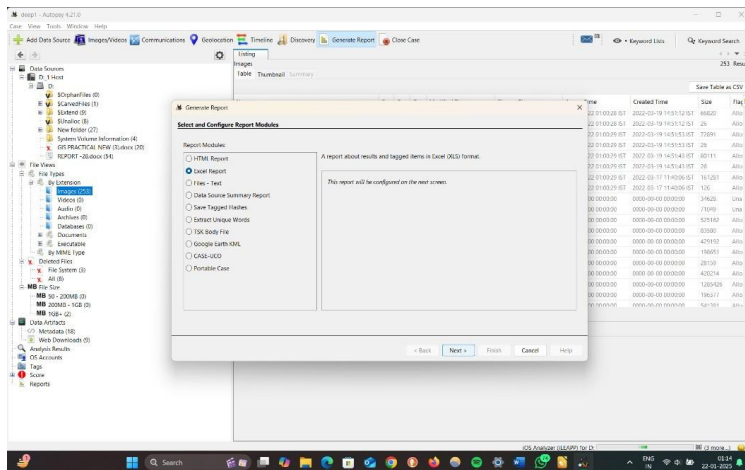
< Back Next > Finish Cancel Help

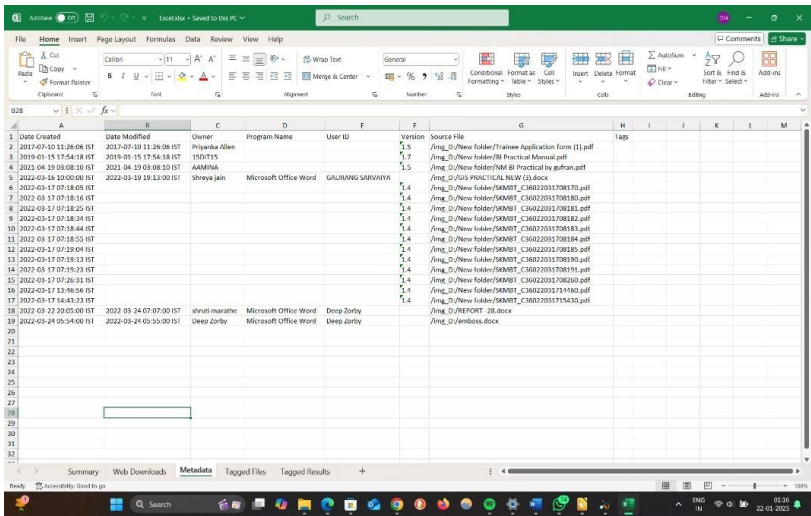
### Step 3 — Choose the path



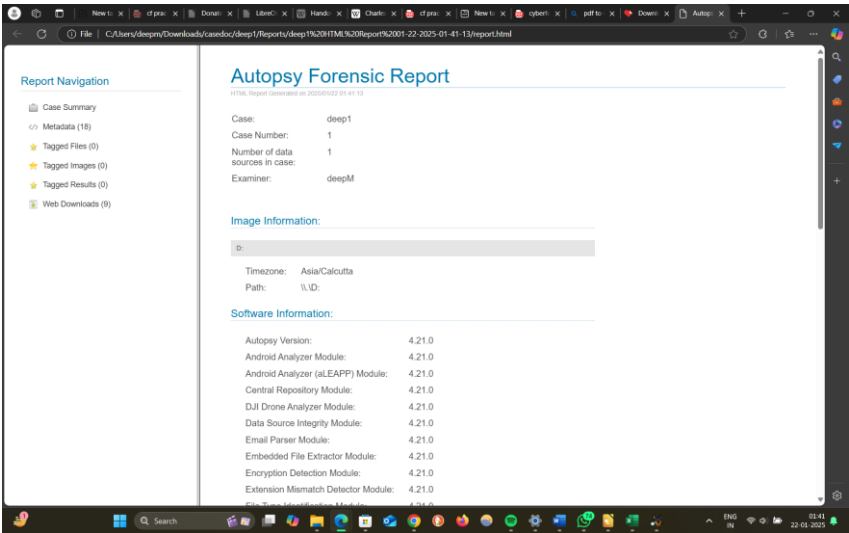
### Step 4 — Retrieve the classified files







|    | A                       | B                       | C              | D                     | E                | F       | G                                                  | H | I | J | K | L | M |
|----|-------------------------|-------------------------|----------------|-----------------------|------------------|---------|----------------------------------------------------|---|---|---|---|---|---|
|    | Date Created            | Date Modified           | Owner          | Program Name          | User ID          | Version | Source File                                        |   |   |   |   |   |   |
| 1  | 2017-07-10 11:26:08 IST | 2017-07-10 11:26:08 IST | Priyanka Allen |                       |                  | 1.5     | /img_0/New folder/Trainee Application form (1).pdf |   |   |   |   |   |   |
| 2  | 2019-01-15 17:54:18 IST | 2019-01-15 17:54:18 IST | 150775         |                       |                  | 1.7     | /img_0/New folder/NM Practical Manual.pdf          |   |   |   |   |   |   |
| 3  | 2021-04-19 09:08:10 IST | 2021-04-19 09:08:10 IST | AAMINA         |                       |                  | 1.5     | /img_0/New folder/NM B Practical by gufran.pdf     |   |   |   |   |   |   |
| 4  | 2022-03-19 10:00:00 IST | 2022-03-19 10:00:00 IST | Shreya Jain    | Microsoft Office Word | GAURANG SARVANYA |         | /img_0/Bus PRACTICAL NEW (1).docx                  |   |   |   |   |   |   |
| 5  | 2022-03-17 07:18:05 IST | 2022-03-17 07:18:05 IST |                |                       |                  | 1.4     | /img_0/New folder/SKMBT_C36022011708170.pdf        |   |   |   |   |   |   |
| 6  | 2022-03-17 07:18:16 IST | 2022-03-17 07:18:16 IST |                |                       |                  | 1.4     | /img_0/New folder/SKMBT_C36022011708180.pdf        |   |   |   |   |   |   |
| 7  | 2022-03-17 07:18:29 IST | 2022-03-17 07:18:29 IST |                |                       |                  | 1.4     | /img_0/New folder/SKMBT_C36022011708181.pdf        |   |   |   |   |   |   |
| 8  | 2022-03-17 07:18:38 IST | 2022-03-17 07:18:38 IST |                |                       |                  | 1.4     | /img_0/New folder/SKMBT_C36022011708182.pdf        |   |   |   |   |   |   |
| 9  | 2022-03-17 07:18:44 IST | 2022-03-17 07:18:44 IST |                |                       |                  | 1.4     | /img_0/New folder/SKMBT_C36022011708183.pdf        |   |   |   |   |   |   |
| 10 | 2022-03-17 07:18:55 IST | 2022-03-17 07:18:55 IST |                |                       |                  | 1.4     | /img_0/New folder/SKMBT_C36022011708184.pdf        |   |   |   |   |   |   |
| 11 | 2022-03-17 07:19:01 IST | 2022-03-17 07:19:01 IST |                |                       |                  | 1.4     | /img_0/New folder/SKMBT_C36022011708185.pdf        |   |   |   |   |   |   |
| 12 | 2022-03-17 07:19:13 IST | 2022-03-17 07:19:13 IST |                |                       |                  | 1.4     | /img_0/New folder/SKMBT_C36022011708190.pdf        |   |   |   |   |   |   |
| 13 | 2022-03-17 07:19:33 IST | 2022-03-17 07:19:33 IST |                |                       |                  | 1.4     | /img_0/New folder/SKMBT_C36022011708191.pdf        |   |   |   |   |   |   |
| 14 | 2022-03-17 07:26:31 IST | 2022-03-17 07:26:31 IST |                |                       |                  | 1.4     | /img_0/New folder/SKMBT_C36022011708200.pdf        |   |   |   |   |   |   |
| 15 | 2022-03-17 13:06:58 IST | 2022-03-17 13:06:58 IST |                |                       |                  | 1.4     | /img_0/New folder/SKMBT_C36022011711160.pdf        |   |   |   |   |   |   |
| 16 | 2022-03-17 14:43:33 IST | 2022-03-17 14:43:33 IST |                |                       |                  | 1.4     | /img_0/New folder/SKMBT_C36022011715430.pdf        |   |   |   |   |   |   |
| 17 | 2022-03-22 20:05:00 IST | 2022-03-24 07:07:00 IST | shruti marathe | Microsoft Office Word | Deep Dorly       |         | /img_0/REPORT_28.docx                              |   |   |   |   |   |   |
| 18 | 2022-03-24 09:54:00 IST | 2022-03-24 09:55:00 IST | Deep Dorly     | Microsoft Office Word | Deep Dorly       |         | /img_0/emboss.docx                                 |   |   |   |   |   |   |



**Report Navigation**

- Case Summary
- Metadata (18)
- Tagged Files (0)
- Tagged Images (0)
- Tagged Results (0)
- Web Downloads (0)

### Autopsy Forensic Report

URL: <http://report.localhost:8080/20200203-01-41-13/report.html>

Case: deep1  
Case Number: 1  
Number of data sources in case: 1  
Examiner: deepM

**Image Information:**

ID: [REDACTED]  
Timezone: Asia/Calcutta  
Path: \\.\D:

**Software Information:**

|                                     |        |
|-------------------------------------|--------|
| Autopsy Version:                    | 4.21.0 |
| Android Analyzer Module:            | 4.21.0 |
| Android Analyzer (aLEAPP) Module:   | 4.21.0 |
| Central Repository Module:          | 4.21.0 |
| DJI Drone Analyzer Module:          | 4.21.0 |
| Data Source Integrity Module:       | 4.21.0 |
| Email Parser Module:                | 4.21.0 |
| Embedded File Extractor Module:     | 4.21.0 |
| Encryption Detection Module:        | 4.21.0 |
| Extension Mismatch Detector Module: | 4.21.0 |







## Practical 2

### Explore Windows forensic tools (OS Forensics).

#### Literature/Theory:

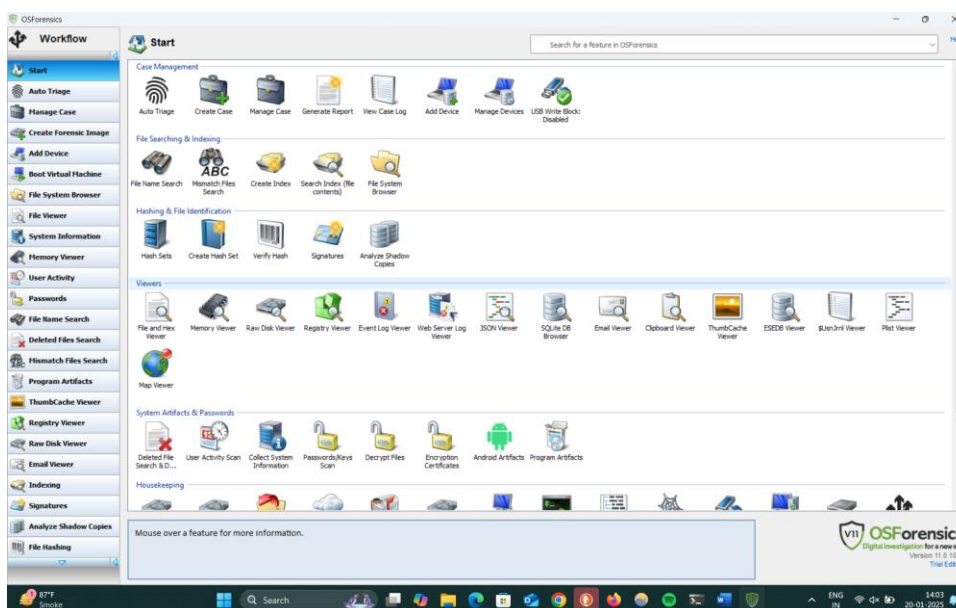
OS Forensics extract forensic data from computers, quicker and easier than ever.

The professional and bootable editions of OSForensics have many features not available in the free edition, including:

- Import and export of hash sets.
- Customizable system information gathering.
- No limits on the number of cases being managed through OSForensics.
- Restoration of multiple deleted files in one operation.
- List and search for alternate file streams.
- Sort image files by color.
- Disk indexing and searching not restricted to a fixed number of files.
- No watermark on web captures.
- Multi-core acceleration for file decryption.
- Customizable System Information Gathering.
- View NTFS directory \$I30 entries to identify potential hidden/deleted files.

#### Code/Demo & Output/Steps:

Go to: <https://www.osforensics.com/download.html>



After Successfully installing OS Forensics this kind of interface should be display on your system.

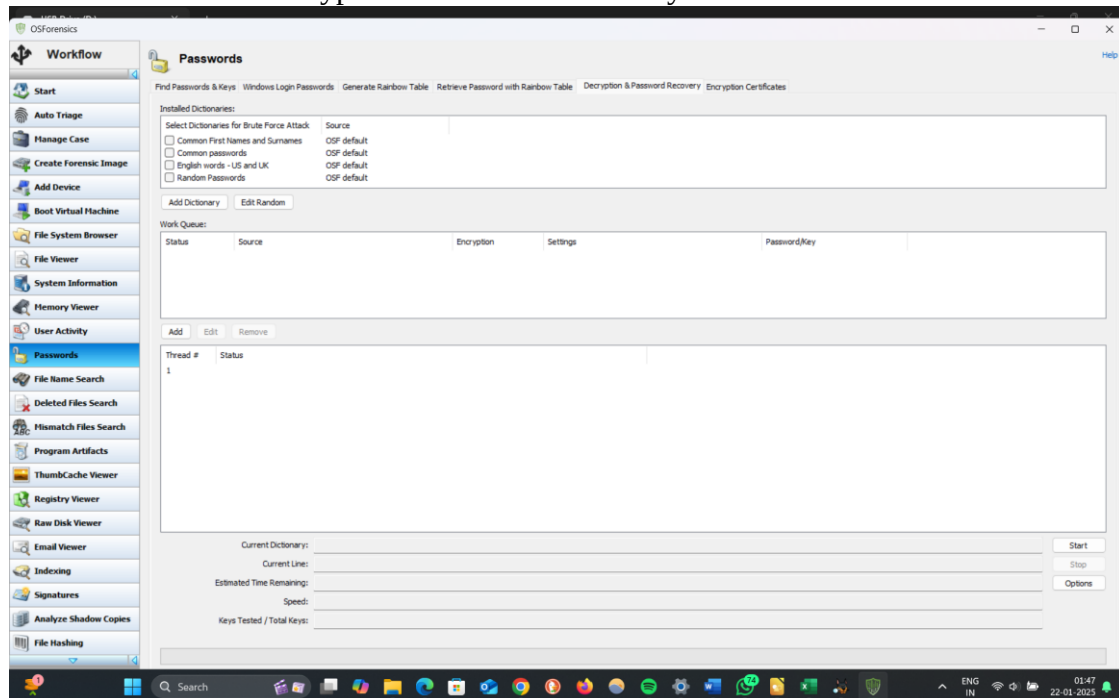
OS Forensics supports two methods of gaining access to encrypted office documents.

The first method is for older documents that use 40bit encryption (old XLS, DOC & PDF files).For these documents is it possible to try all possible keys to decrypt the document, with the output being an unencrypted file.

By using this tool we crack password protected files by simply importing the file

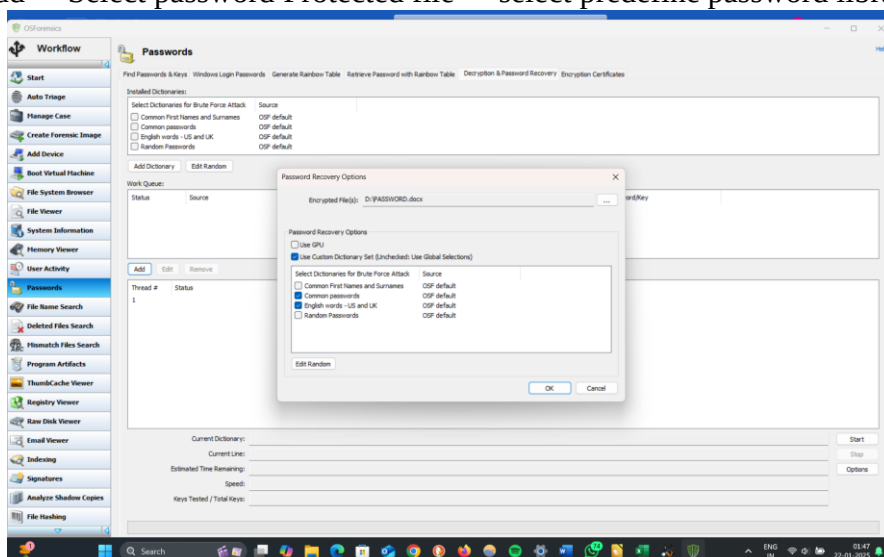
Select

### 1. Password -> Decryption & Password recovery



This tool having some preloaded decrypted password file by using unique combination of alpha-numeric. Password combination it is encrypting password protected files.

Add -> Select password Protected file -> select predefine password libraries -> Start



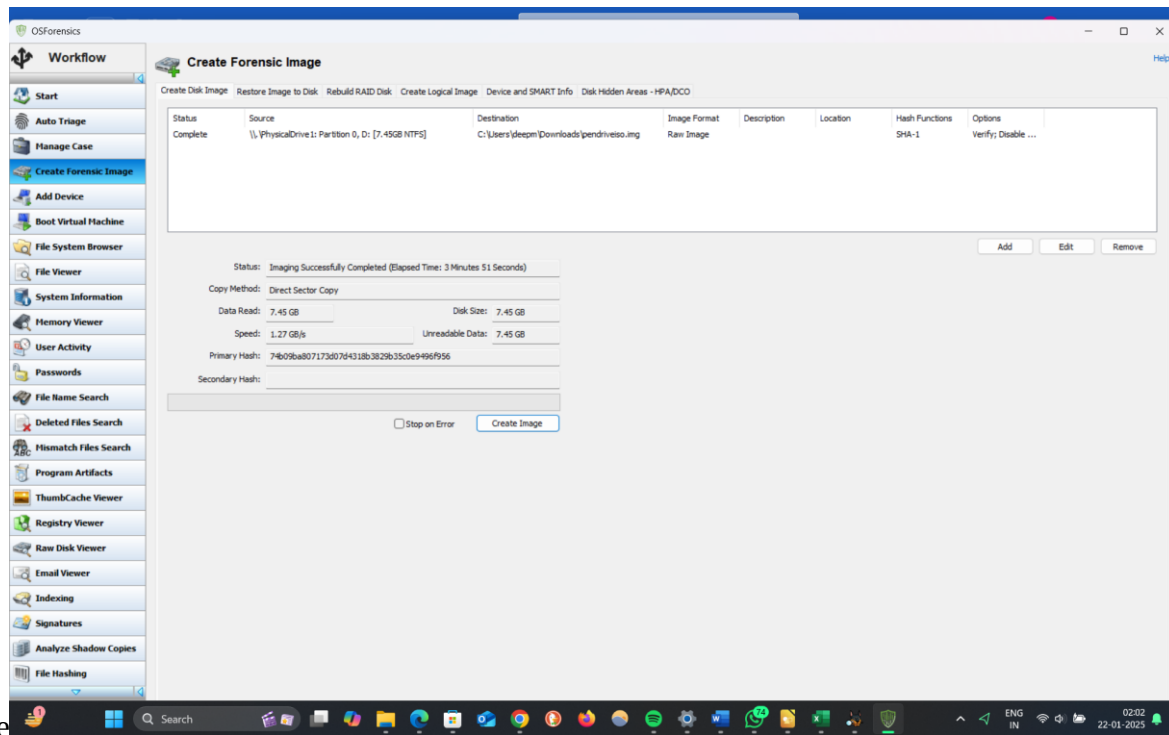
Password is Display in this Prompt



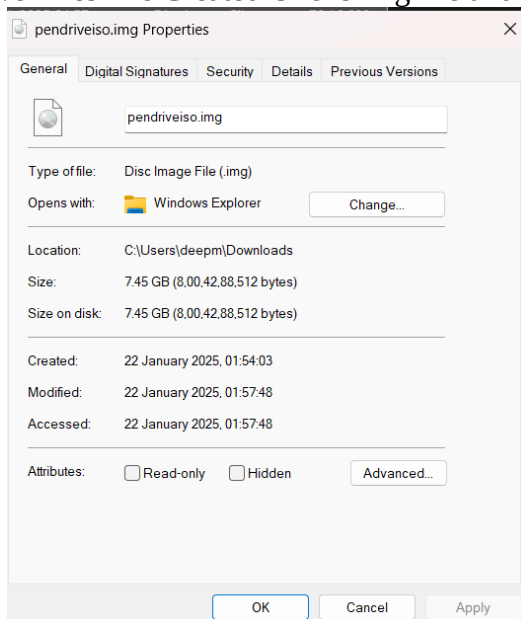
16



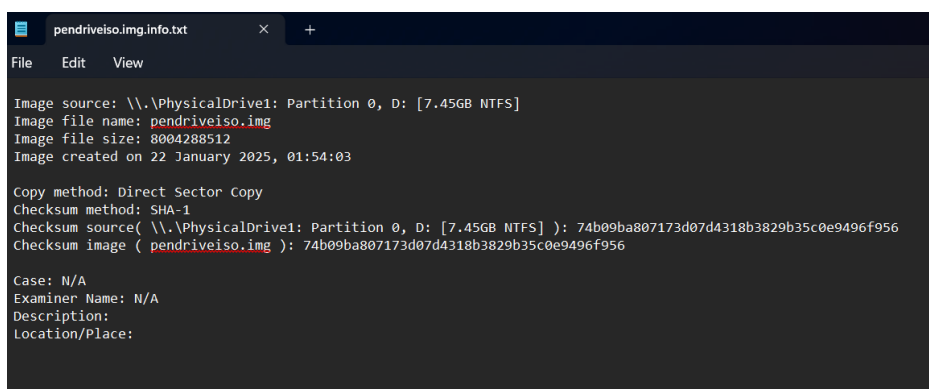




Two Files Are Created One is img file and other is text file.Img file is created as size of hard drive.



Text file is for verifying the source and image file checksum value.



**A. Forensics Investigation Using Encase.****Literature/Theory:**

EnCase Forensic enables you to collect forensically sound data and conduct complex large scale investigations from beginning to end.

EnCase Forensic is designed to be used by:

- Those responsible for collecting evidence.
- Forensic examiners and analysts.
- Forensic examiners who develop and use EnScript code to automate repetitive or complex tasks.

With EnCase Forensic these types of investigators can:

- Acquire data in a forensically sound manner using software with an unparalleled record in courts worldwide.
- Investigate and analyze data from multiple platforms—Windows, Linux, AIX, OS X, Solaris, and more—using a single tool.
- Find information despite efforts to hide, cloak, or delete.
- Easily manage large volumes of computer evidence, viewing all relevant files, including deleted files, file slack, and unallocated space.
- Create exact duplicates of original data, verified by hash and Cyclic Redundancy Check (CRC) values.
- Transfer evidence files directly to law enforcement or legal representatives.
- Review options that allow non-investigators, such as attorneys, to review evidence with ease.
- Use reporting options for quick report preparation.

**System Requirement:**

- An EnCase security key (dongle), or an electronic license and connection information.
- An optional certificate file for users who want to activate an EnCase Version 6 dongle to run EnCase Version 8.
- Installation files for the current release of EnCase.

**B. Using Mobile Forensics software tools.**

## Literature/Theory:

Cellebrite Frontliner collect digital evidence in real-time. Capture select digital evidence at the scene to help solve cases faster and improve community trust.

With a user-friendly mobile application, field officers increase witness and victim cooperation by selecting and transferring only relevant pictures, videos and documents from their mobile devices.

Cellebrite Frontliner key capabilities:

- Witnesses and victims can share key evidence from their devices using a wireless, non-intrusive format.
- Gather all information into a single case.
- Record audio of witness testimony, take pictures or videos, and automatically tag them with meta data that includes location details, time stamps and hash values — all admissible in court.
- Share HTML or UFDR\* forensic reports with the wider team in real-time from the scene.
- Password protect captured data.



### C. Exploring Mobiledit Forensics.

#### Literature/Theory:

MOBILedit is a unique and universal tool that supports phones with any system, such as Android, iPhone, Blackberry, Windows Phone, Bada, Symbian and Meego in addition to many feature phones including even those that are practically obsolete.

It is the ultimate PC suite for all phones with features including an organizer, rich phonebook functionality, smart phonebook optimizer, battery life indicator, signal indicator, memory status, SMS sending and management and much more.

Different phones require many different tools in order to connect to your PC. But with MOBILedit, it can be easily managed almost any type of phone.

Following are the main functions that are provided by Mobiledit Forensics:

- Complete contacts management
- Text messaging center
- Phone data transfer
- Control your phone from PC
- Central tool for backup and restore
- Import and export your phone data
- Supports many different phones and brands

#### Functionalities

**Import:** Files that are supported to import are vCard vcf files, csv files, Outlook live connection, iTunes backup, Android ADB backup, MOBILedit backup, Clipboard paste from a list of contacts such as Excel or Word, Import contacts and messages through Bluetooth from almost any phone

**Export:** Files that it can be extracted to are vCard vcf files, csv, xml, xls, MOBILedit backup file, Clipboard copy of contacts and messages

**Contacts:** It provides various contacts information like Supports contacts stored on SIM, PhoneMemory and other sources (WhatsApp, Cloud, Gmail, Facebook), Import vCard files, Robust editor of structured contacts such as phone number, email address, home address, birthday, employer, etc., Contact profile pictures management, Printing of whole phonebook is the best backup, especially for travelers, Search & Replace across all contacts, Comfortable internationalization - automatically adds prefix of selected country - great feature for travelers, Swap the first name for the last name with one click

**Messages:** It provides various messages information like Complete SMS management and printing, Sort, read, and archive your SMS messages on your PC, Compose a message from a PC keyboard, Automatically stores them to Sent Items on the PC, Backup or delete your old messages, Handling of incoming texts through MOBILedit, Automatic conversion of SMS characters into GSM compliant character set, Switch between different views

**Conversations:** Efficiently read through entire uninterrupted conversation streams. It separates entire conversations by contact and are ordered by time sequence. Conversations include all SMS, MMS, iMessages and applications messages including encrypted-messaging apps.

**Call Logs:** The Calls Report provides details of all mobile service and applications-based phonecalls, See

the time and date of when you made or received calls, the duration of the call and the contact and phone number associated with each call.

**Applications:** Manage your applications regardless of platform. MOBILedit has a widget app viewer which makes accessing all your apps straight forward. View or make backups of app data and delete unwanted or unused apps. Access to file sharing for iOS.

**Phone Status Information:** It provides various information like Picture of your phone, Name of the phone - manufacturer and model, Signal strength, battery status, Current network operator, connection type, hardware and software revision, IMEI, SIM card information, Remaining storage space, Phone's display resolution, Phone's platform/OS

**Multimedia:** Create and edit ringtones, Edit, resize or crop your photos with our integrated Photo Editor, Convert your music into various formats

**Organizer:** Switch between different calendar views, Create or manage existing calendar events, Easily manage notes and tasks

**Cloud Connectivity:** New way of copying phones to the internet cloud, Access your backup from anywhere, Use our mobile app to upload/download and migrate your contacts and messages using our MOBILedit Cloud, Transfer your contacts wirelessly and import them to MOBILedit

**Sim Card:** SIM card status information (IMSI, ICCID, LAI, PIN, PUK, call costs), SIM phonebook, sent-received-draft SMS, call log, SIM card reader support

**Phone Data transfer:** Transfer contacts, messages, files from other phones with a single click, Intelligent or manual mapping of fields in phonebook, Wipe and backup before you copy option, SIM card data transfer available, Copy data also to/from Outlook, a file, MOBILedit Cloud or Google contacts (Gmail), Copy from Offline Devices or local/transferred backup file

**File System:** With MOBILedit you will gain access to your File System. It includes internal system files as well as an in-depth look at your application data.

## Practical 3

### A. Using Forensic Tool Kit (FTK) & Writing report using FTK (AccessData FTK).

Steps:

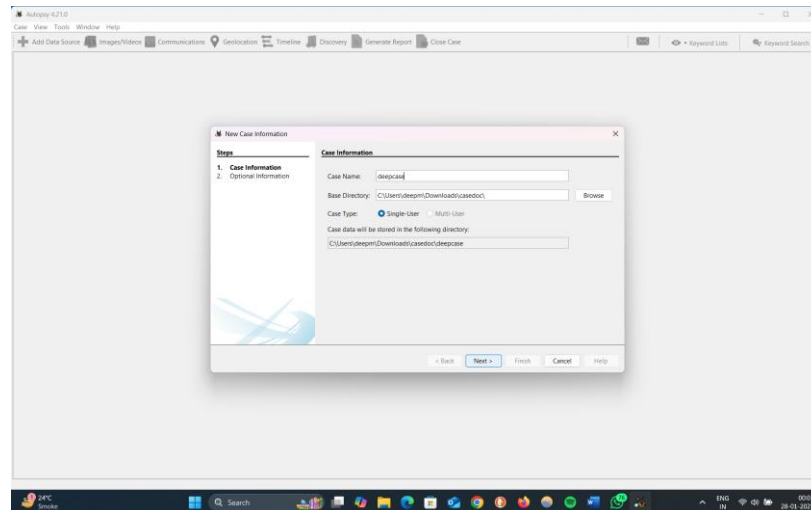
**Step 1:** Open Autopsy



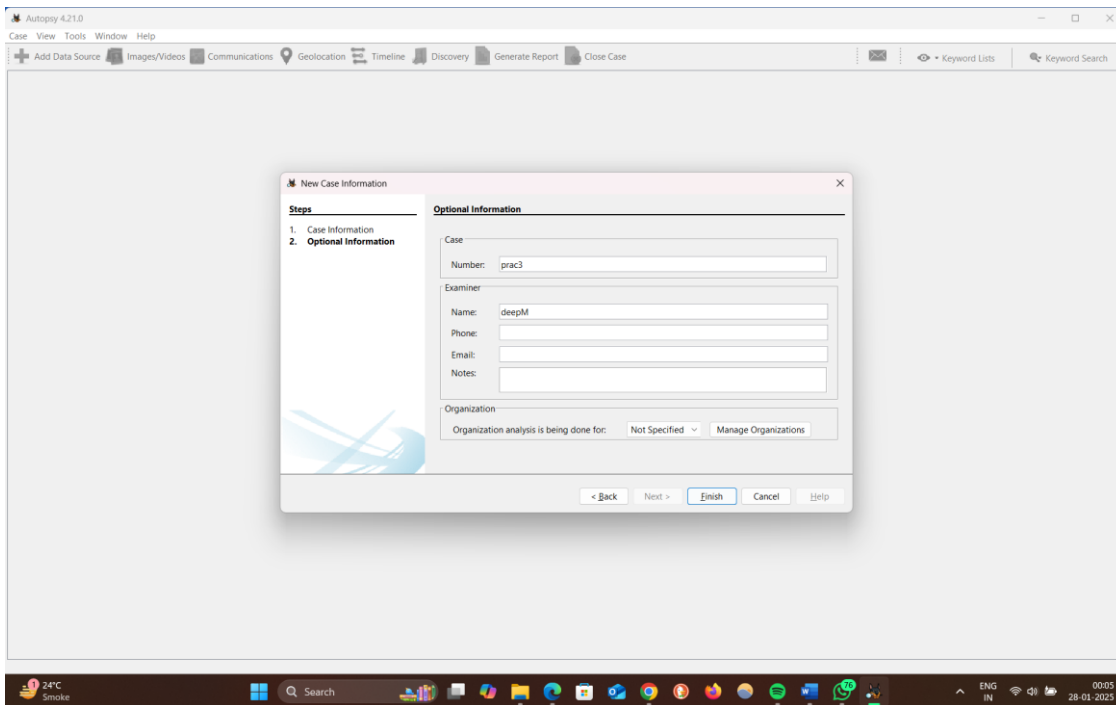
**Step 2:** Click on new case



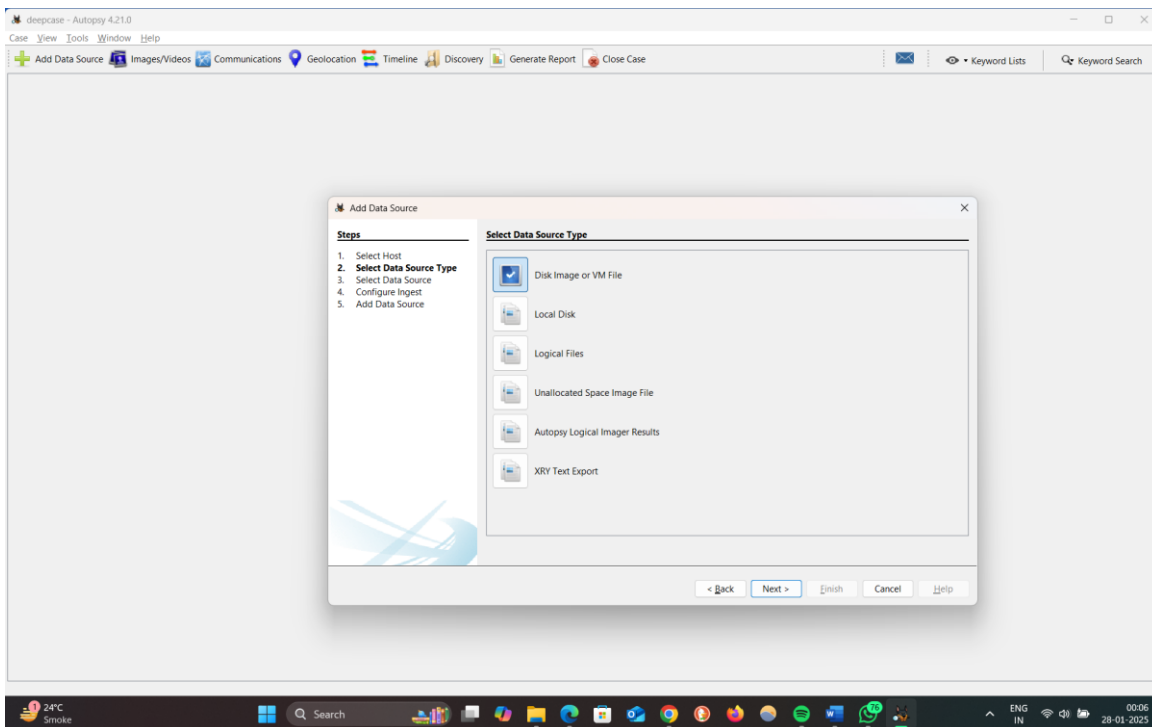
**Step 3:** Enter details regarding the case and click on the next button.



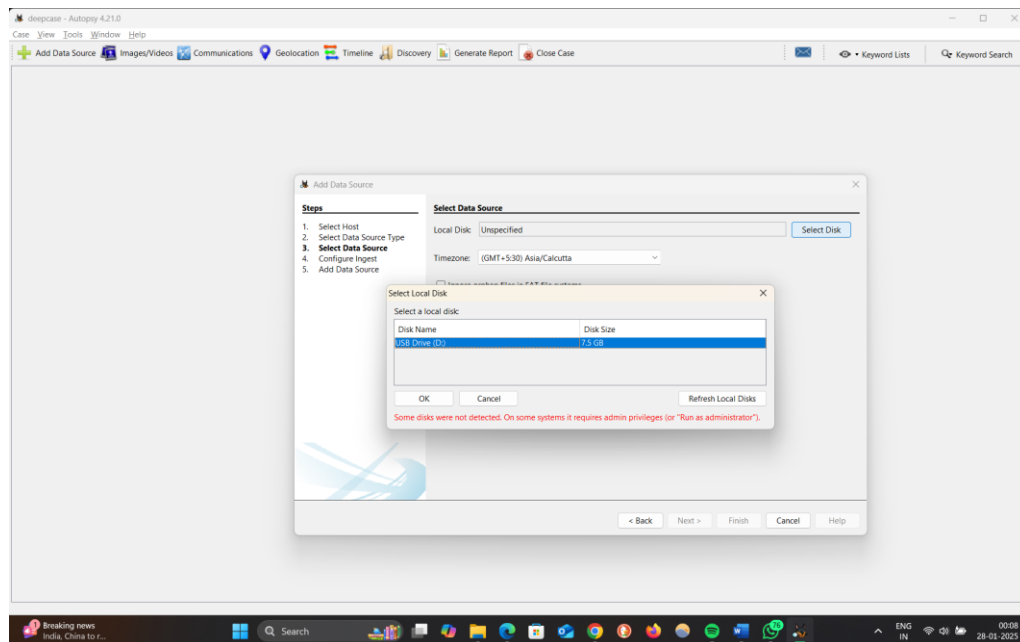
Step 4: Enter further details and click on next button.



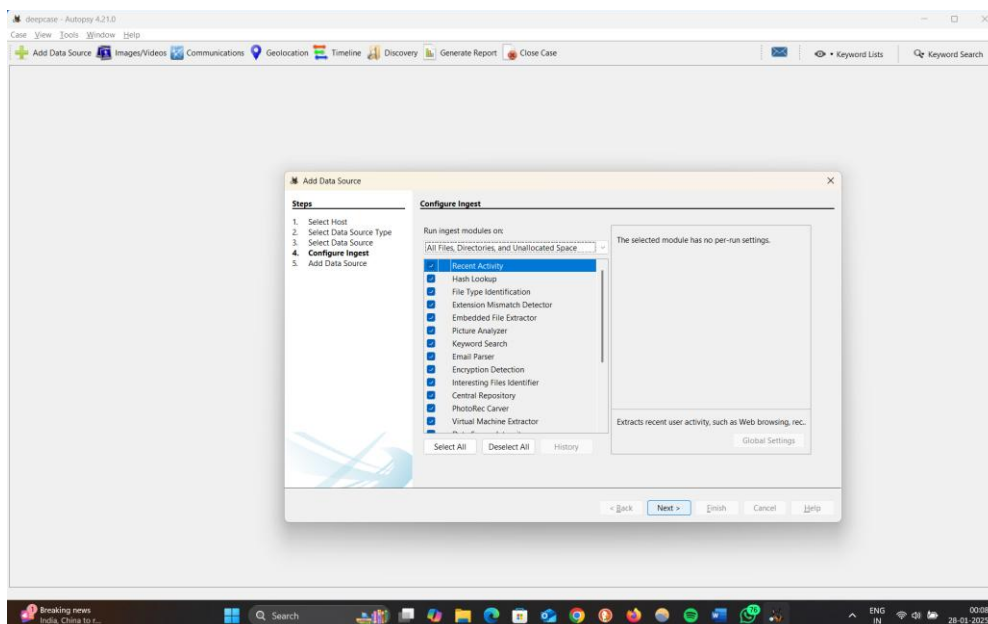
Step 5: Now here we have to select Type of data source to add, in our case disk image or VMfile and click on next.



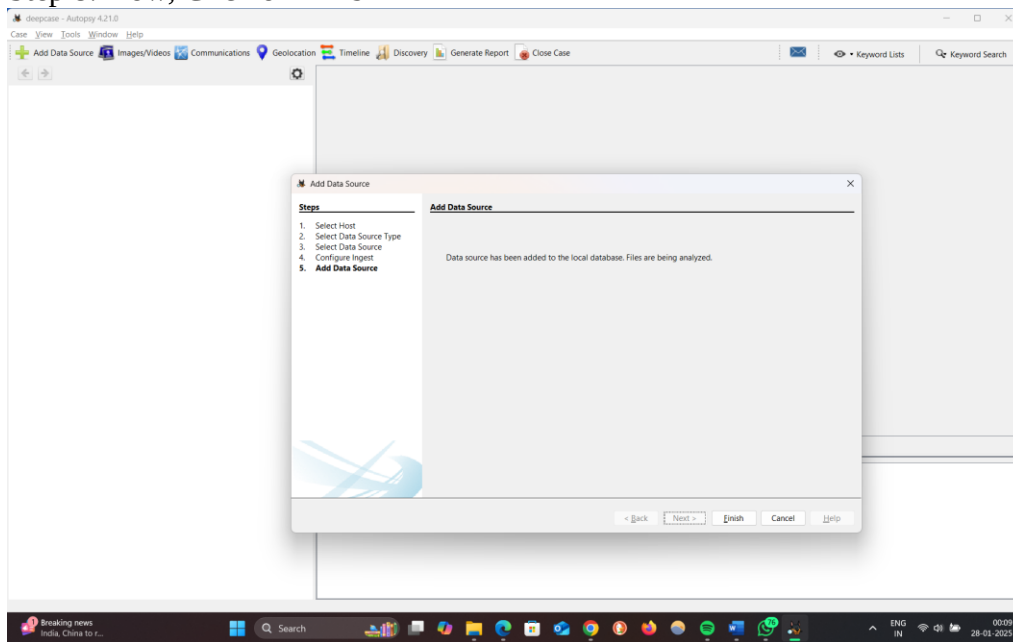
Step 6: Now we have to select the image file and click on the next button



Step 7: Now click on select all in order to run ingest modules on and click on next.



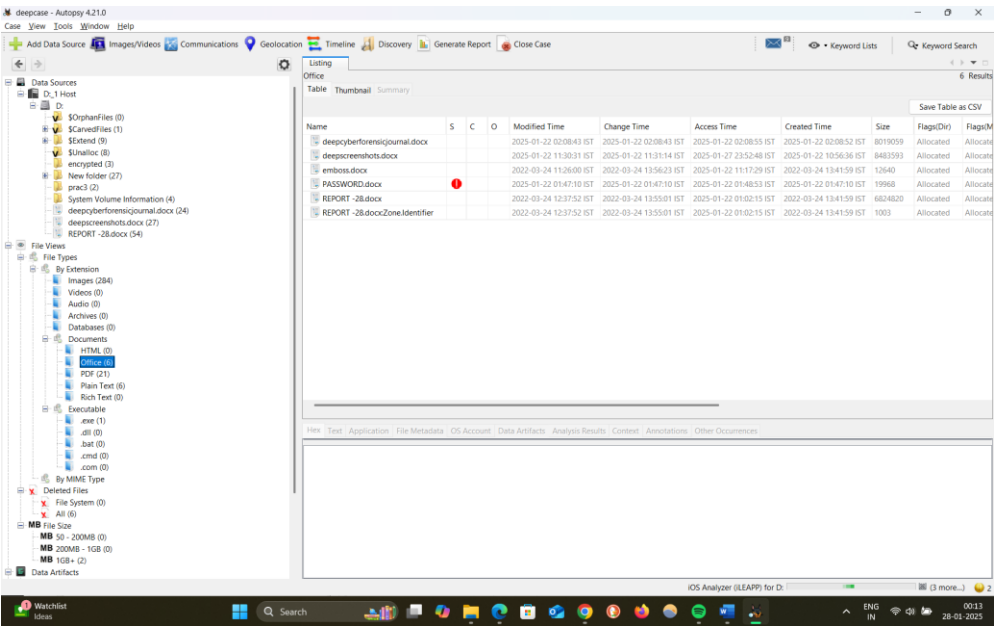
## Step 8: Now, Click on Finish



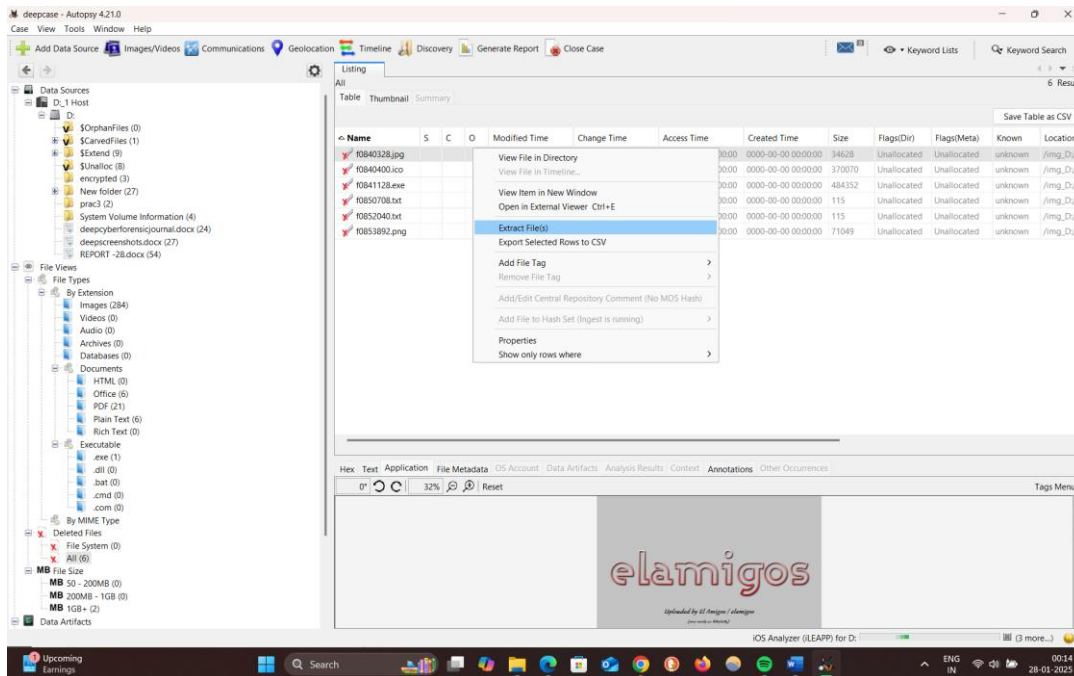
[illegible]

28

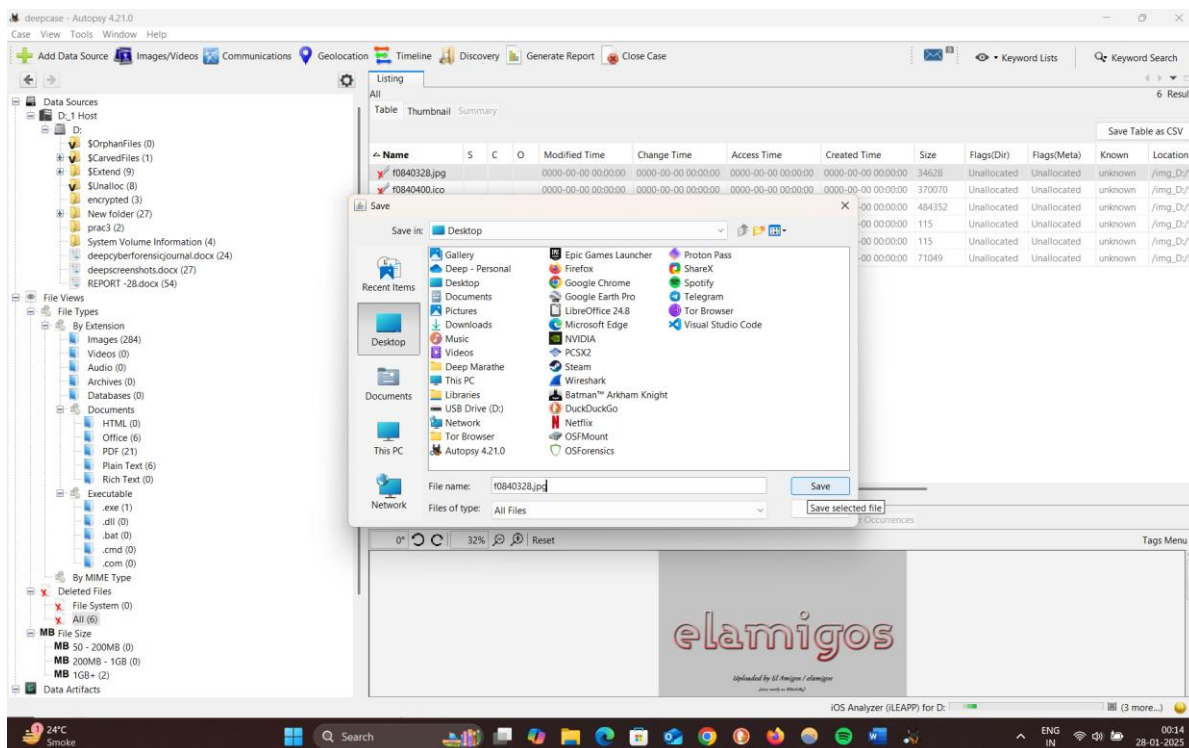




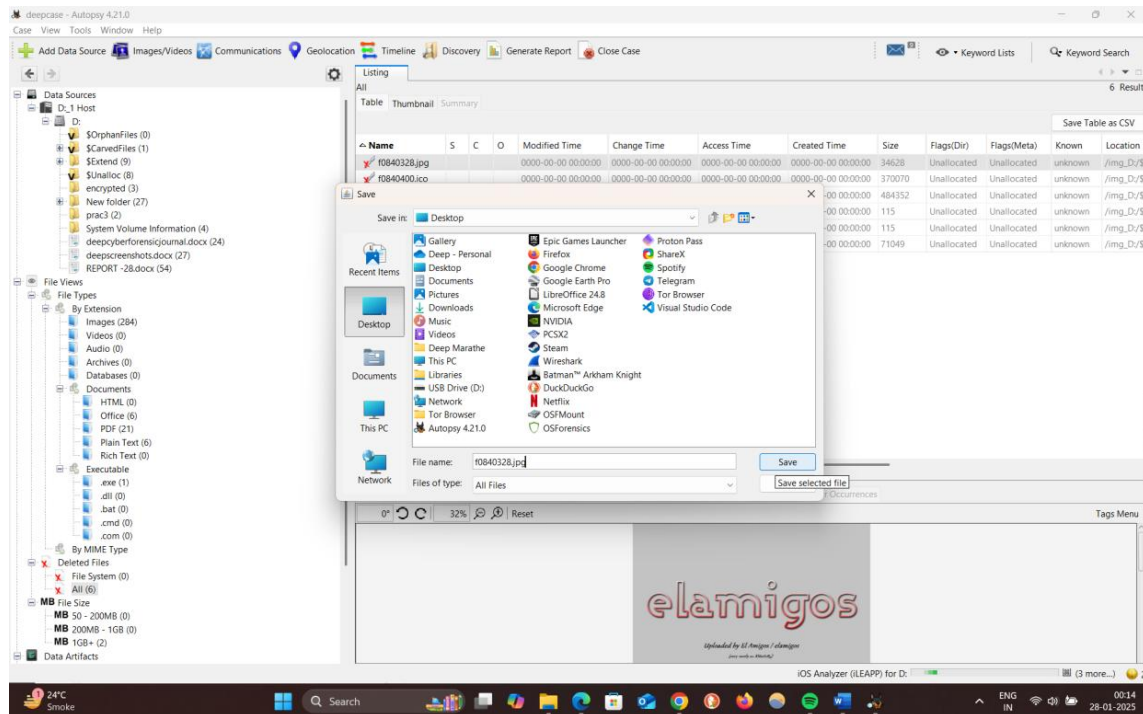
Step 12: To recover the files, go to view code Deleted files node, here select any file and rightclick on it then select Extract files option.

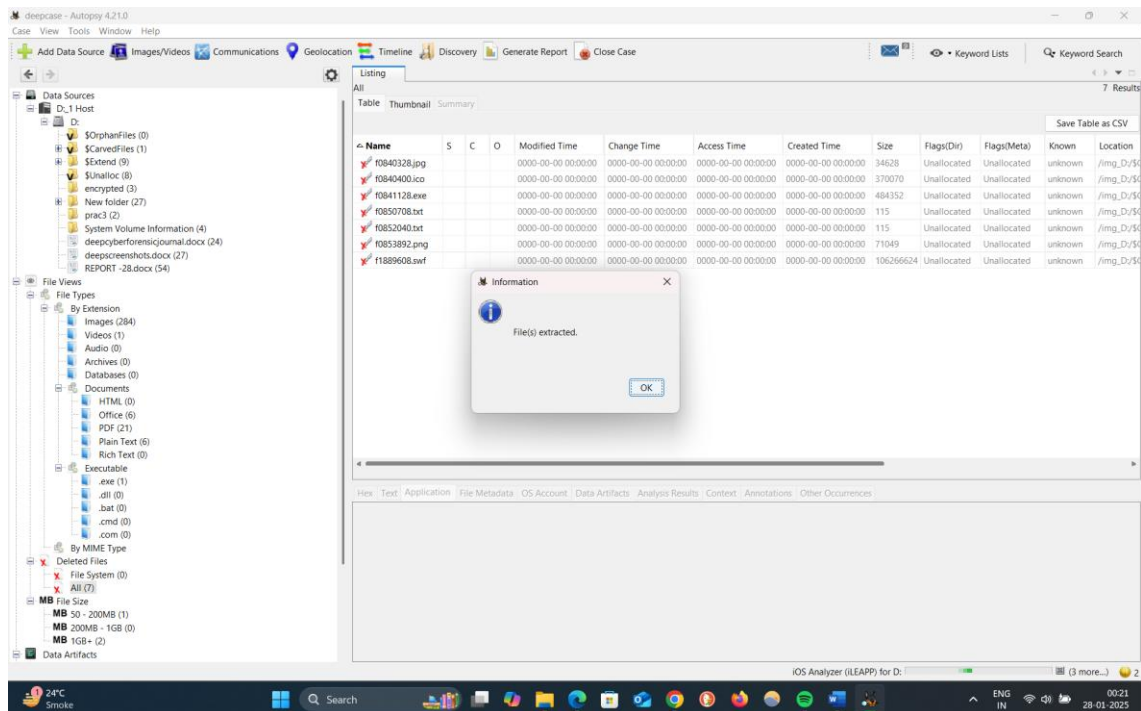


Step 13: Select Path where you want to save the extracted file and click on save.

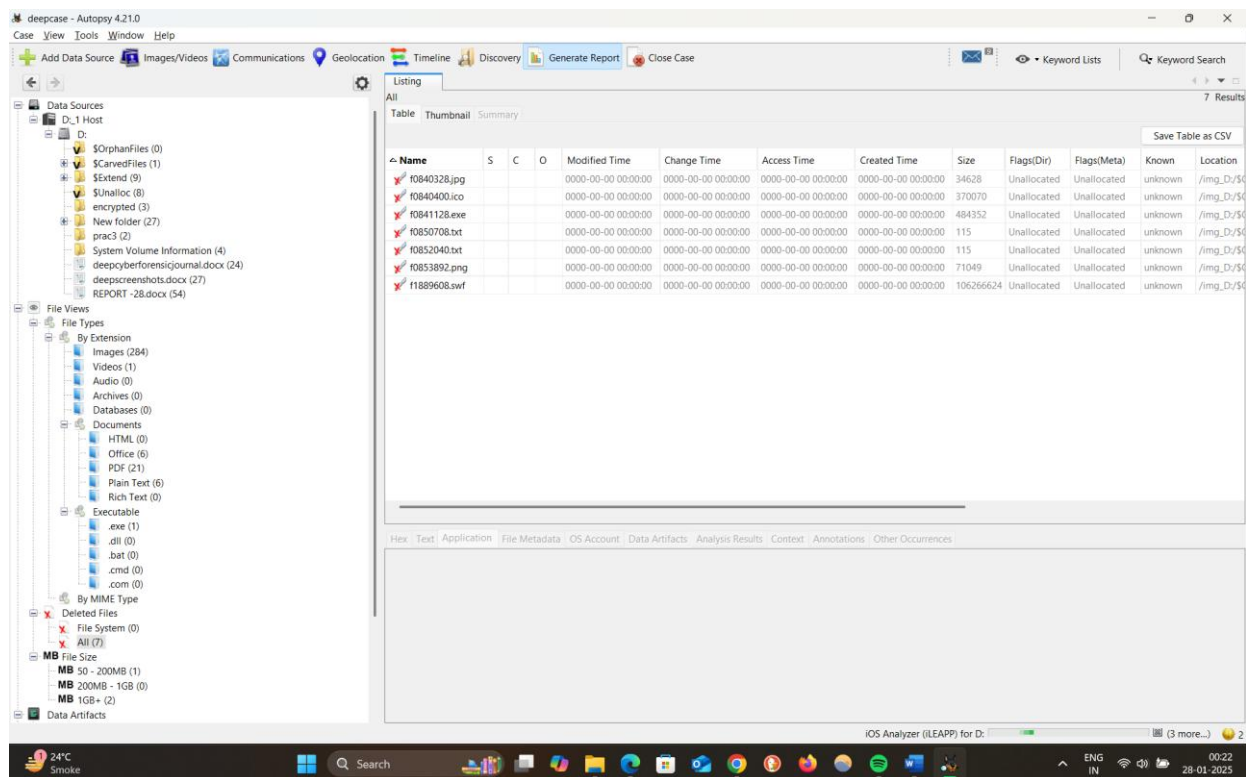


## Step 14: Now click on Save Button

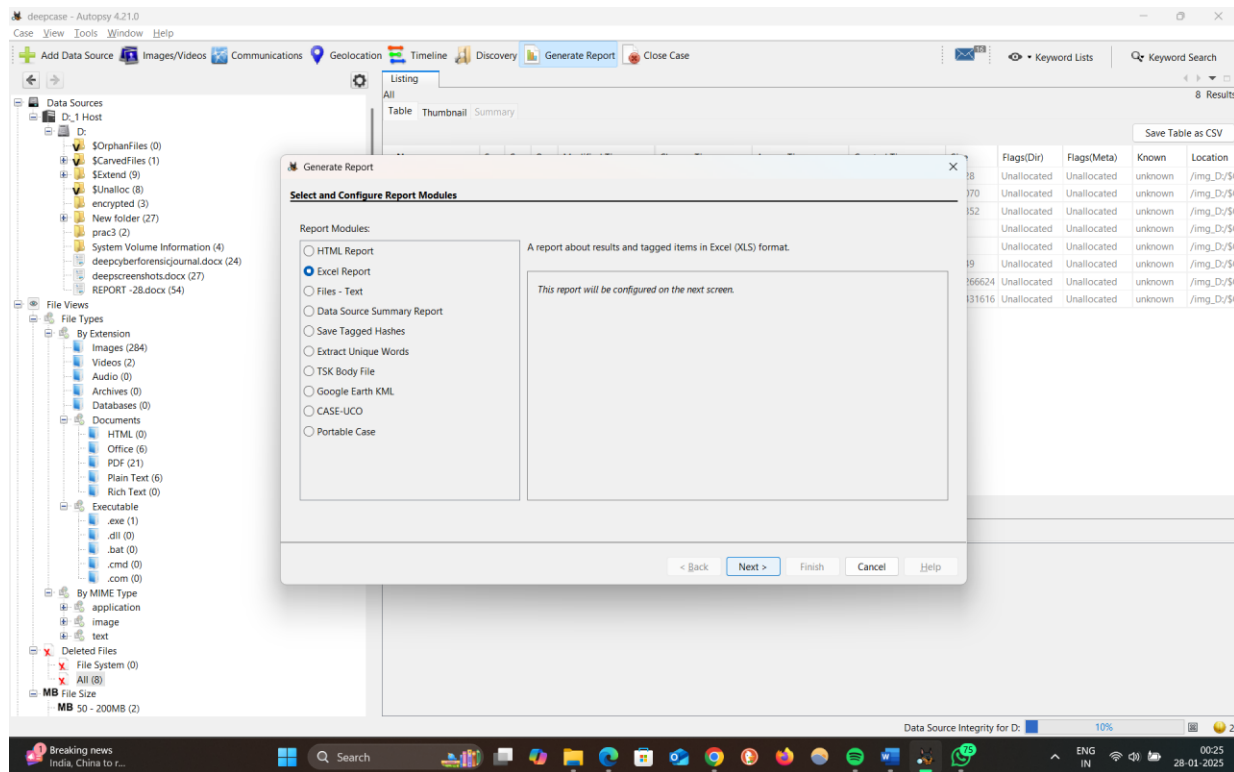




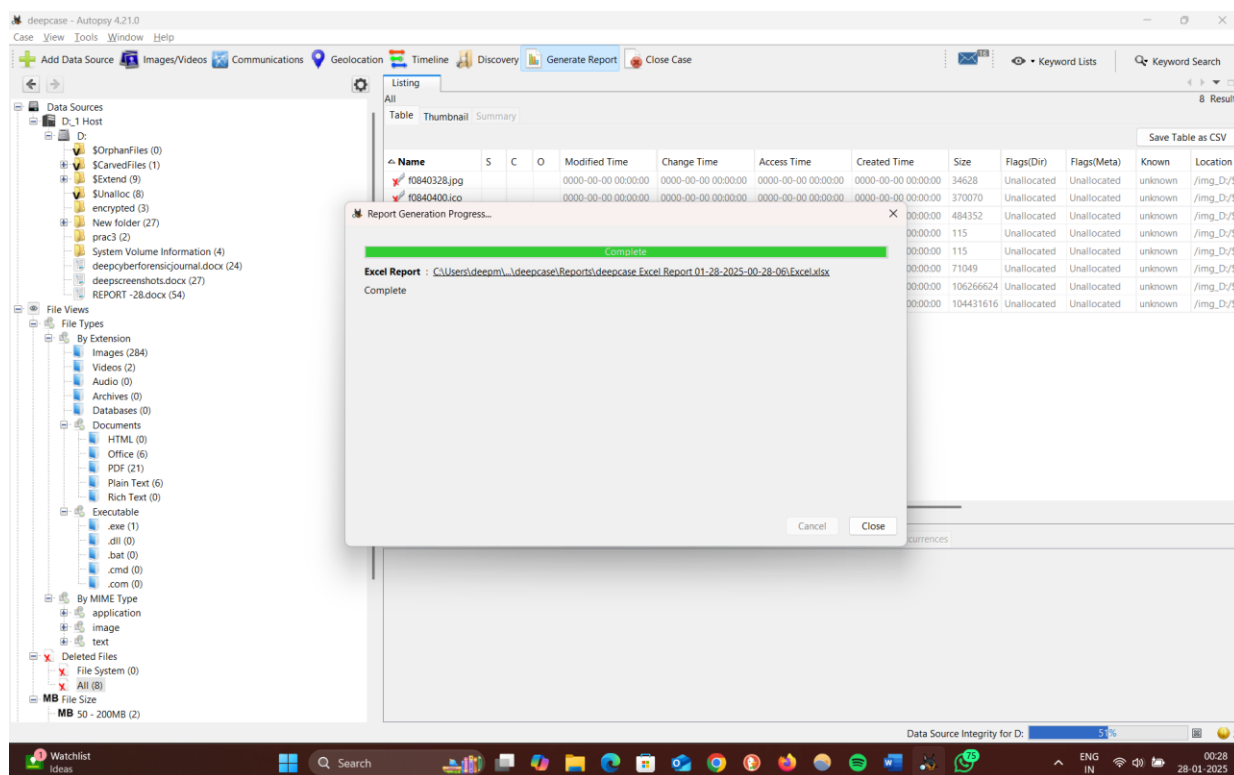
Step 16: Click on generate report from Autopsy window and select the Excel format and click on next.



Step 17: This window will appear.



Step 18: Now the report is generated so click on the close button. We can see the Report on Report Node



## Step 19: Click on report

The screenshot shows the DeepCase Autopsy 4.21.0 interface. The 'Listing' tab is active, displaying a table of reports. The table has columns: Source Module Name, Report Name, Created Time, and Report File Path. One report is listed: 'Excel Report' with a created time of '2025-01-28 00:28:08 IST' and a file path of 'C:\Users\deepm\Downloads\causedoc\deepcase\Reports\deepcase Excel Report 01-28-2025-00-28-08\Excel.xlsx'. The bottom pane shows the 'Hex' tab with a hex dump of data.

| Source Module Name | Report Name | Created Time            | Report File Path                                                                                        |
|--------------------|-------------|-------------------------|---------------------------------------------------------------------------------------------------------|
| Excel Report       |             | 2025-01-28 00:28:08 IST | C:\Users\deepm\Downloads\causedoc\deepcase\Reports\deepcase Excel Report 01-28-2025-00-28-08\Excel.xlsx |

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

```
0x00000000: 50 48 03 04 14 00 08 08 08 00 84 03 3C 5A 00 00 FF.....<2..
0x00000010: 00 00 00 00 00 00 00 00 00 00 13 00 00 00 5B 43 .....[C
0x00000020: 6F 4E 74 45 4E 74 5F 54 79 70 65 73 5D 2E 78 6D content_Type].xm
0x00000030: 4C CD 56 D0 4A 02 31 10 55 5F 65 C9 6D D9 44 6D 1...j.L...e-m.Dm
0x00000040: 28 A5 B0 7A D1 9F CB 56 A3 7D 80 34 99 75 83 F9 +...V...u...
0x00000050: 23 89 56 DF BE 93 55 2F 6A 15 2C 22 EC 55 36 3B #.V...0/j...06j
0x00000060: 67 CE F9 32 04 32 9A AC 8D 2E 56 10 A2 72 B6 22 g..2...V...e.*
0x00000070: 7D 2A 23 05 58 E1 A4 B2 F3 6A 7C CE BE CB 07 52 )#.X.....).R
0x00000080: C4 C4 A0 E4 28 55 A8 89 75 64 32 1E CD 36 1E 62 .....V...m2...b
0x00000090: 81 AD 36 56 A4 49 C9 3F 32 16 45 03 86 47 EA 3C ..67.1.72.E..G.C
0x000000A0: 58 AC D4 2E 18 9E 70 18 E6 CC 73 B1 E0 73 60 83 X....p...8..8'.
0x000000B0: 5E 6F C9 84 B3 09 6C 2A 53 F6 20 83 D1 33 D4 7C ^0....3*S...3.i
0x000000C0: 54 53 81 84 7D 9F 8D 78 C9 8D 77 48 7D 84 54 6C %...a...T.T3
```

## Practical 4

### A. Using File Recovery Tool (FTK) Creating Image. AIM: Creating a Forensic Image using FTK Imager.

#### **DESCRIPTION:**

A Forensic Image is most often needed to verify the integrity of the image after an acquisition of a Hard Drive has occurred. This is usually performed by law enforcement for court because, after a forensic image has been created, its integrity can be checked to verify that it has not been tampered with. The image is an identical copy of all the drive structures and contents.

#### Need for a Forensic Image

In today's world of crime, many cases have been solved by using this technique, as evidence apart from what is available through an operating system, has been found using this technique, as incriminating data might have been deleted to prevent discovery during the investigation. Unless that data is overwritten and deleted securely, it can be recovered.

One of the advantages includes the prevention of the loss of critical files.

When you suspect a custodian of deleting or altering files. A complete forensic image will, to a certain extent, allow you to recover deleted files. It can also potentially be used to identify files that have been renamed or hidden.

#### What Is FTK Imager?

FTK Imager is a tool for creating disk images and is absolutely free to use. It was developed by The Access Data Group. It is a tool that helps to preview data and for imaging.

With FTK Imager, you can:

Create forensic images or perfect copies of local hard drives, floppy and Zip disks, DVDs, folders, individual files, etc. without making changes to the original evidence. Preview files and folders on local hard drives, network drives, floppy diskettes, Zip disks, CDs, and DVDs.

You can also preview the contents of the forensic images that might be stored on a local machine or drive. View and recover files that have been deleted from the Recycle Bin, but have not yet been overwritten on the drive.

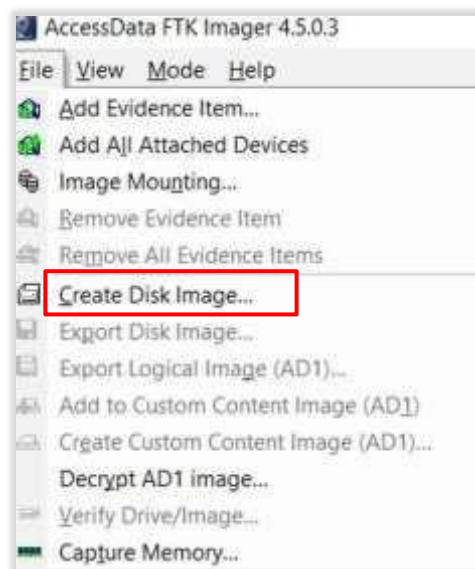


To create a forensic image using FTK imager we need to carry out following steps:-

**Step 1:** Open the AccessData FTK Imager.

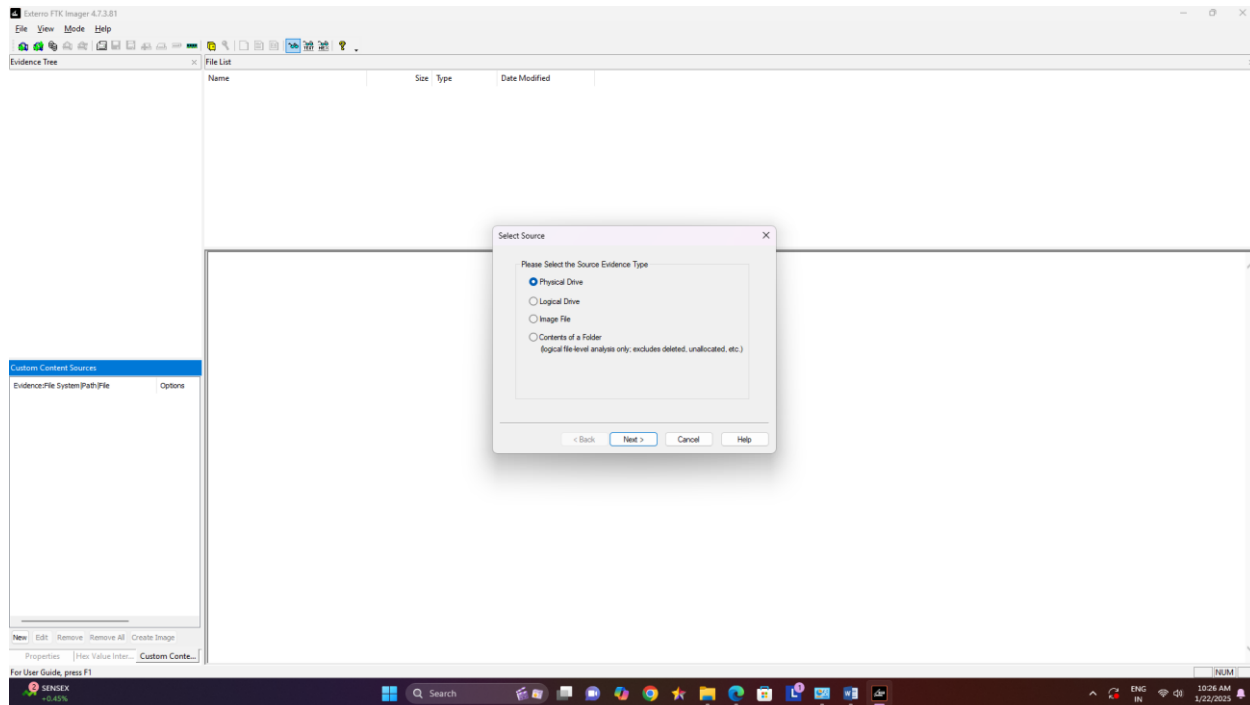


**Step 2:** Go to File → Click on Create Disk Image.

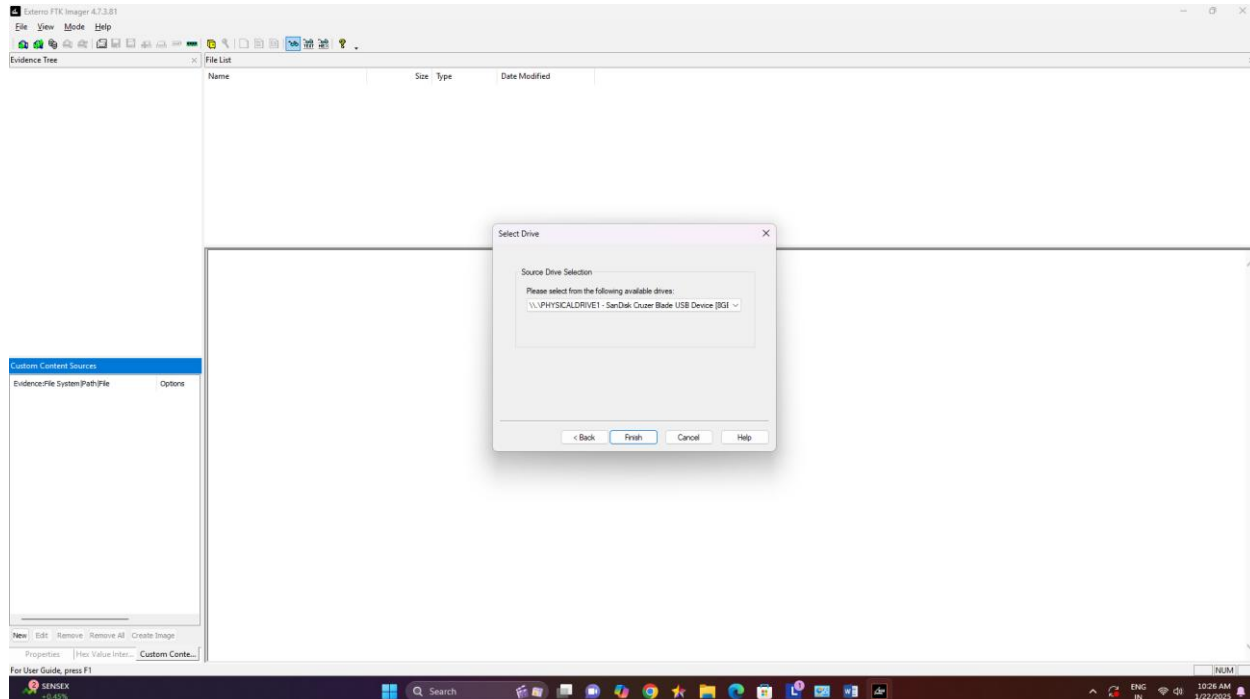




**Step 3:** After clicking on the Create Disk Image you will need to select the source evidence type you want to make an image of and click Next. So here we will select the Physical Drives as shown below.

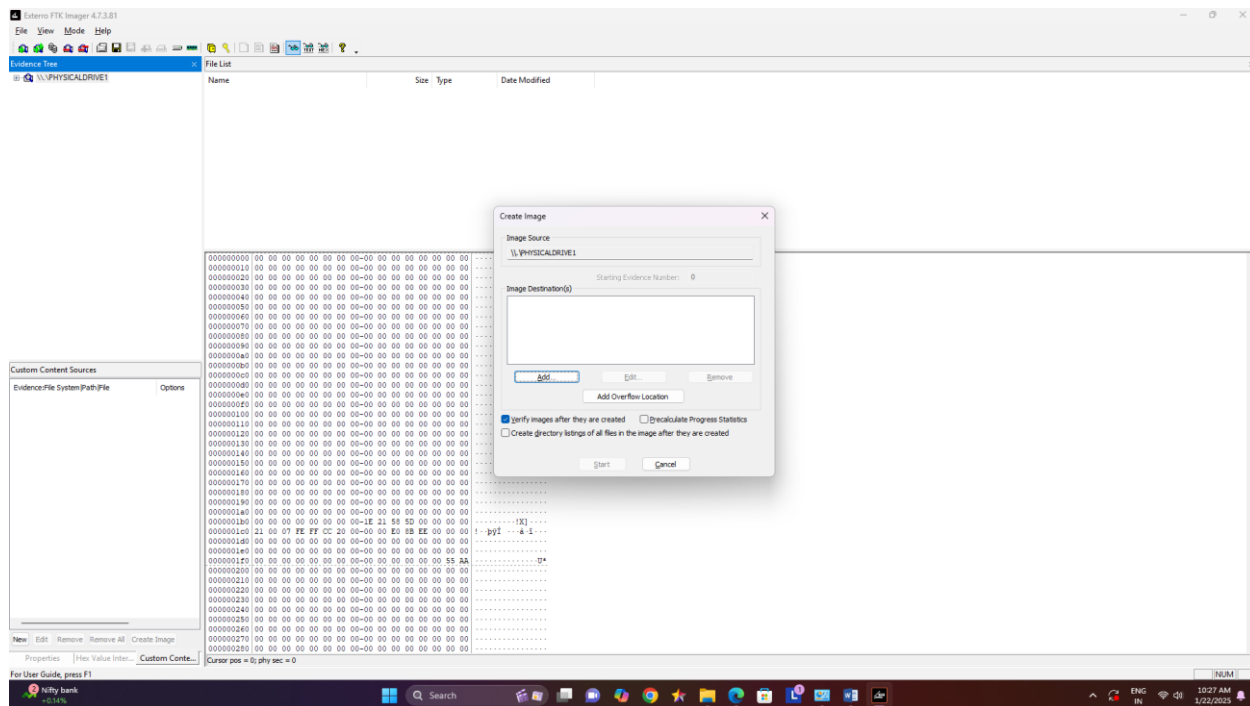


**Step 4:** So here we are going to select the source drive selection. Here you need to select USB Device → Click on Finish.

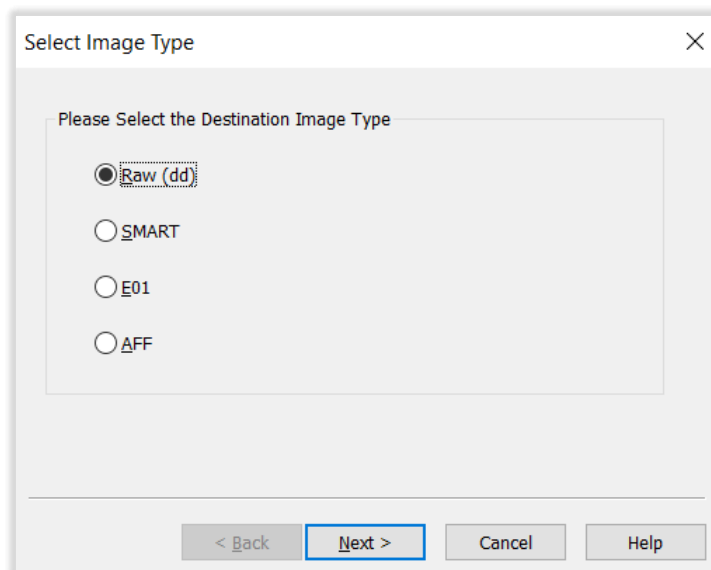


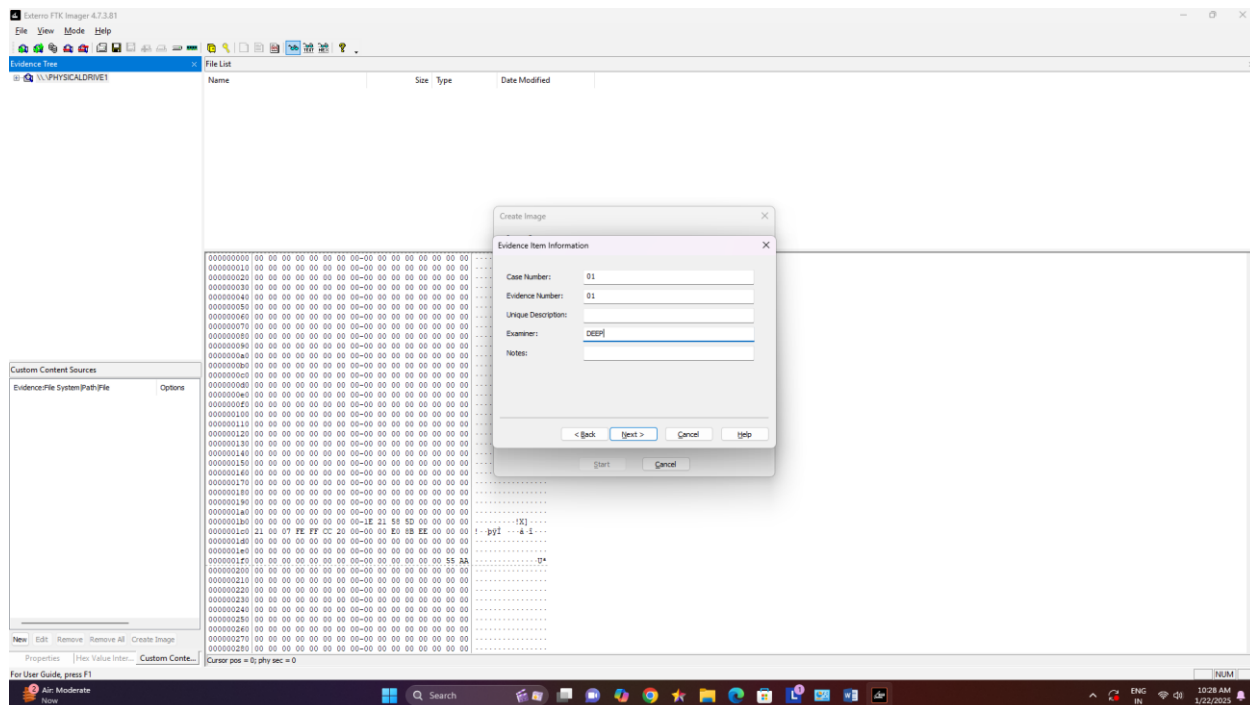
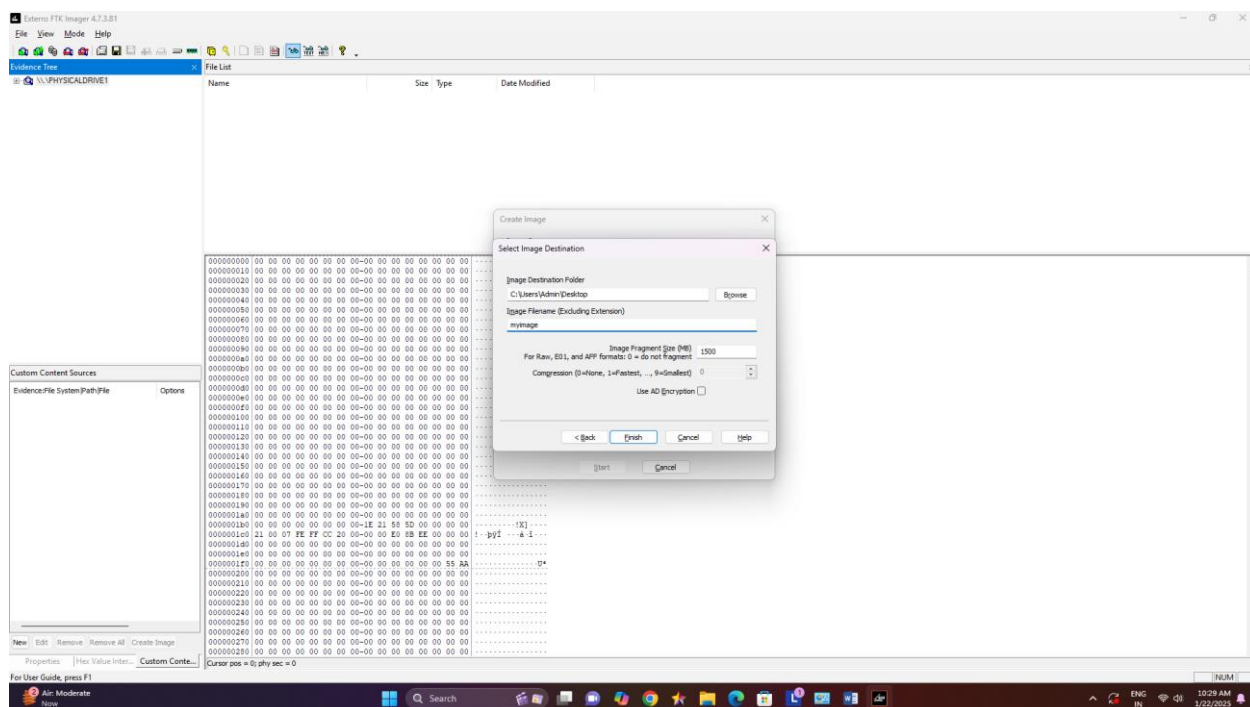
In the below Screenshot you can see the data in Physical Drive

**Step 5:** After clicking on finish, Pop-up window will be displayed as shown below simply click on Add and do the following changes.

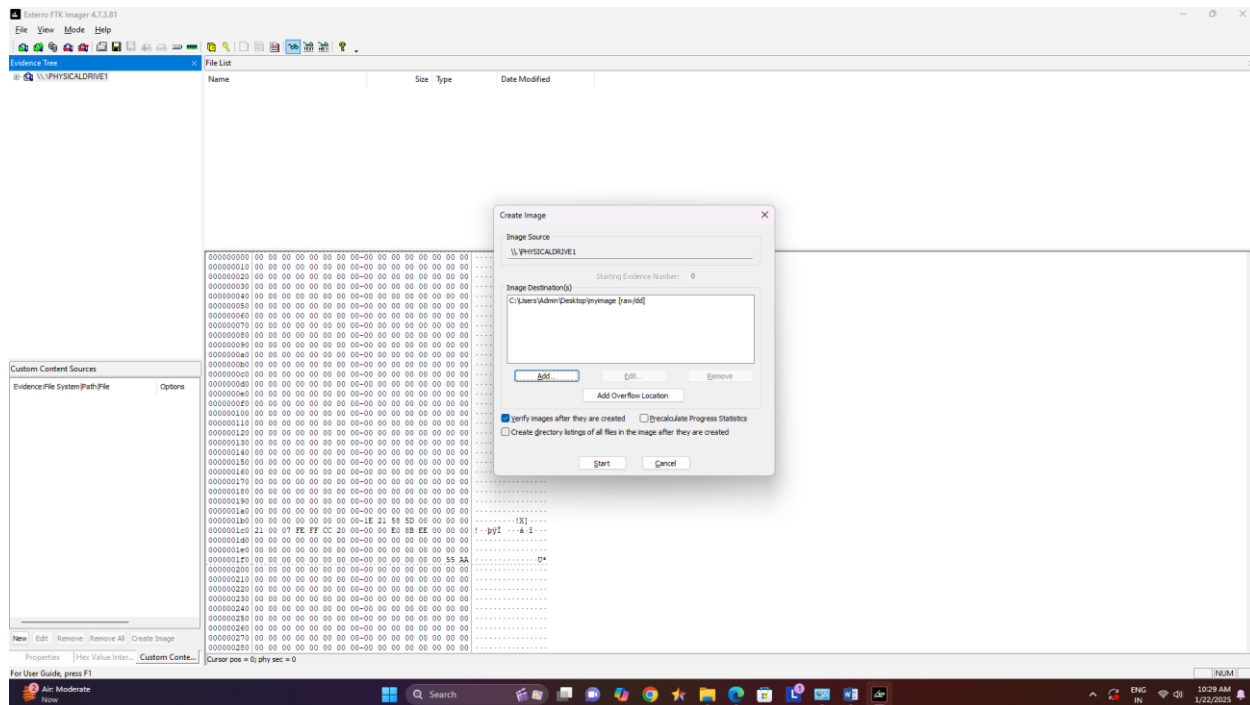


**Step 6:** Select Raw as shown below and click Next.

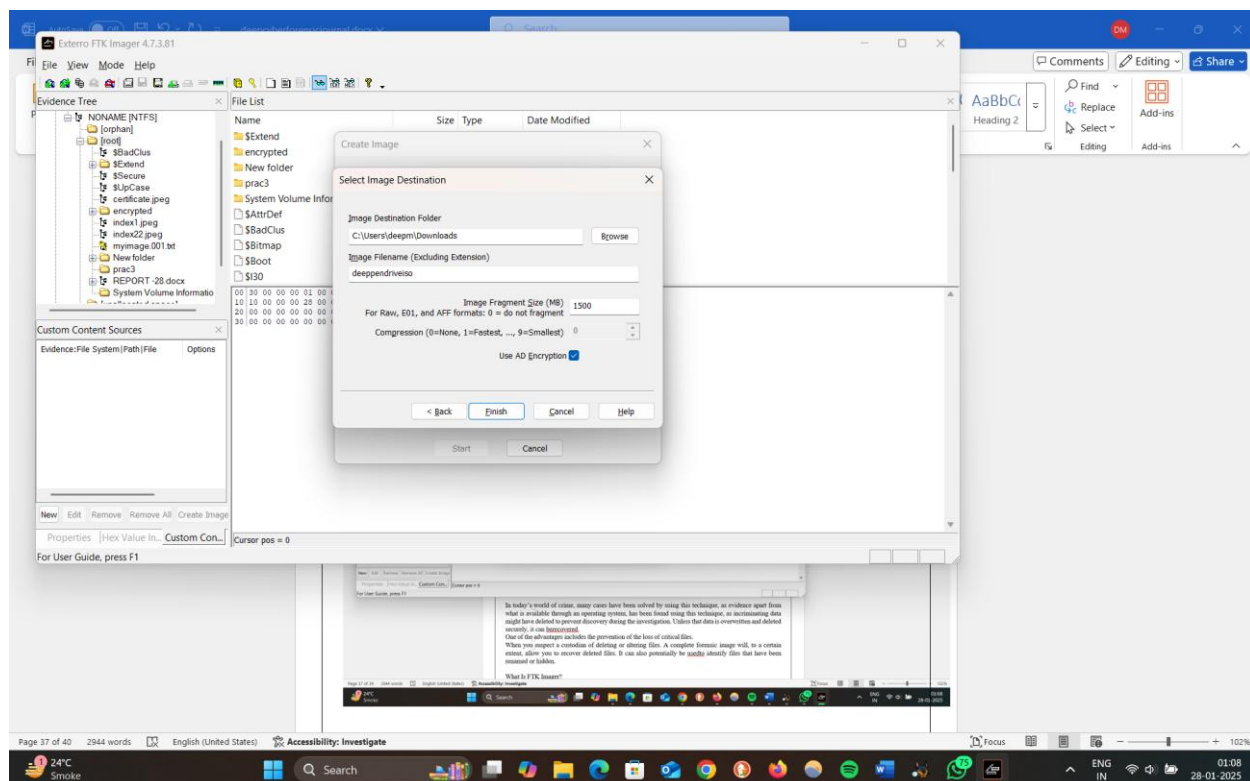


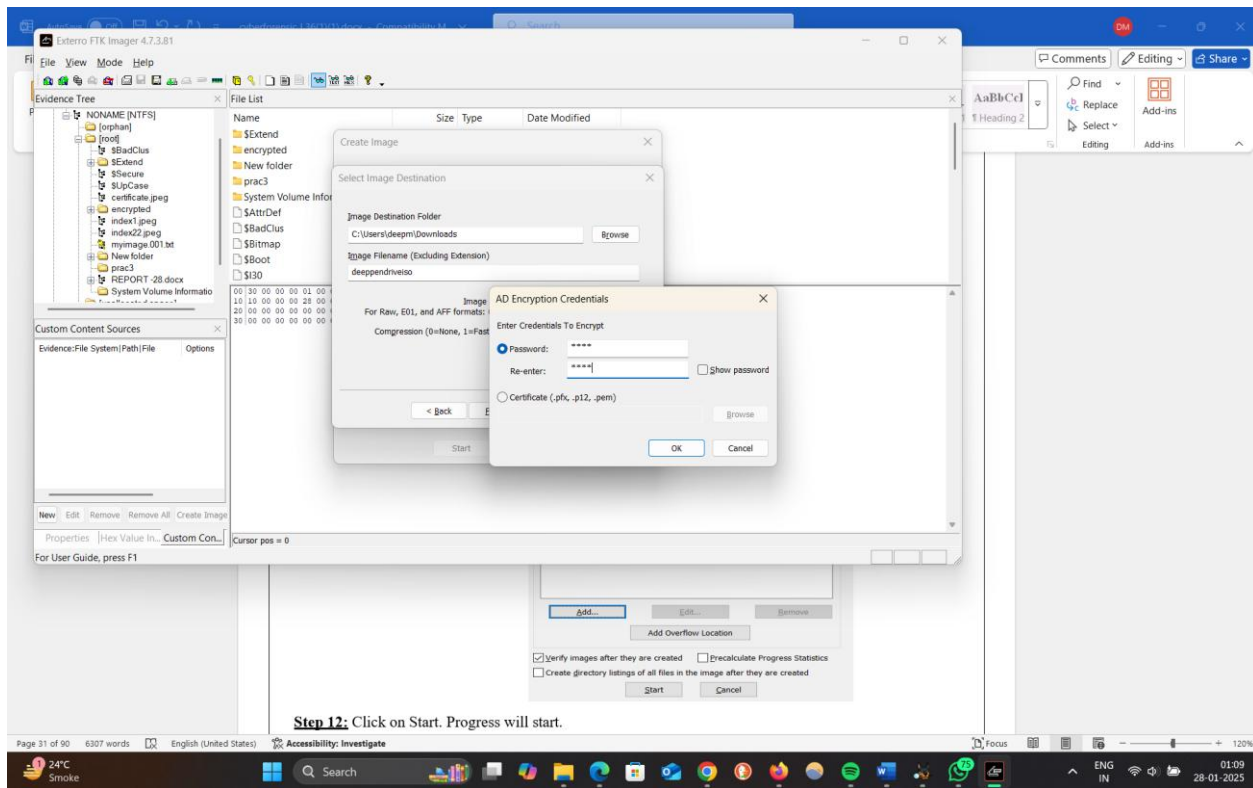
**Step 7:** Do the following changes as shown below. Click Next.**Step 8:** Clicking Next you next you need to make the new folder as shown below and clickok.

**Step 9:** Browse the folder and write the Image Filename as shown below. Click on Start.

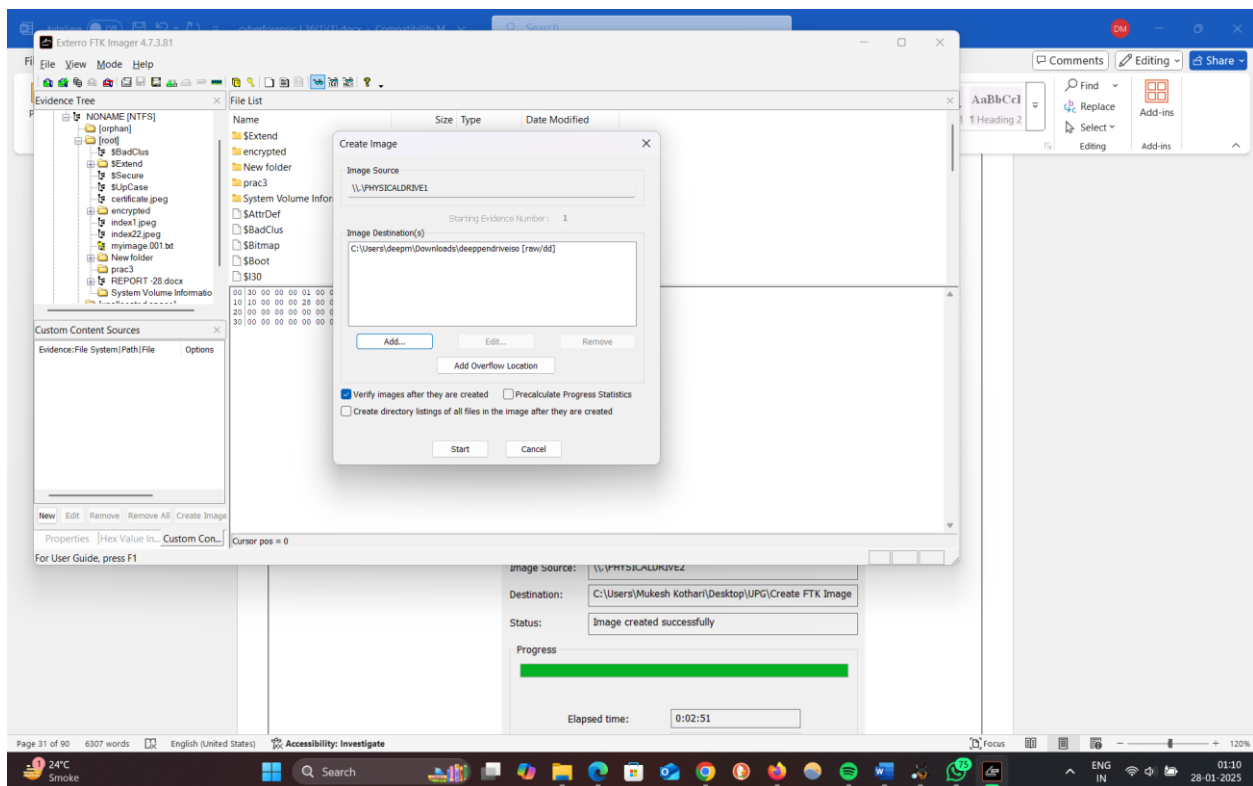


**Step 10:** Select on Use AD Encryption and type the password and Click Ok.

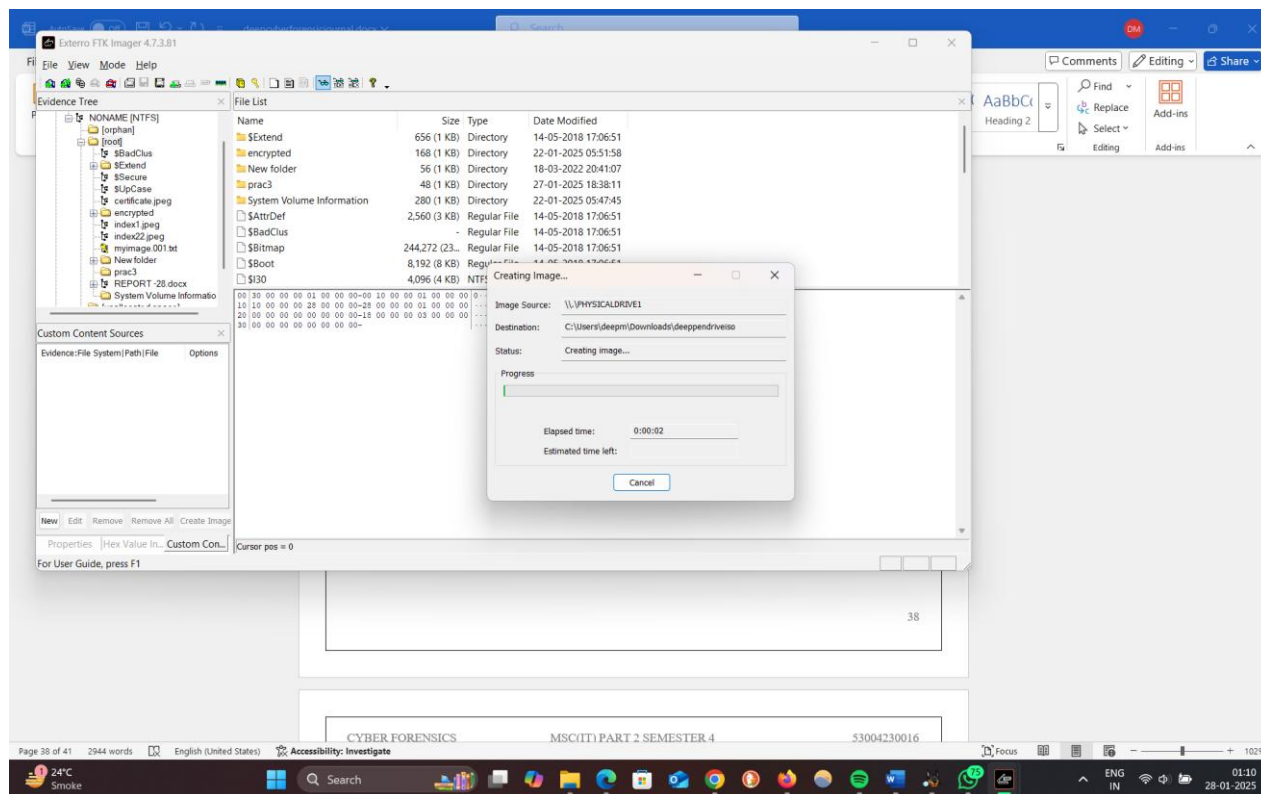




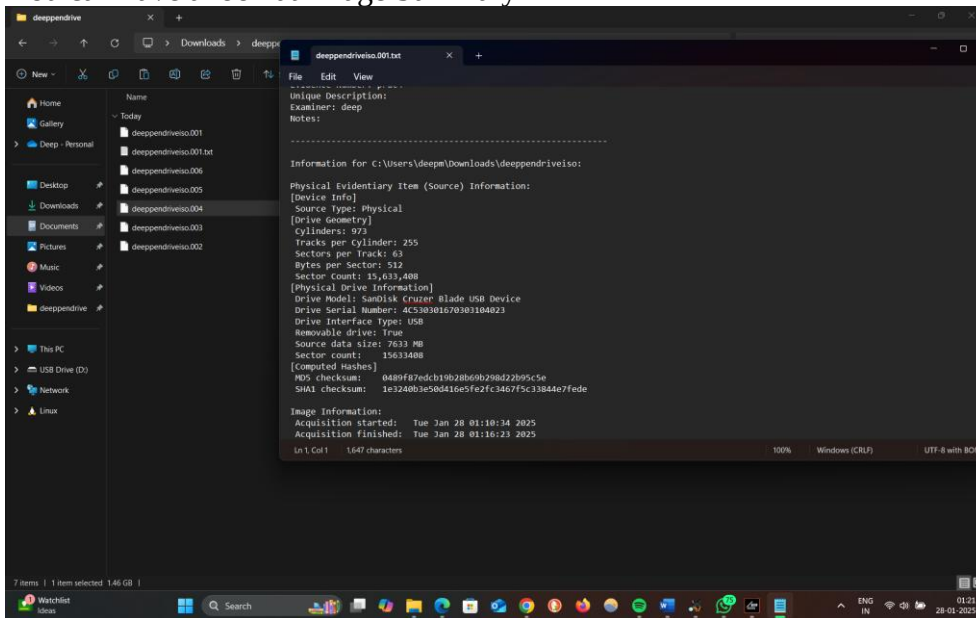
**Step 11:** You can see the Image Destination Path



**Step 12:** Click on Start. Progress will start.



You can have a look at Image Summary



### **Step 13:**

Go to the destination where you have created the folder. In our case since we have created the folder up on the desktop, open it and the image will be created.

## Practical 5

### A. Web Attack Detection

#### Problem Statement: Detecting web attack on the data packets.

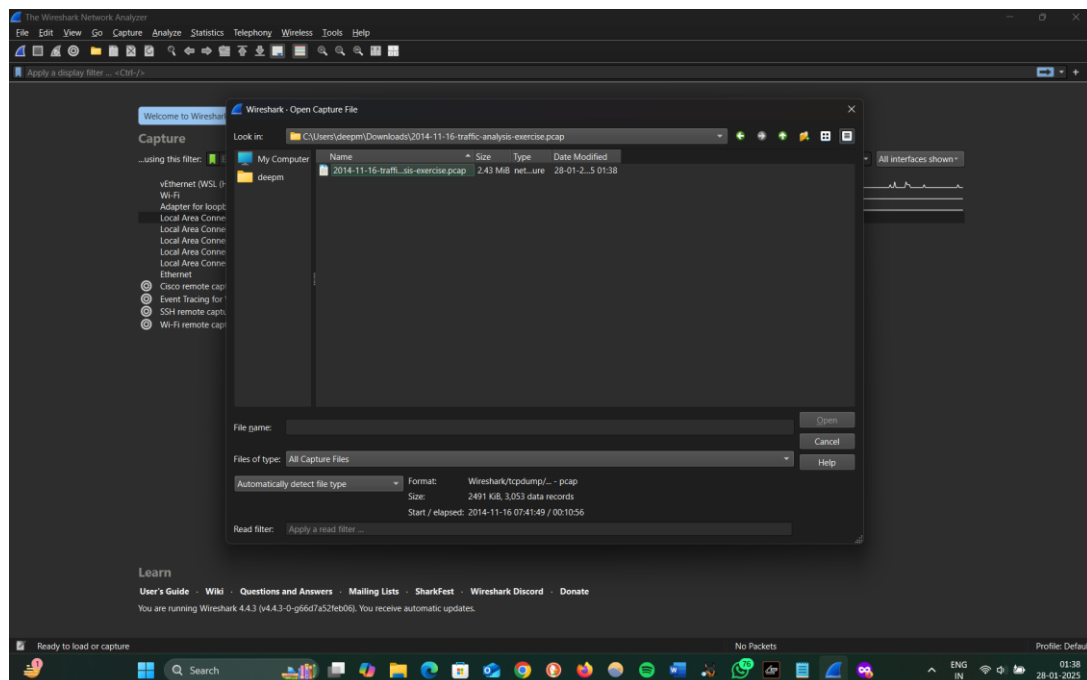
#### Theory:

Wireshark is a packet sniffer and analysis tool. It captures network traffic on the local network and stores that data for offline analysis. Wireshark captures network traffic from Ethernet, Bluetooth, Wireless (IEEE.802.11), Token Ring, Frame Relay connections, and more.

Wireshark allows you to filter the log either before the capture starts or during analysis, so you can narrow down and zero in on what you are looking for in the network trace.

#### Steps:

Here we have used Sample Packets for Detection from <https://www.malware-traffic-analysis.net/>





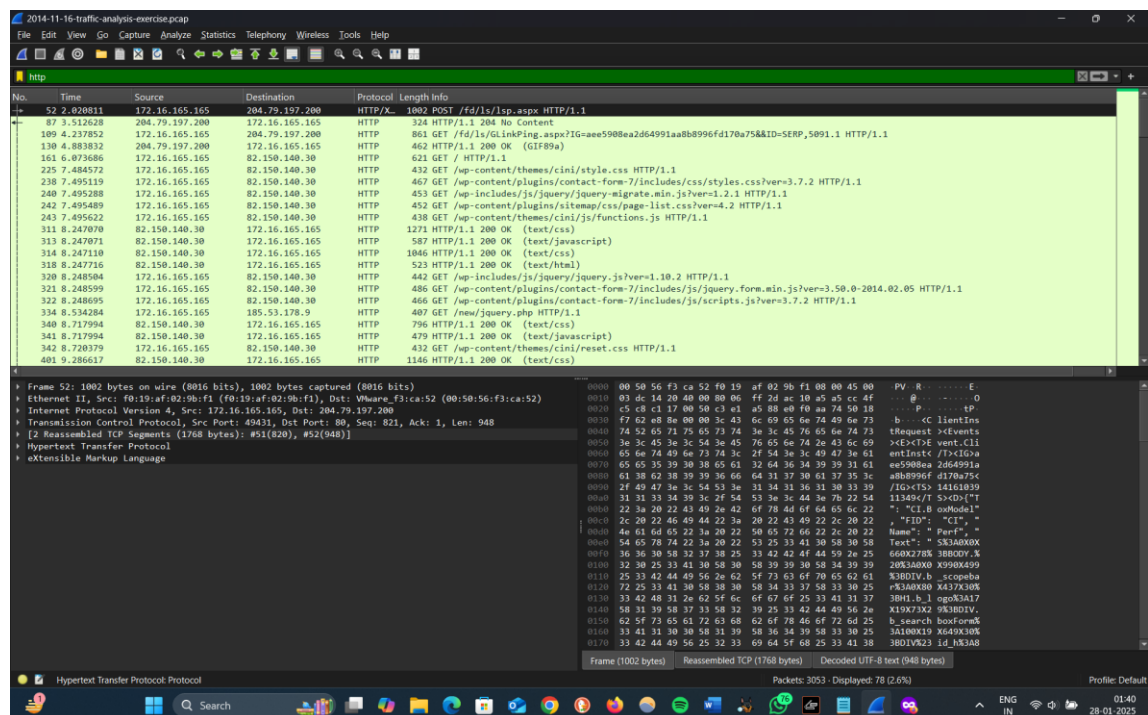
The image shows a Wireshark packet capture analysis of a network traffic file named '2014-11-16-traffic-analysis-exercise.pcap'. The main display area shows a list of captured packets. The first packet (No. 1) is a TCP Reset (RST) from source 204.79.197.200 to destination 172.16.165.165, with sequence number 60 and window size 0. The subsequent packets (No. 2-22) are TCP Retransmissions from the same source to the same destination, with sequence numbers ranging from 60 to 220. The packets are filtered by 'Protocol' and 'Length'.

The packet details pane shows the following information for the selected packet (No. 1):

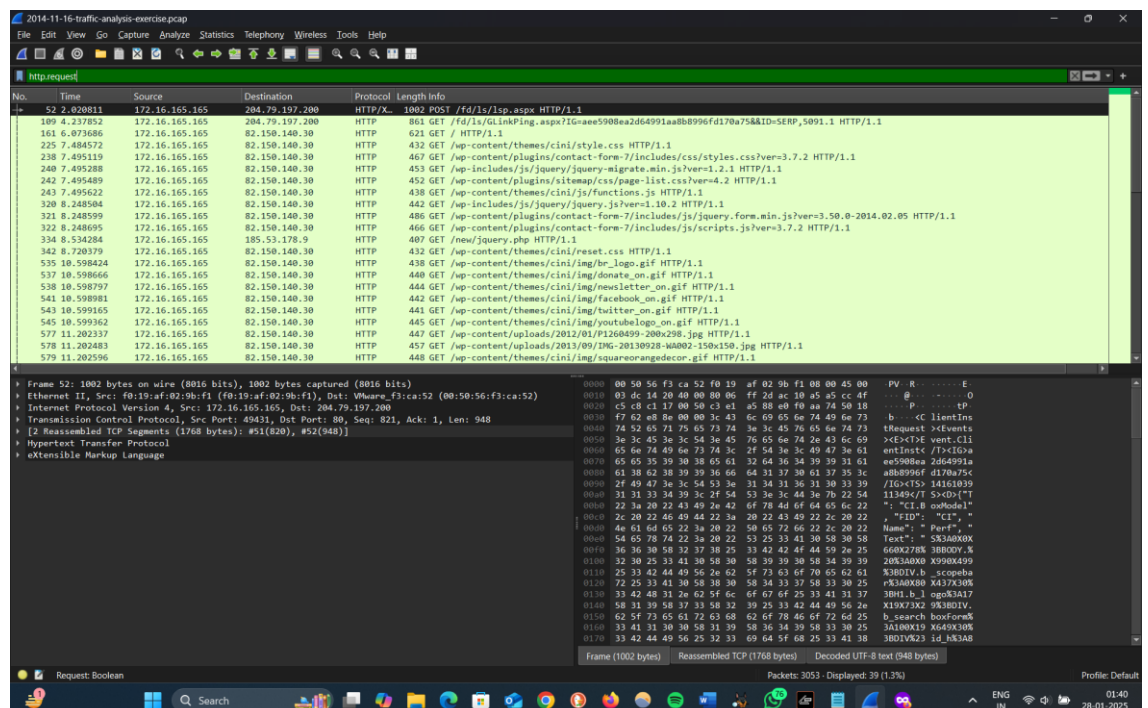
- Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits)
- Ethernet II, Src: VMware\_Flco52 (00:50:56:93:ca:52), Dst: f0:19:af:02:9b:f1 (f0:19:af:02:9b:f1)
- Internet Protocol Version 4, Src: 204.79.197.200, Dst: 172.16.165.165
- Transmission Control Protocol, Src Port: 80, Dst Port: 49433, Seq: 0, Ack: 1, Len: 0

The packet bytes pane shows the raw data in hexadecimal and ASCII format. The first few bytes are 0000, 0020, 01c2, 0000, 0000, 551c, cc4f, c5c8, ac10, a5a5, 0050, c119, 2044, 312c, 3700, 4d64, 6012, faf0, 21c9, 0000, 0204, 05b4, 0000.

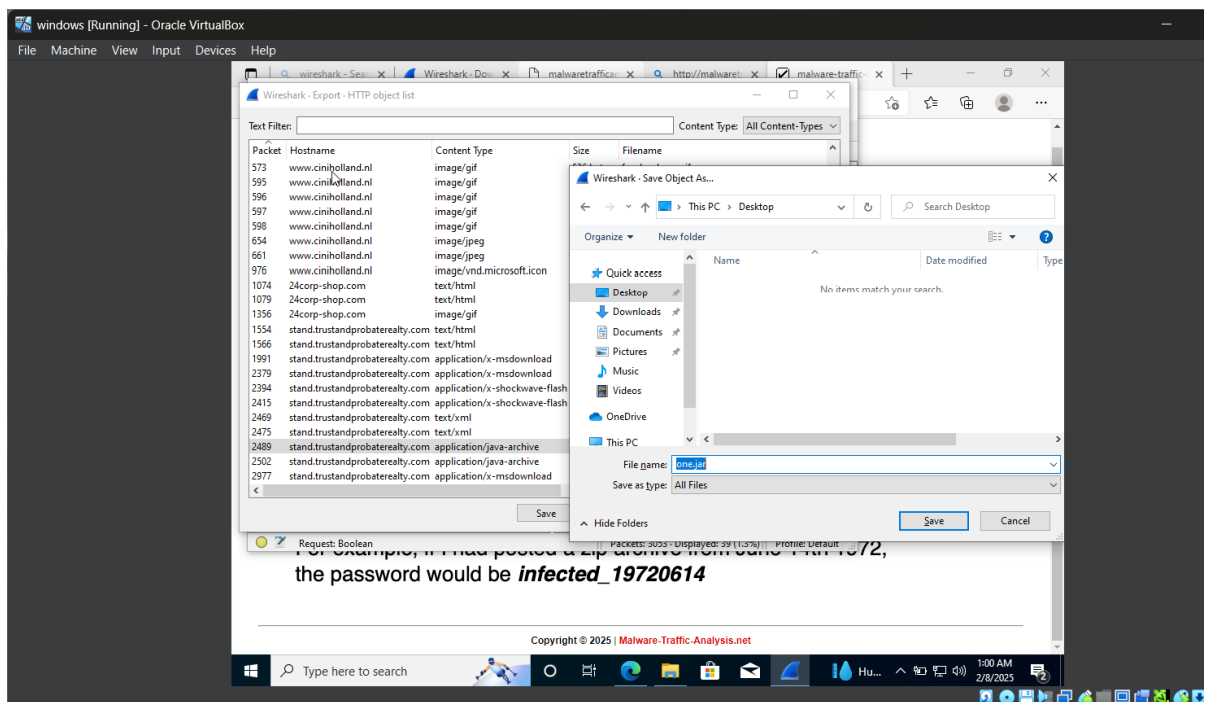
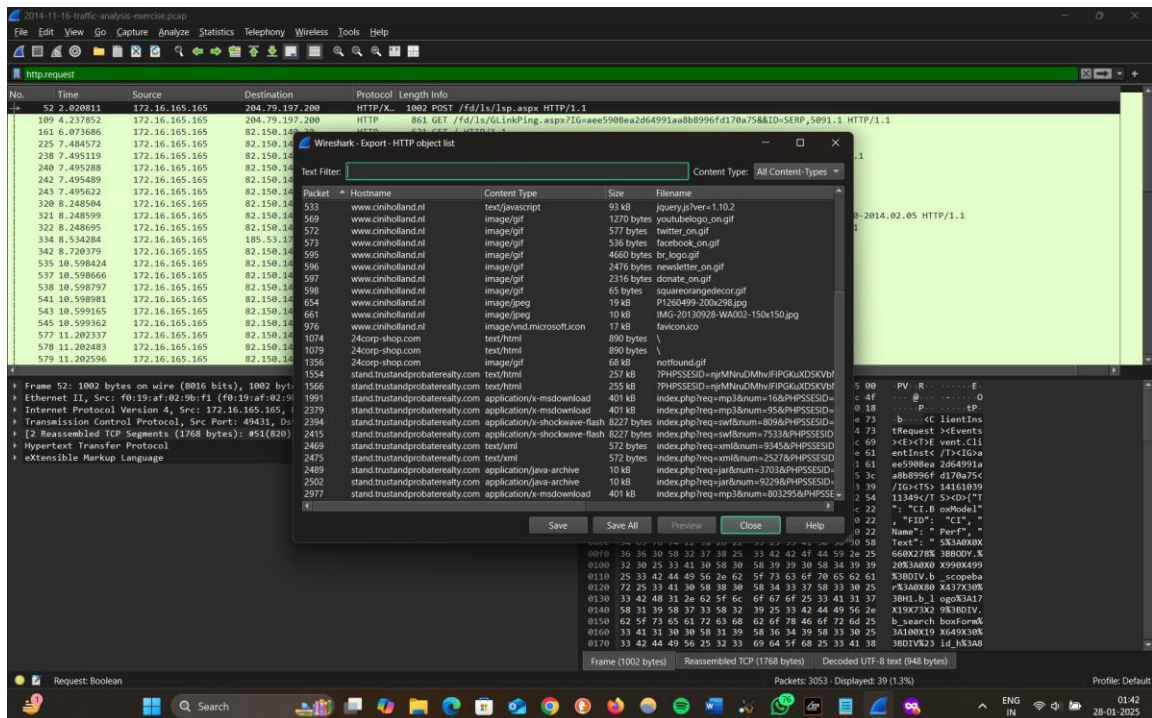
We have filter the packets.  
Now only http packets are only been displayed.

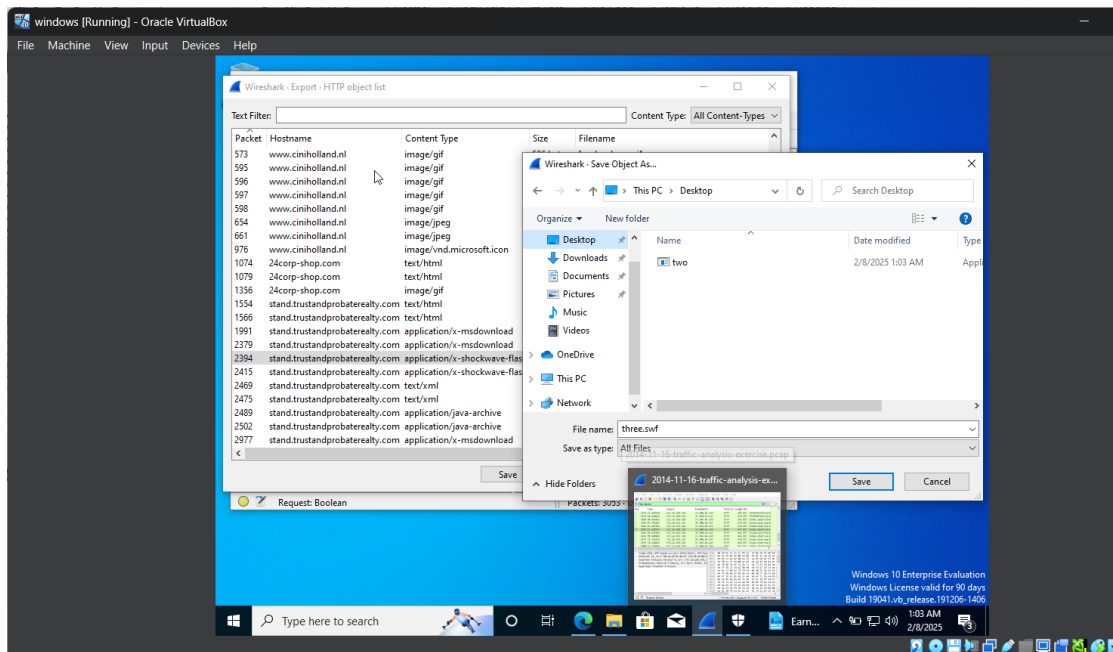
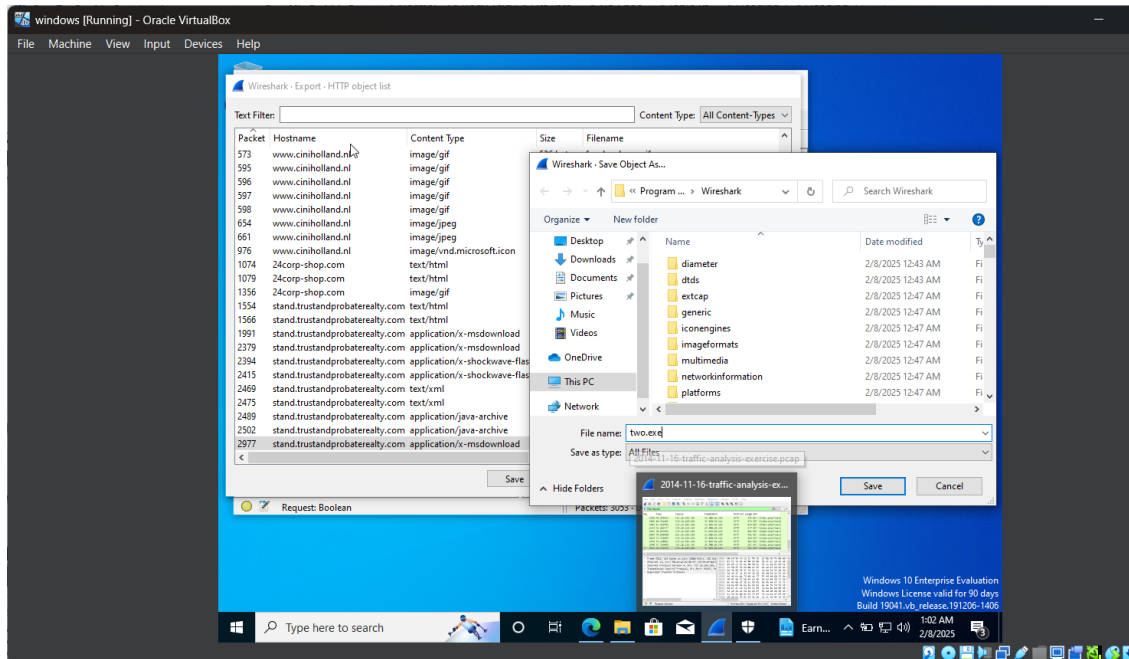


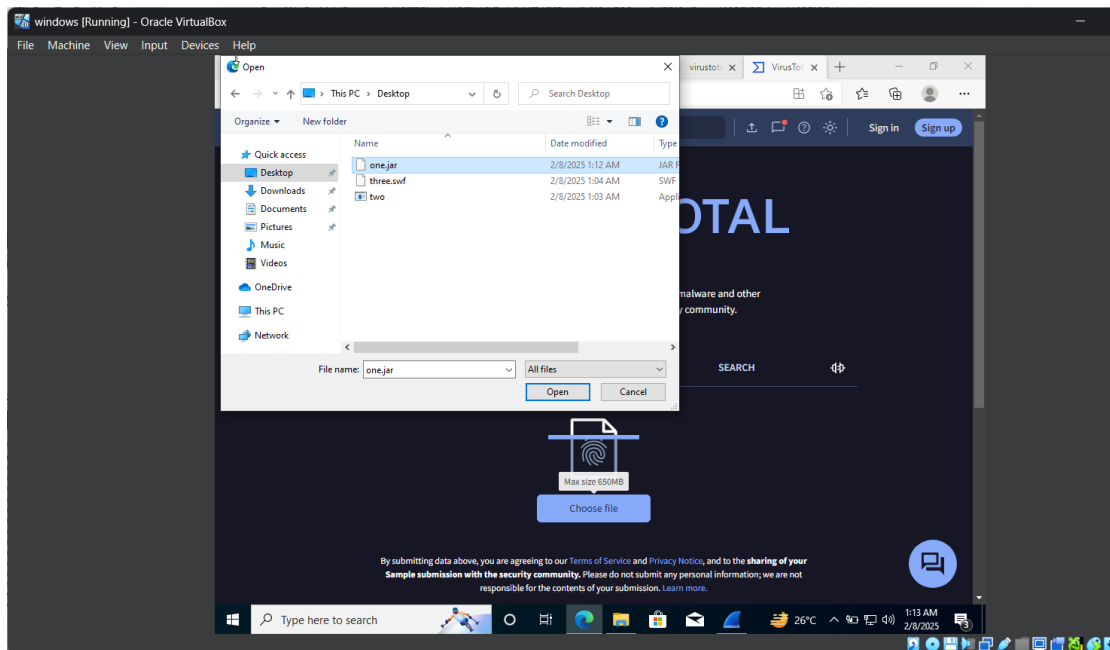
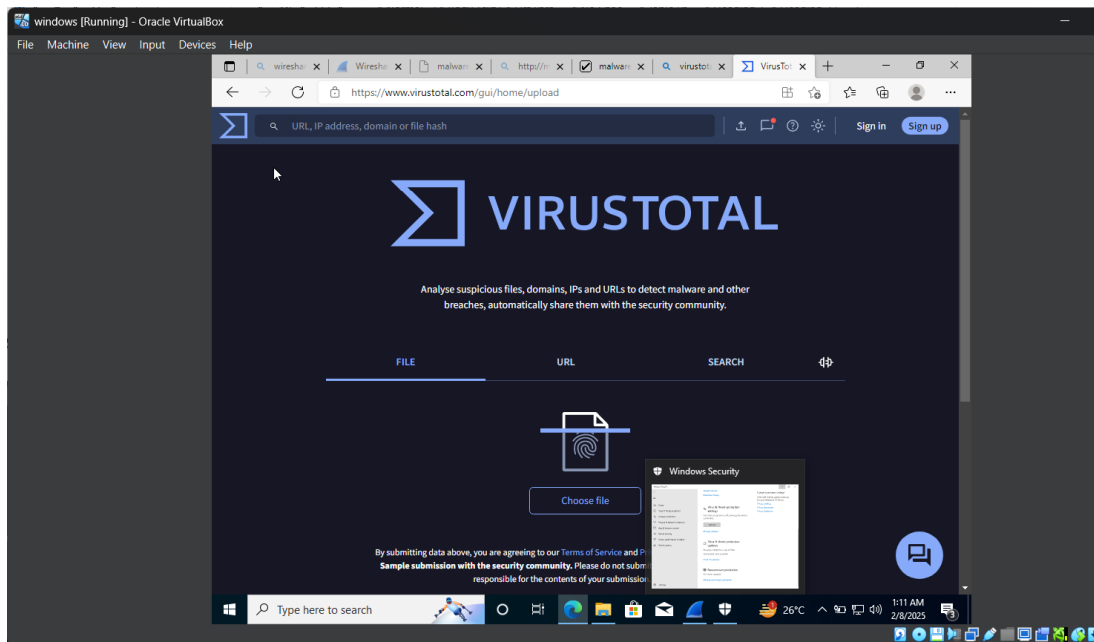
Narrow down the search to http.request which will show only the GET/POST requests

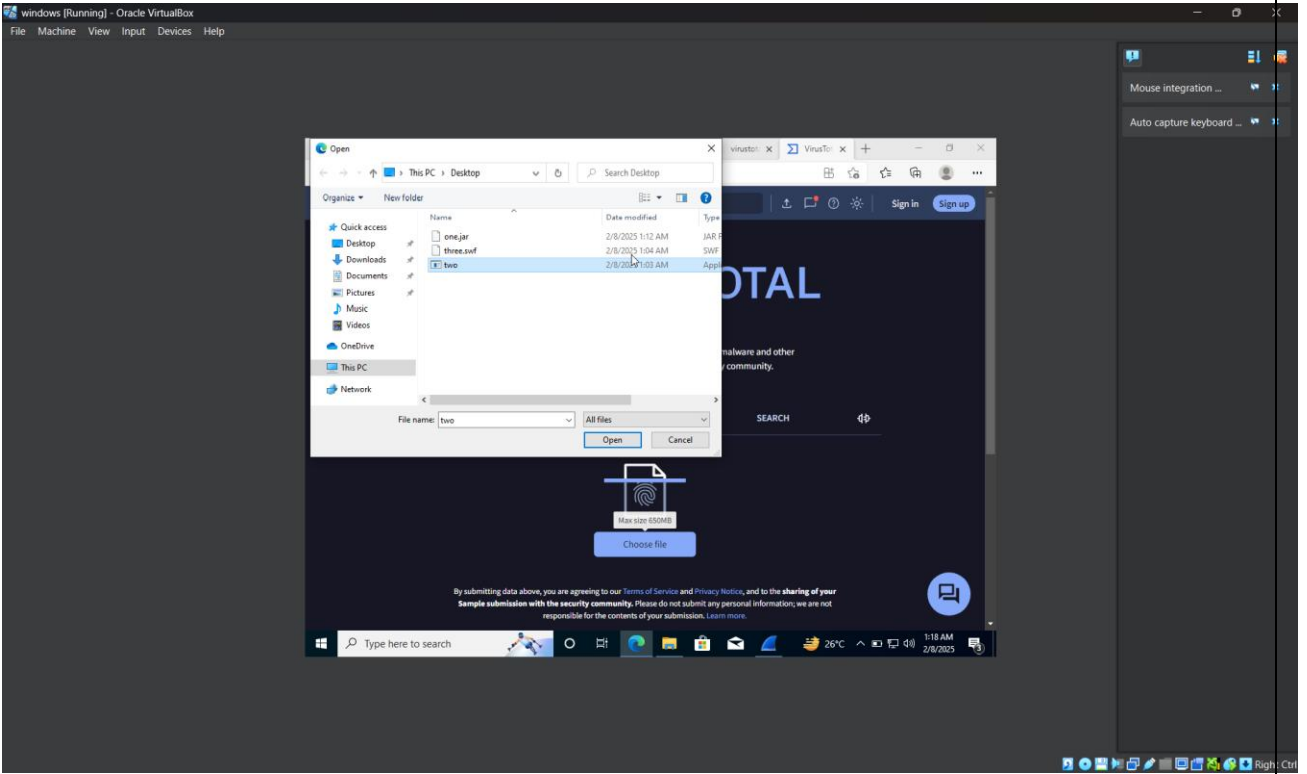
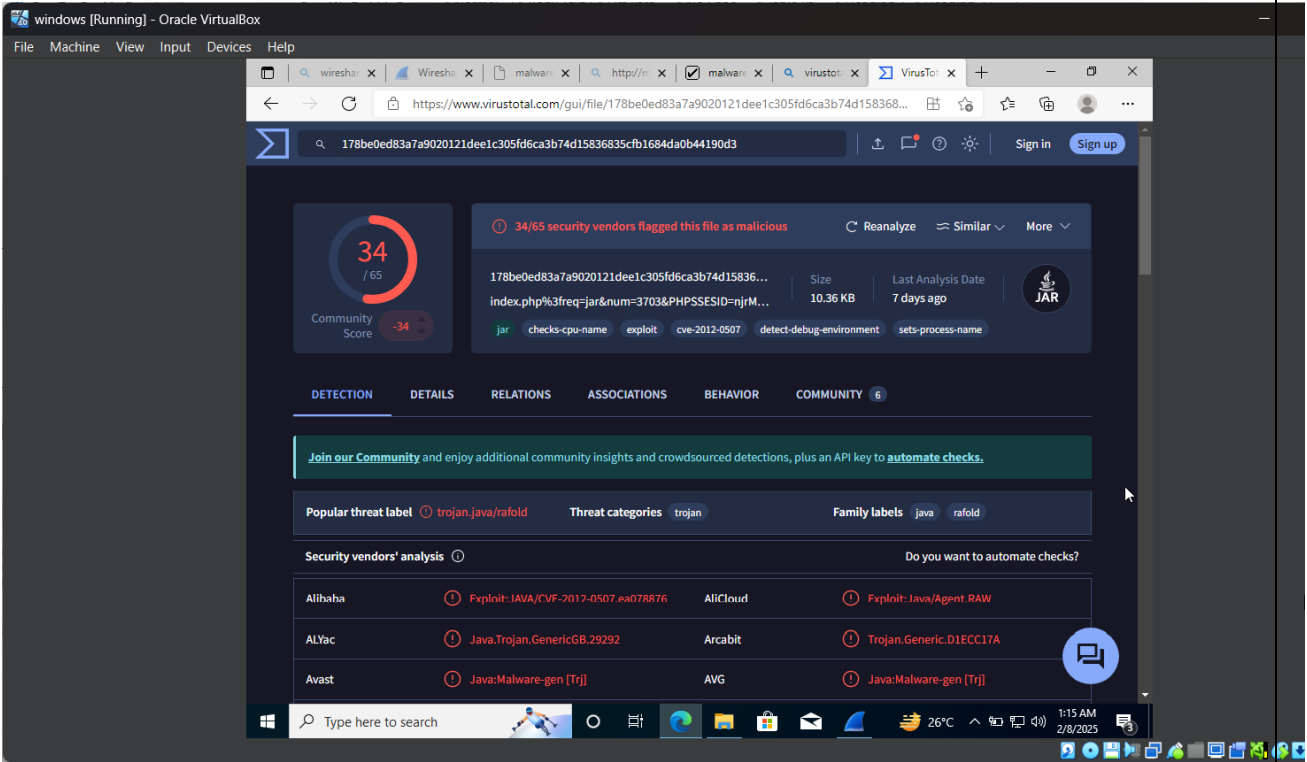


Now go to File -> Export Objects -> HTTP You will get a Window like this Here you will see all the HTTP objects which are downloaded in this Packet

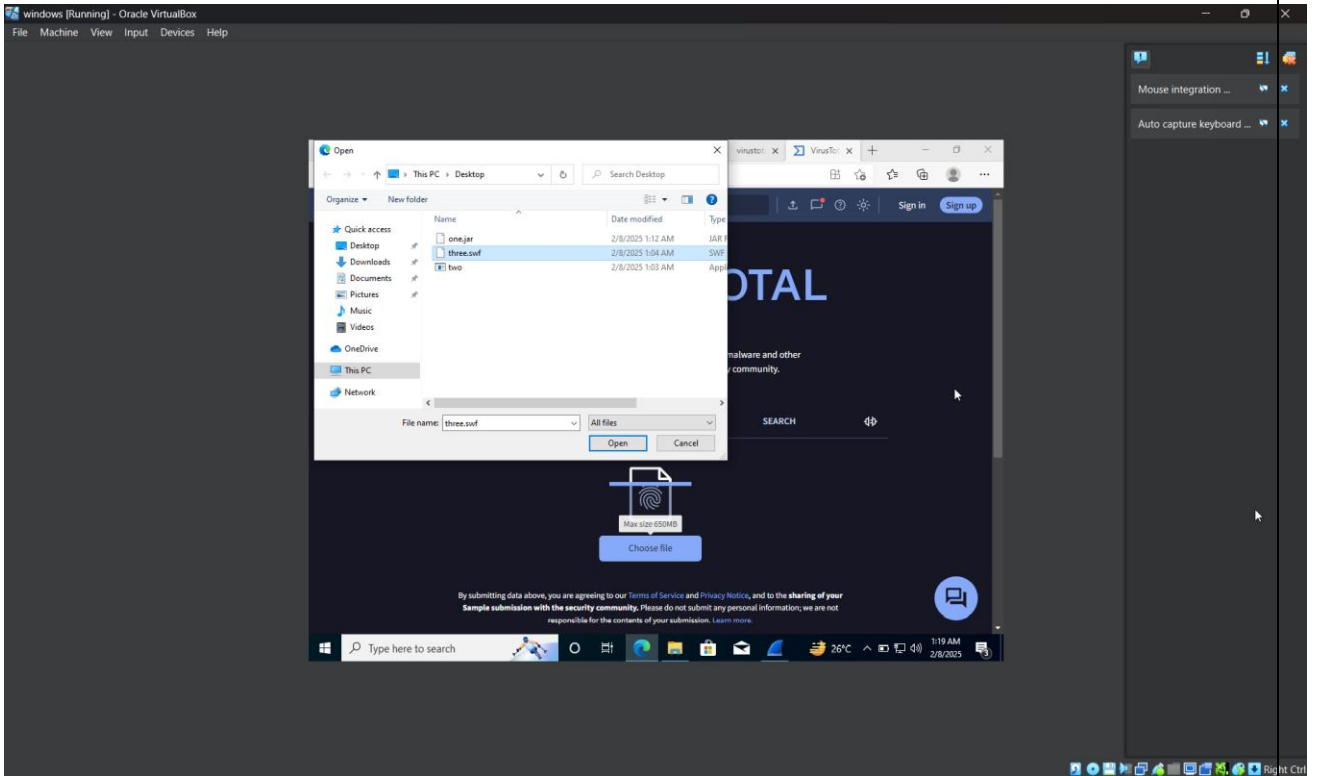
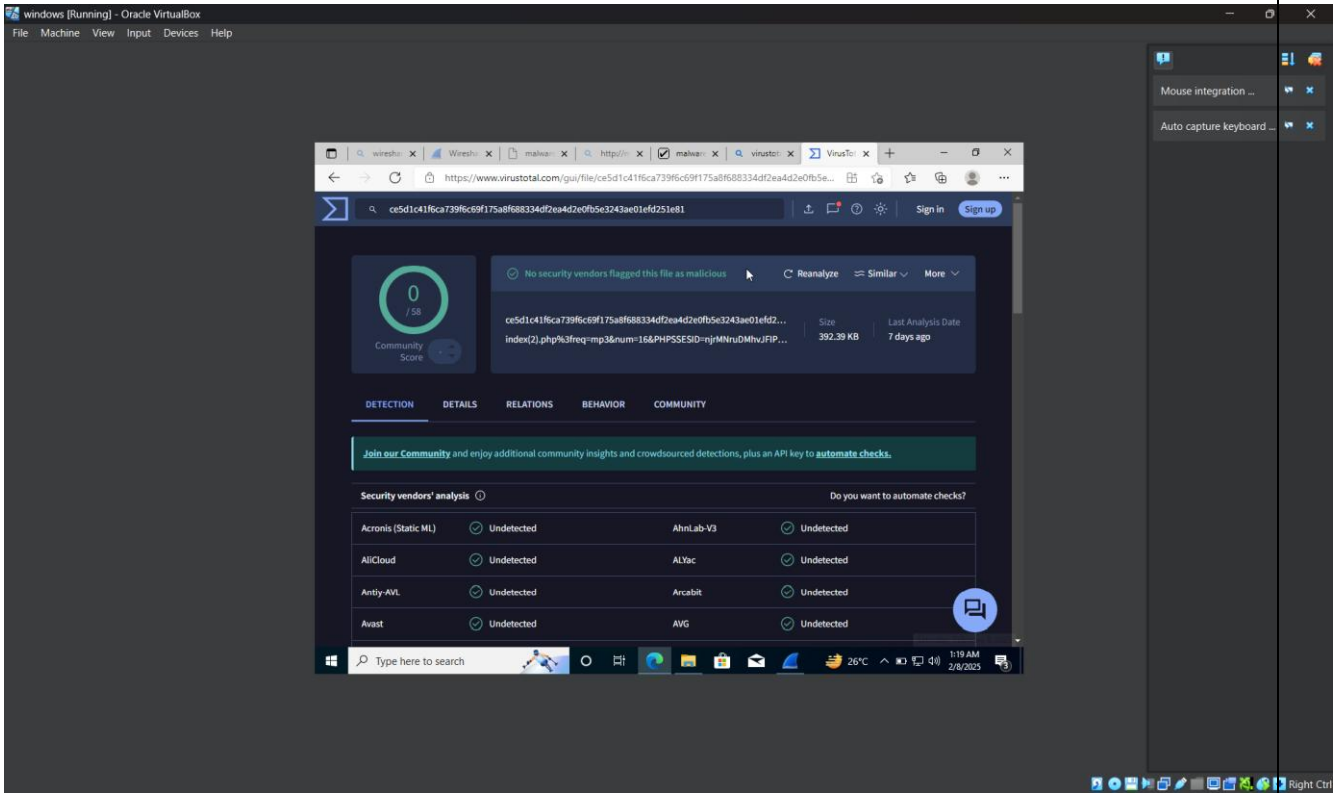












The screenshot shows a Windows VirtualBox environment. The top window is a web browser displaying the VirusTotal analysis page for the file `e2e33b802a0d939d07bd8291f23484c2f68ccc33dc065...`. The page indicates that 36 out of 60 security vendors flagged this file as malicious. The file size is 8.03 KB, and the last analysis was performed 7 days ago. The page also shows a community score of 36/60 and a threat label of `trojan.exhit/expl`. Below the analysis results, there is a table of security vendors' analysis, including AhnLab-V3, AlYac, Avast, and others, with their respective detection results.

The bottom window is Wireshark, showing a network capture. The packet list on the left shows several HTTP packets. The selected packet is a GET request from `37.200.69.143` to `172.16.165.165` on port 80. The packet details pane shows the following information:

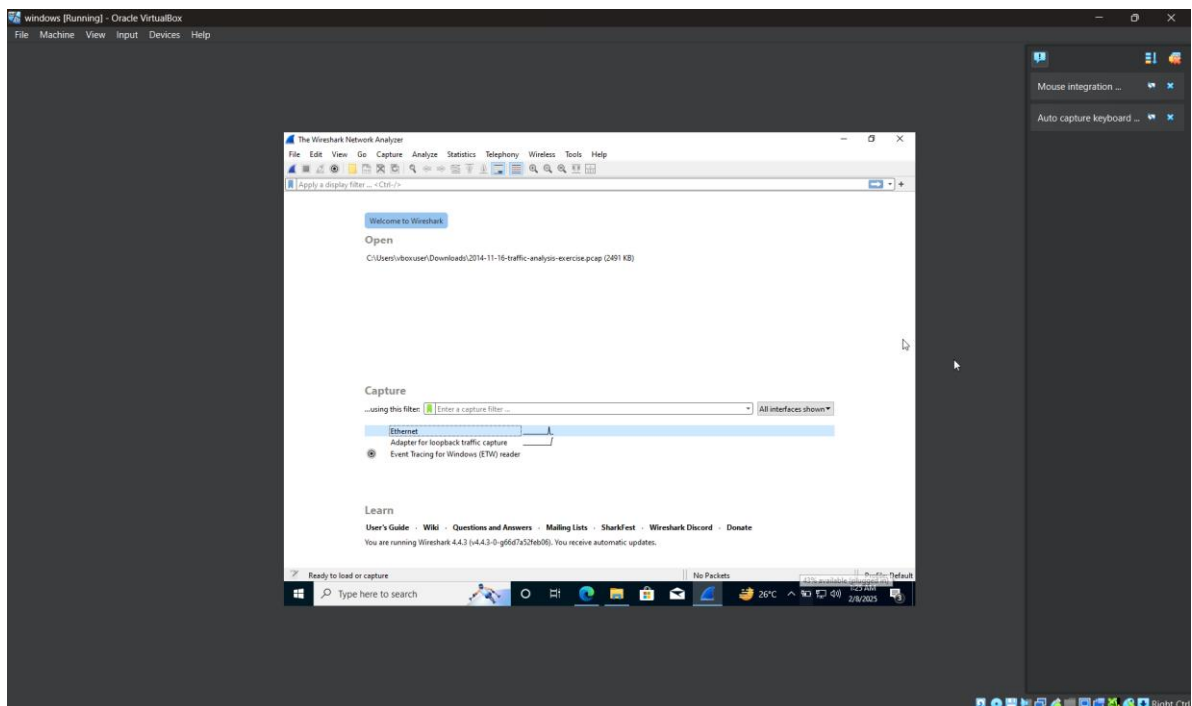
- Frame 2515: 259 bytes on wire (2072 bits), 259 bytes captured (2072 bits) on interface 0
- Ethernet II, Src: Vmware\_F3 (ca:52:00:50:56:f3:ca:52), Dst: f0:19:af:02:9b:f1 (f0:19:af:02:9b:f1)
- Internet Protocol Version 4, Src: 37.200.69.143, Dst: 172.16.165.165
- Transmission Control Protocol, Src Port: 80, Dst Port: 49457, Seq: 1, Ack: 298, Len: 205
- Hypertext Transfer Protocol

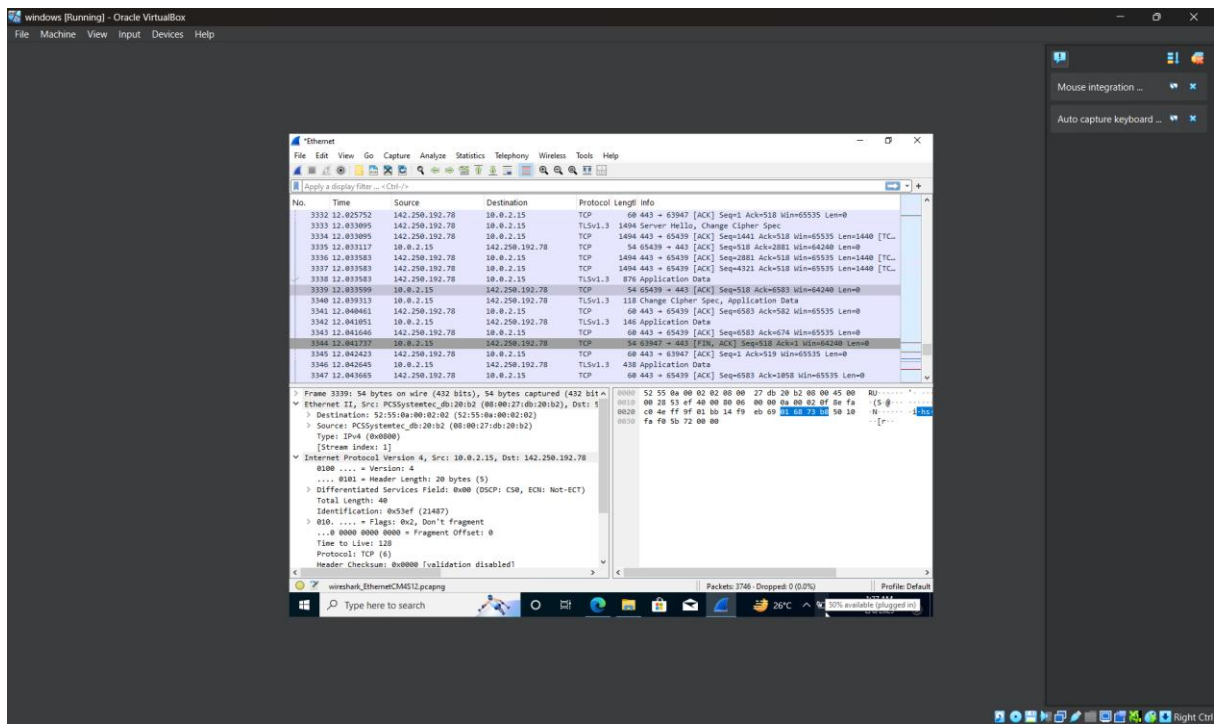
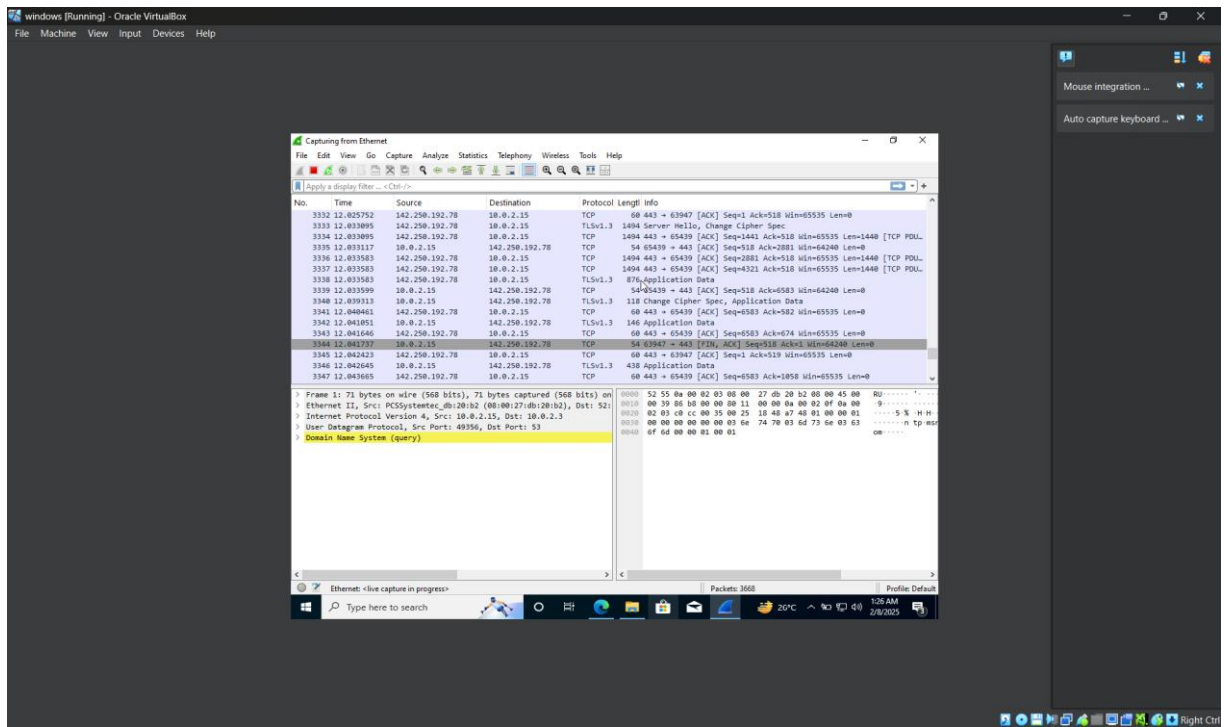
The packet bytes pane shows the raw data of the packet, including the HTTP request line: `GET /index.php?req=swf&num=809&PHPSESSID=njnm... HTTP/1.1`.

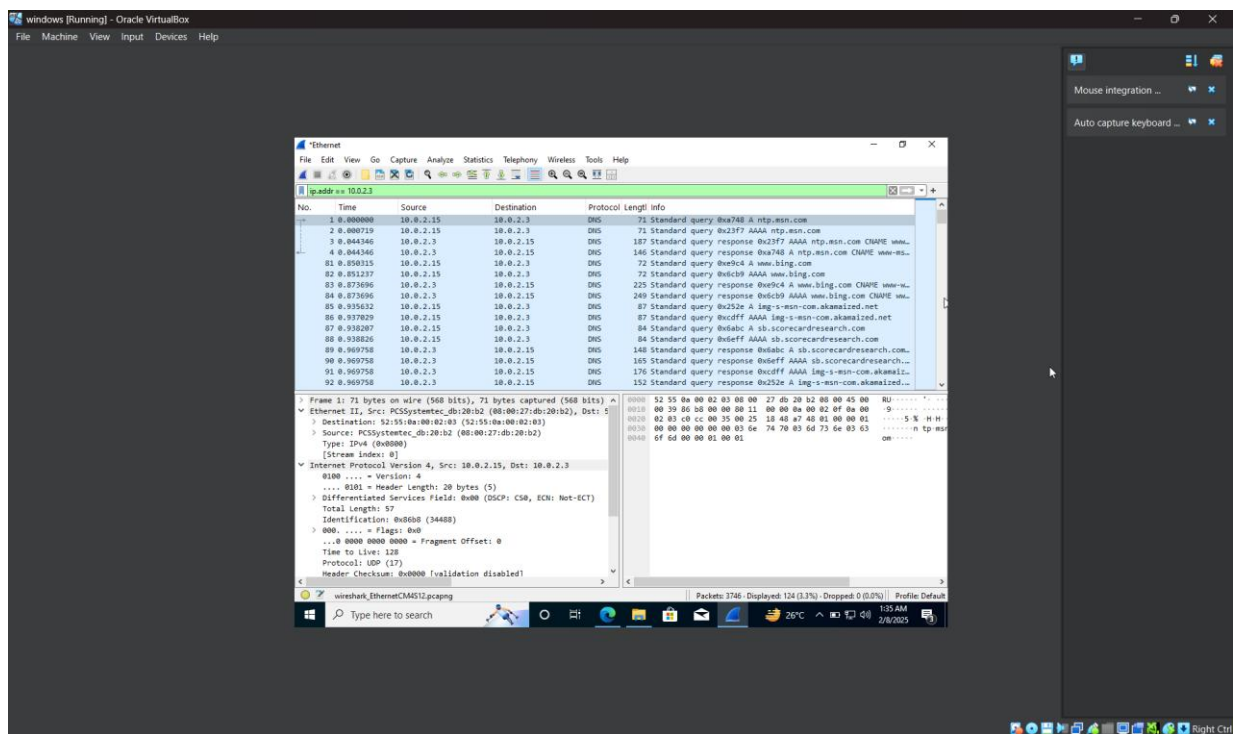
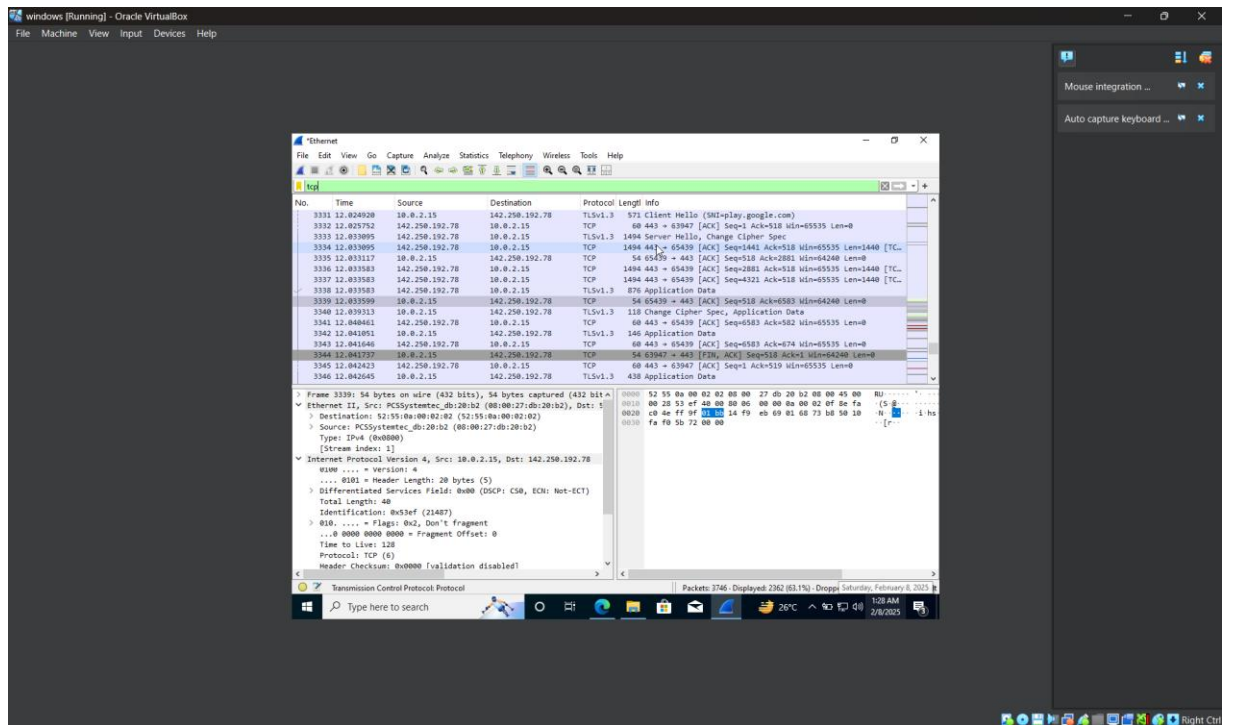


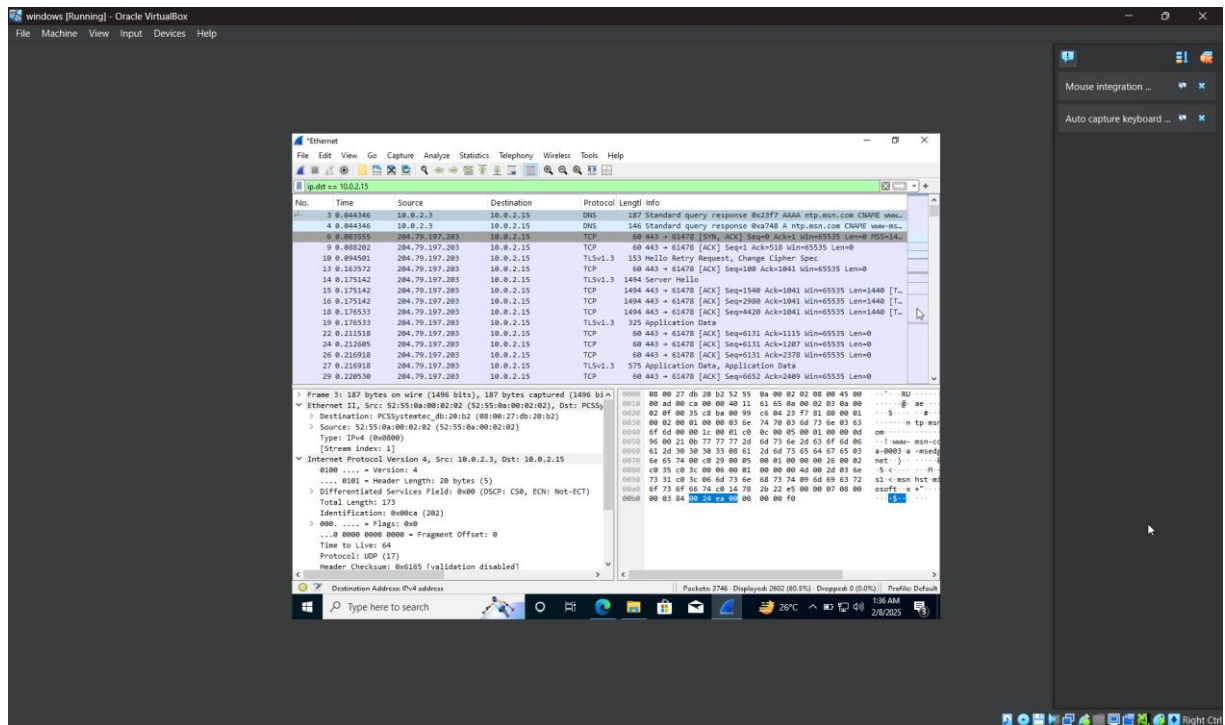
|                                                                                                          |           |               |                |      |                                                |
|----------------------------------------------------------------------------------------------------------|-----------|---------------|----------------|------|------------------------------------------------|
| 2515                                                                                                     | 75.499480 | 37.200.69.143 | 172.16.165.165 | HTTP | 259 [TCP Retransmission] HTTP/1.1 200 OK       |
| 2977                                                                                                     | 84.464154 | 37.200.69.143 | 172.16.165.165 | HTTP | 941 HTTP/1.1 200 OK (application/x-msdownload) |
| Frame 2515: 259 bytes on wire (2072 bits), 259 bytes captured (2072 bits)                                |           |               |                |      |                                                |
| Ethernet II, Src: vmware_f3:ca:52 (00:50:56:f3:ca:52), Dst: f0:19:af:02:9b:f1 (f0:19:af:02:9b:f1)        |           |               |                |      |                                                |
| Internet Protocol Version 4, Src: 37.200.69.143 (37.200.69.143), Dst: 172.16.165.165 (172.16.165.165)    |           |               |                |      |                                                |
| Version: 4                                                                                               |           |               |                |      |                                                |
| Header length: 20 bytes                                                                                  |           |               |                |      |                                                |
| Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00: Not-ECT (Not ECN-Capable Transport)) |           |               |                |      |                                                |
| Total Length: 245                                                                                        |           |               |                |      |                                                |
| Identification: 0x0920 (2336)                                                                            |           |               |                |      |                                                |
| Flags: 0x00                                                                                              |           |               |                |      |                                                |
| Fragment offset: 0                                                                                       |           |               |                |      |                                                |
| Time to live: 128                                                                                        |           |               |                |      |                                                |
| Protocol: TCP (6)                                                                                        |           |               |                |      |                                                |
| Header checksum: 0x73d6 [correct]                                                                        |           |               |                |      |                                                |
| Source: 37.200.69.143 (37.200.69.143)                                                                    |           |               |                |      |                                                |
| Destination: 172.16.165.165 (172.16.165.165)                                                             |           |               |                |      |                                                |
| [Source GeoIP: Unknown]                                                                                  |           |               |                |      |                                                |
| [Destination GeoIP: Unknown]                                                                             |           |               |                |      |                                                |
| Transmission Control Protocol, Src Port: http (80), Dst Port: 49457 (49457), Seq: 1, Ack: 298, Len: 205  |           |               |                |      |                                                |
| Hypertext Transfer Protocol                                                                              |           |               |                |      |                                                |

B









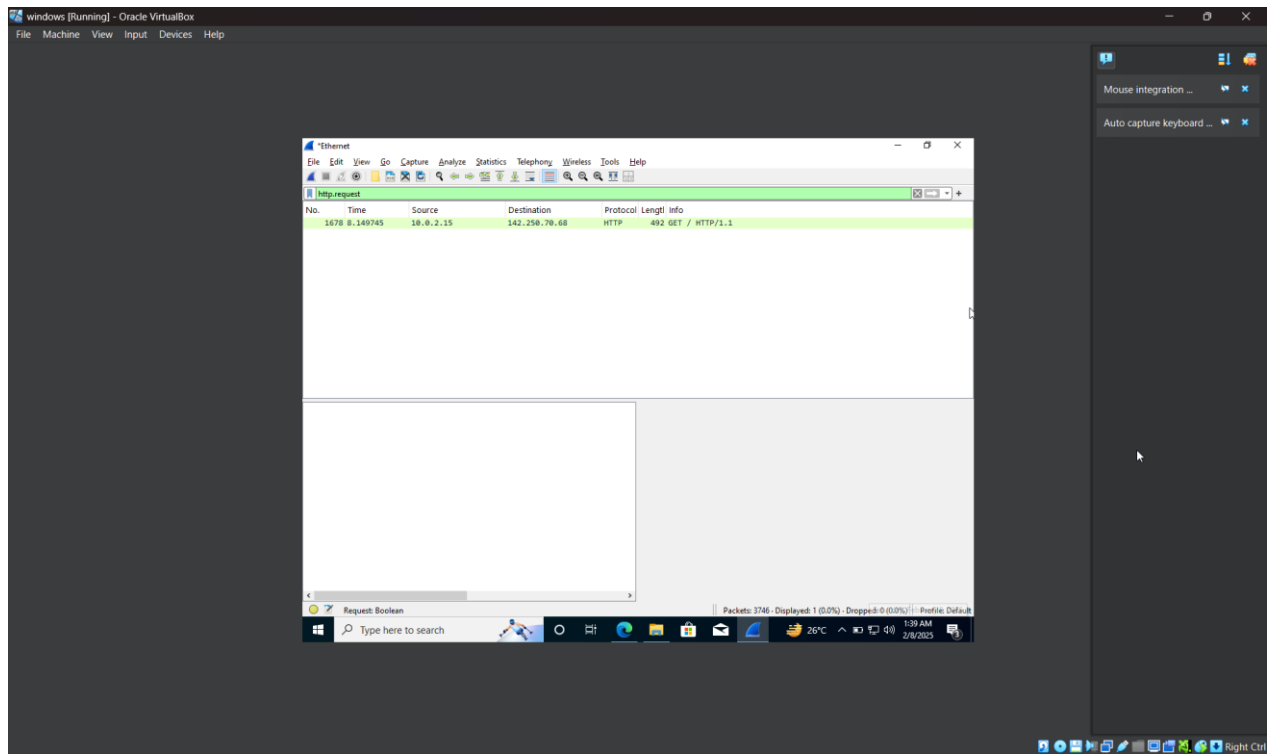
\*Wi-Fi [Wireshark 1.10.8 (v1.10.8-2-g52a5244 from master-1.10)]

File Edit View Go Capture Analyze Statistics Telephony Tools Internals Help

Filter: `http.response.code==200` Expression... Clear Apply Save

| No.  | Time       | Source      | Destination   | Protocol | Length | Info                         |
|------|------------|-------------|---------------|----------|--------|------------------------------|
| 319  | 10.9289020 | 23.56.97.66 | 10.200.117.23 | HTTP     | 241    | HTTP/1.1 200 OK (text/plain) |
| 1282 | 41.1071580 | 23.56.97.66 | 10.200.117.23 | HTTP     | 241    | HTTP/1.1 200 OK (text/plain) |

+ Frame 1282: 241 bytes on wire (1928 bits), 241 bytes captured (1928 bits) on interface 0  
 + Ethernet II, Src: Cisco\_ff:fc:50 (00:08:e3:ff:fc:50), Dst: 90:65:84:19:93:12 (90:65:84:19:93:12)  
 + Internet Protocol Version 4, Src: 23.56.97.66 (23.56.97.66), Dst: 10.200.117.23 (10.200.117.23)  
 + Transmission Control Protocol, Src Port: http (80), Dst Port: 53264 (53264), Seq: 1, Ack: 112, Len: 187  
 + Hypertext Transfer Protocol  
 + Line-based text data: text/plain



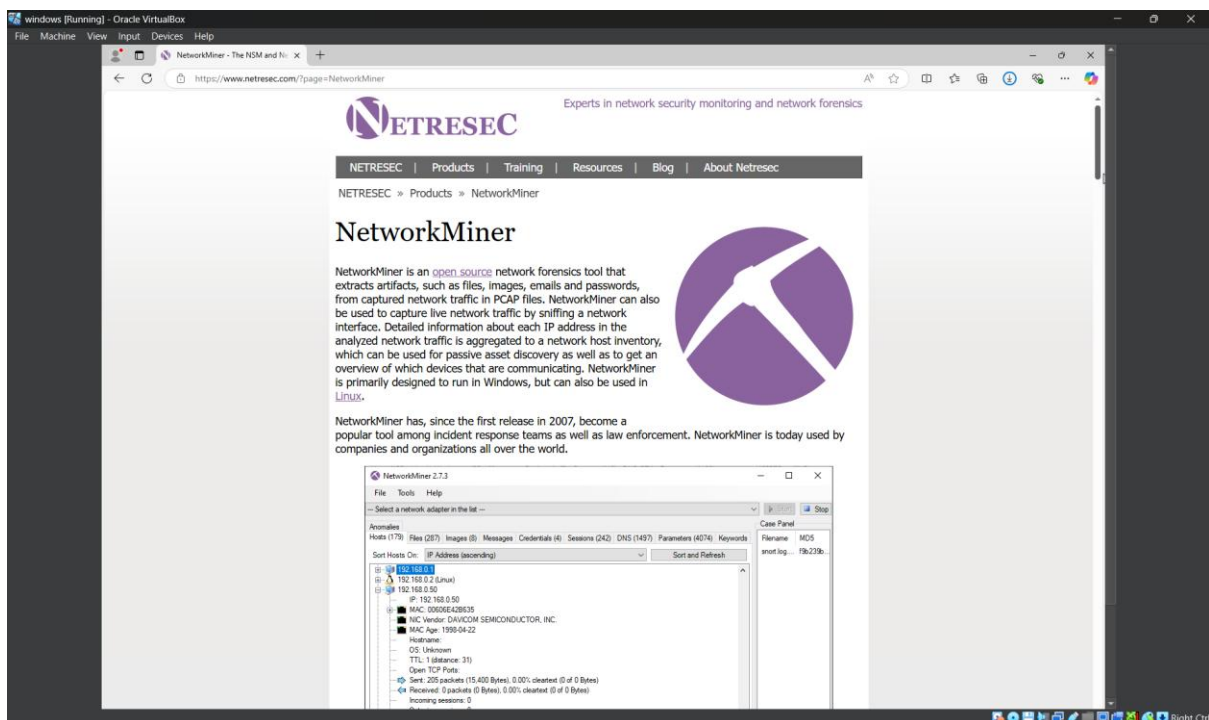
## C. Perform Network Forensic Analysis tool using NetworkMiner.

### Theory:

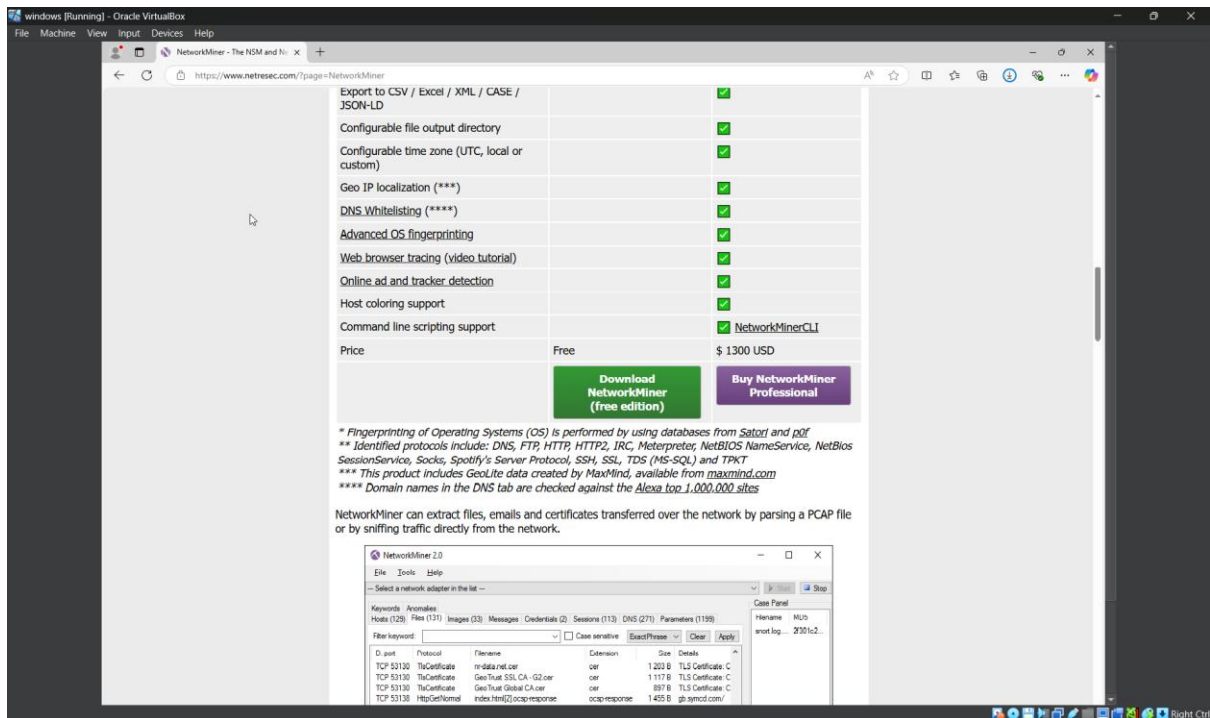
NetworkMiner is an open source Network Forensic Analysis Tool (NFAT) for Windows (but also works in Linux / Mac OS X / FreeBSD). NetworkMiner can be used as a passive networksniffer/packet capturing tool in order to detect operating systems, sessions, hostnames, open ports etc. without putting any traffic on the network. NetworkMiner can also parse PCAP files for off-line analysis and to regenerate/reassemble transmitted files and certificates from PCAP files.

NetworkMiner makes it easy to perform advanced Network Traffic Analysis (NTA) by providing extracted artifacts in an intuitive user interface. The way data is presented not only makes the analysis simpler, it also saves valuable time for the analyst or forensic investigator.

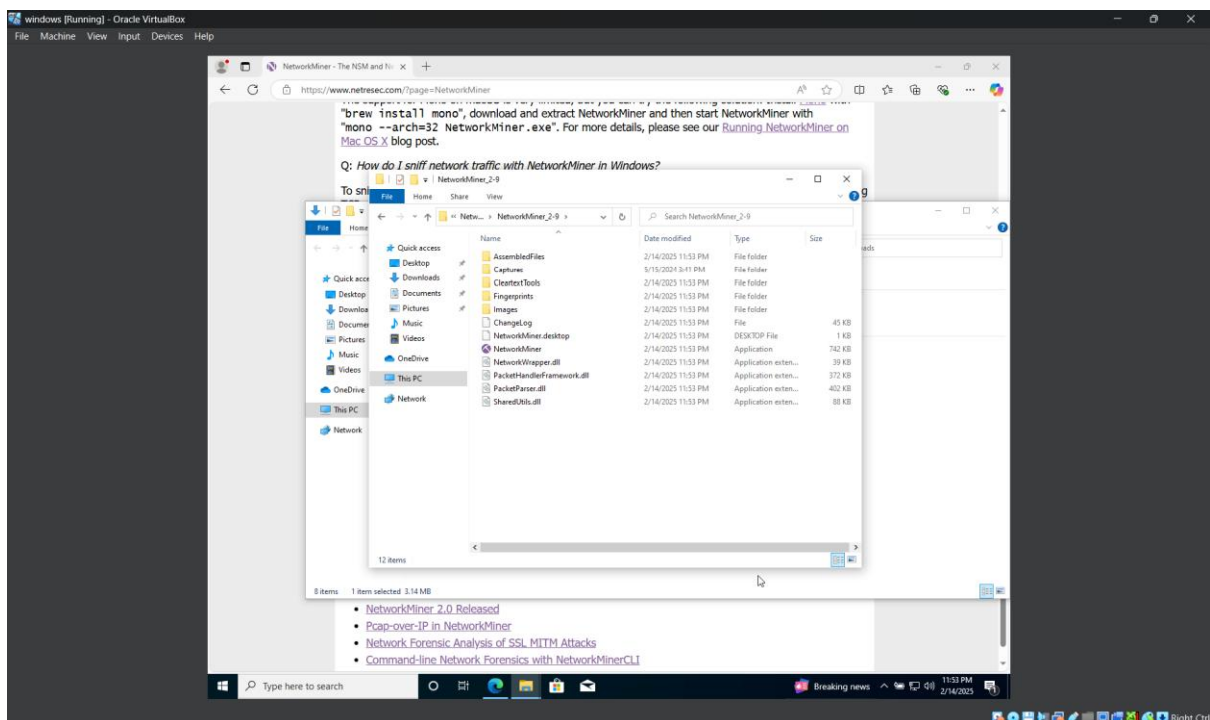
Step 1: Download Networkminer from <https://www.netresec.com/?page=NetworkMiner>



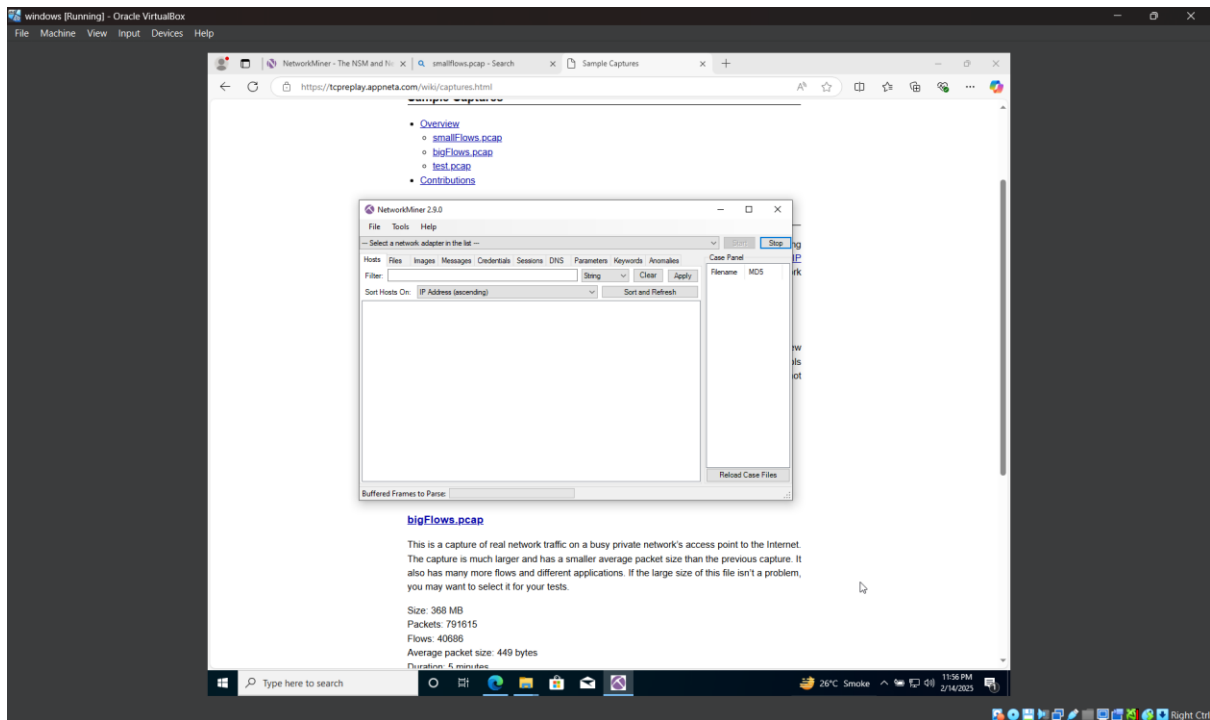




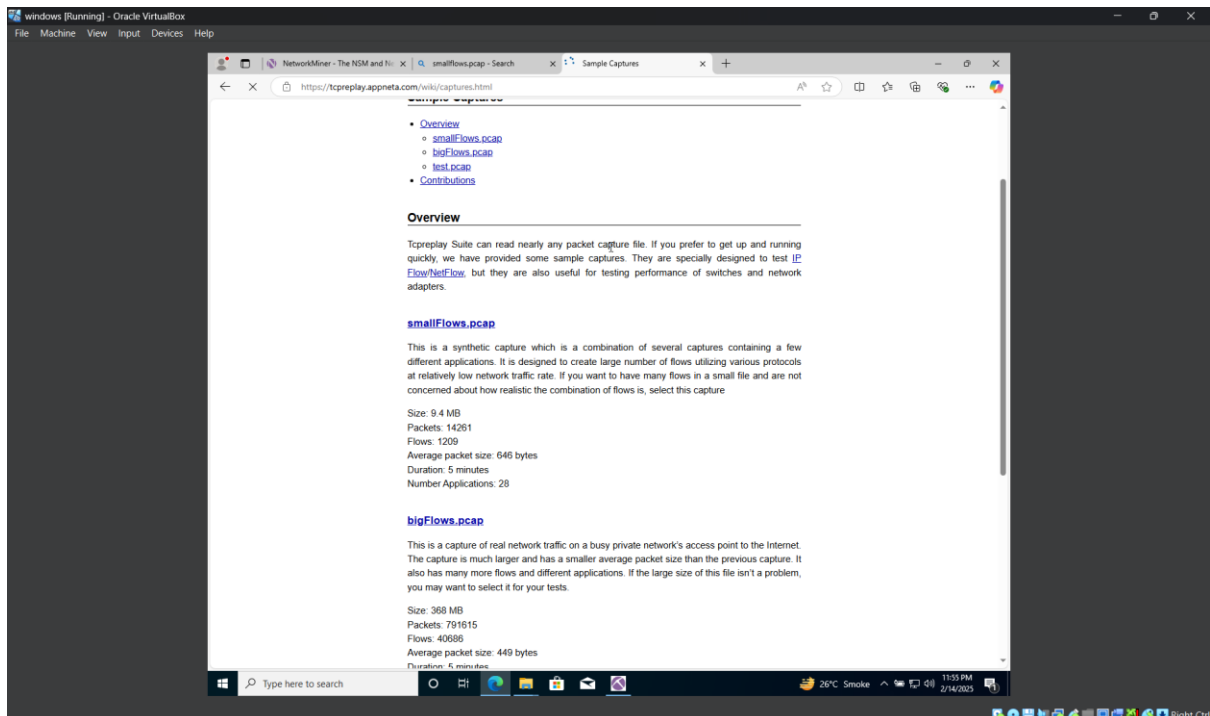
Step 2: Extract the file from zip file contents in download folder. Open C drive- Program Files and Create a new folder- networkminer2. Copy all folder contents in this folder.



Step 3: Run the Application.exe file from the folder to Open NetworkMiner.



Step 4: Download pcap file from the URL mentioned in the screenshot below <https://tcpreplay.appneta.com/wiki/captures.html>





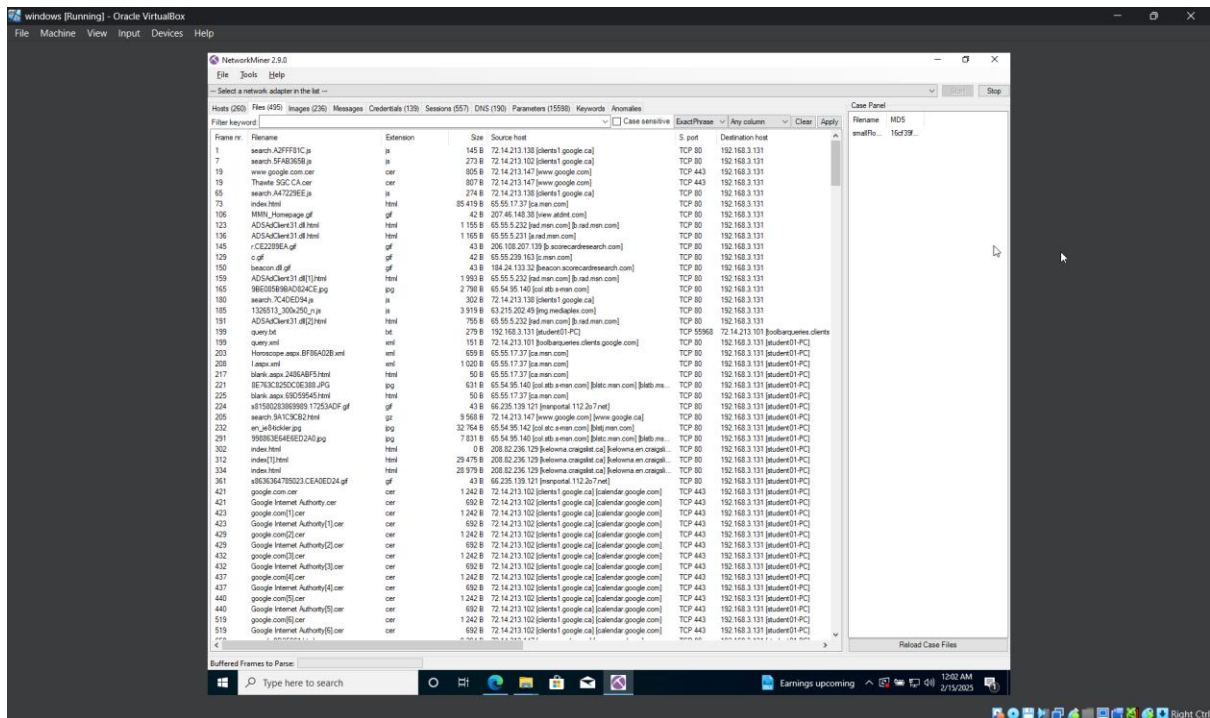
The screenshot shows a Windows VirtualBox environment. The main window is titled "NetworkMiner - The NGM and h..." and displays a web page for "smallflows.pcap". The page has a "Download" button and a list of files: "smallflows.pcap" and "NetworkMiner\_2-9 (1).zip". A file explorer window is open, showing the "Downloads" folder. It contains a table of files:

| Name                                 | Date modified      | Type        |
|--------------------------------------|--------------------|-------------|
| Today (2)                            |                    |             |
| smallflows                           | 2/14/2023 11:58 PM | Wireless    |
| NetworkMiner_2-9 (1)                 | 2/14/2023 11:53 PM | File folder |
| Last week (2)                        |                    |             |
| 2014-11-16 traffic analysis exercise | 2/8/2023 12:54 AM  | Wireless    |
| NetworkMiner_2-9 (2)                 | 2/8/2023 9:33 PM   | File folder |

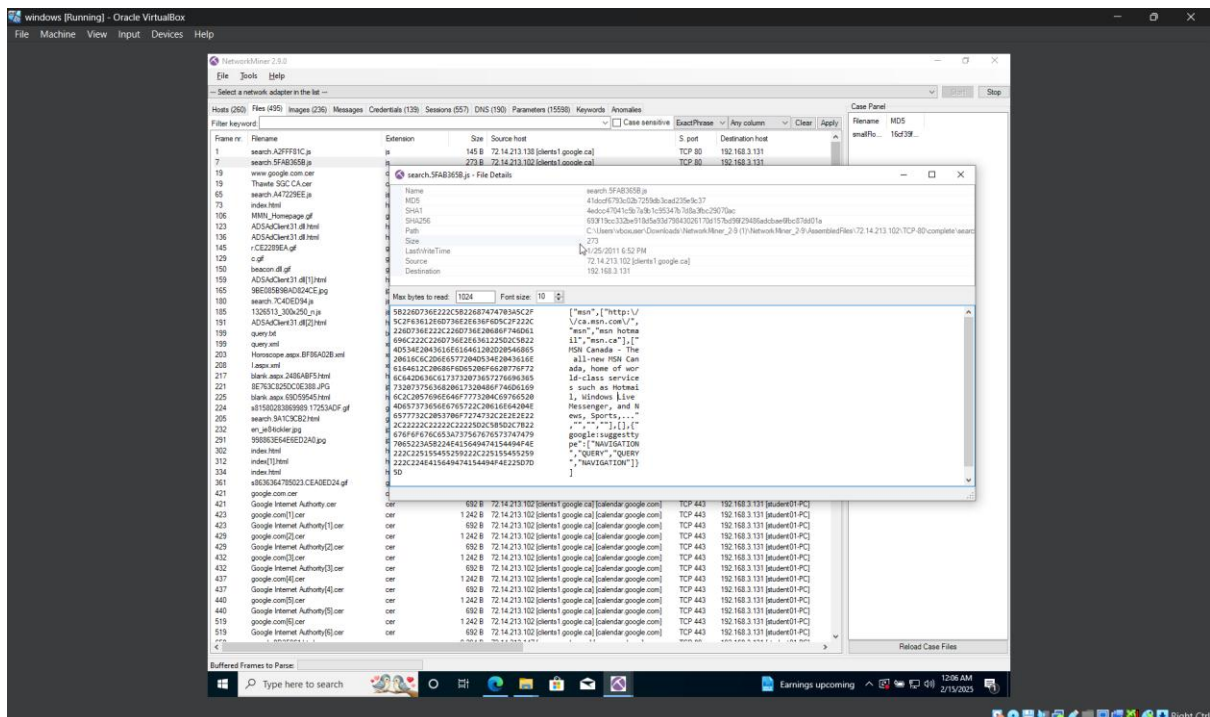
The file explorer window also shows a "File name" field with "smallflows" and a "File type" dropdown set to "Capture Files (\*.pcap, \*.cap, \*.d)". The taskbar at the bottom shows the Windows Start menu and various application icons.

[illegible]

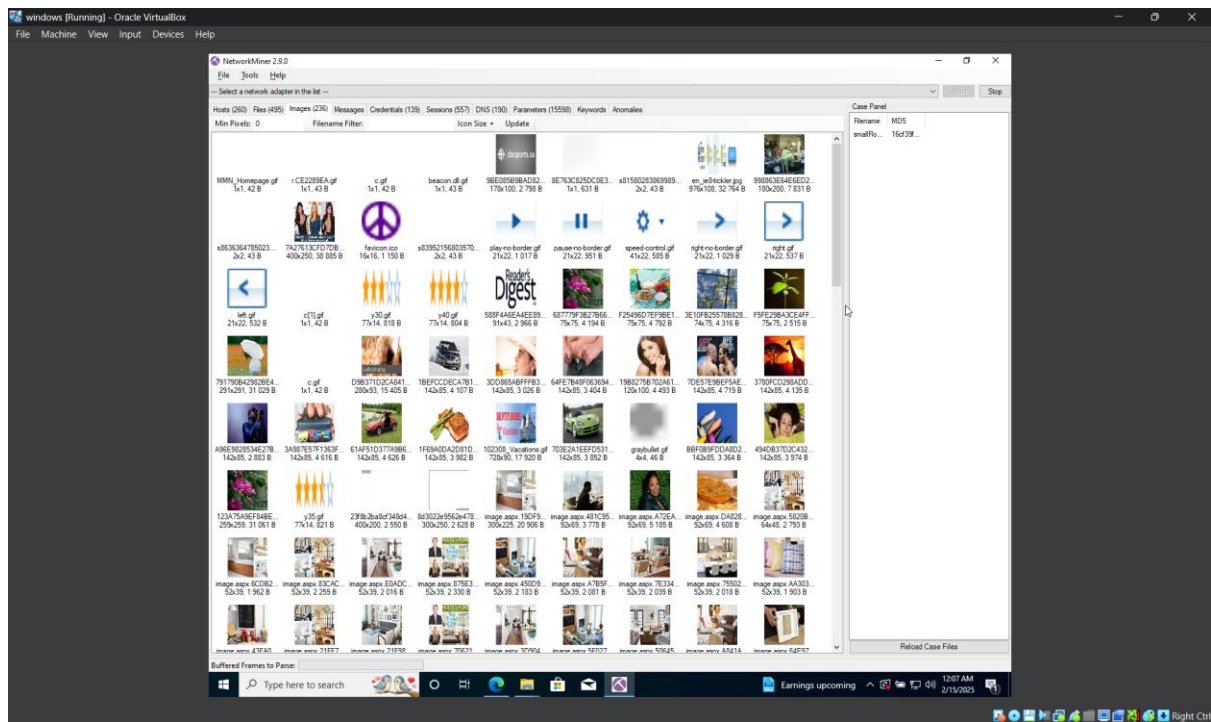
61



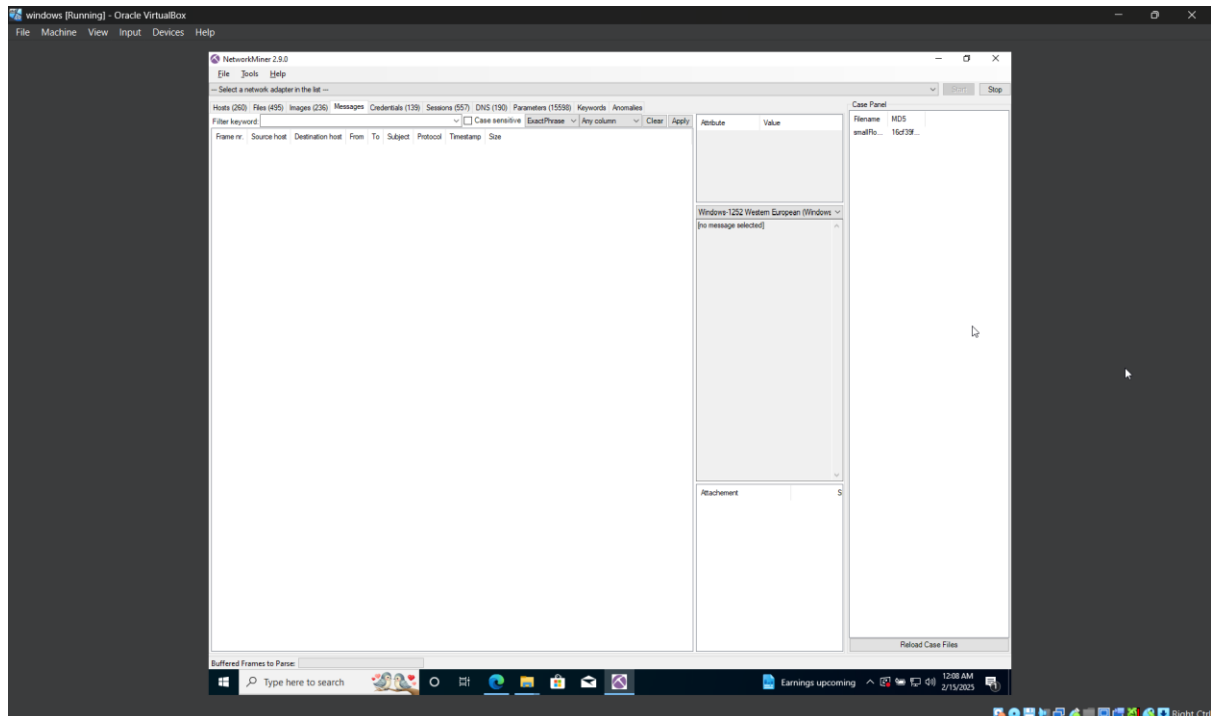
### Step 8: Right click select file details



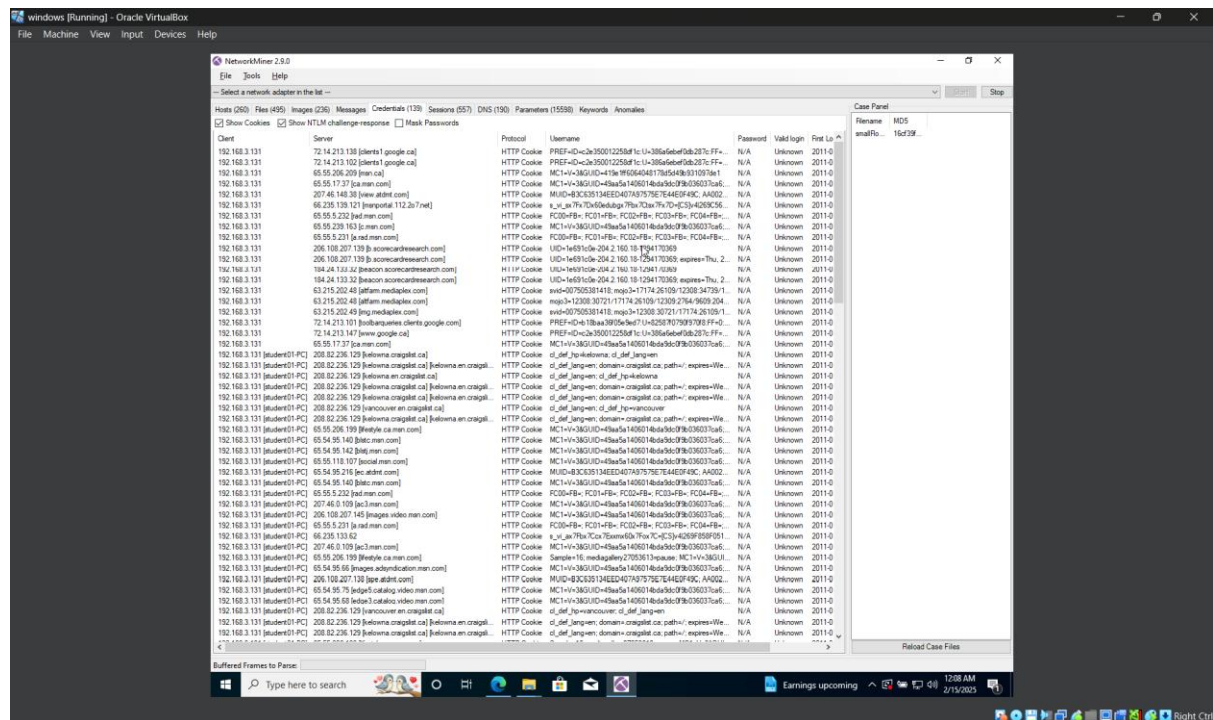
Step 9: You can view all image below



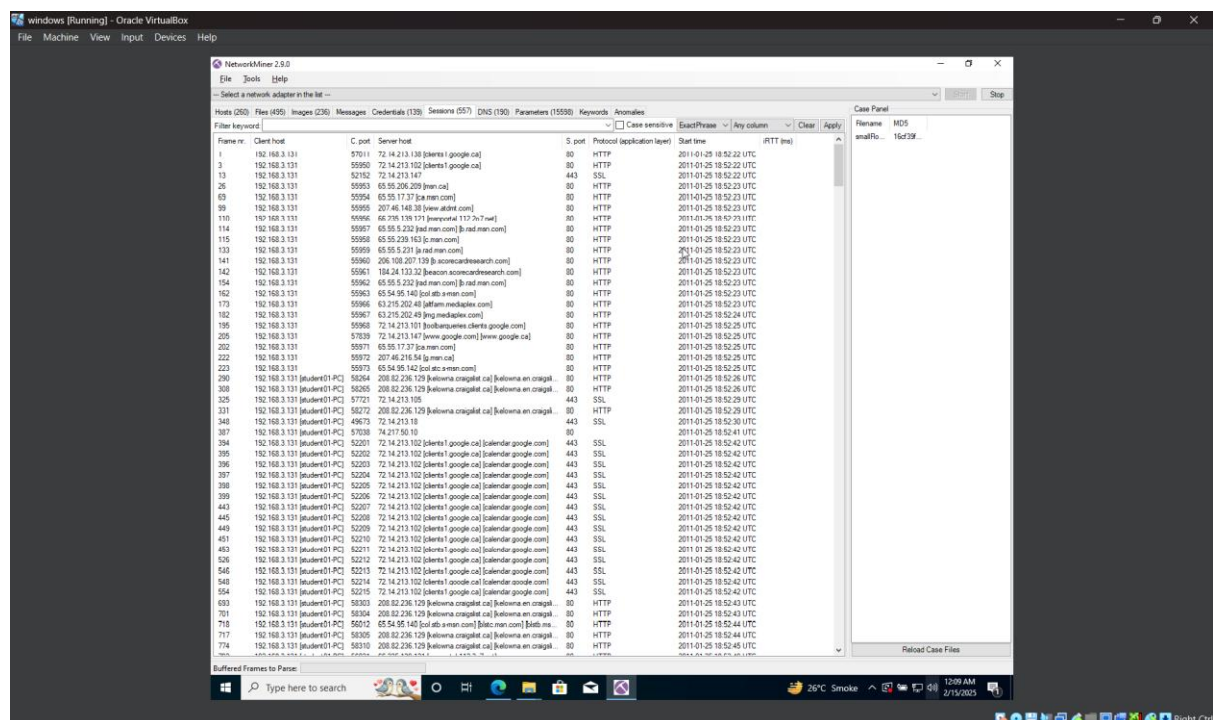
Step 10: You can view all messages below



## Step 11: You can view all credentials below

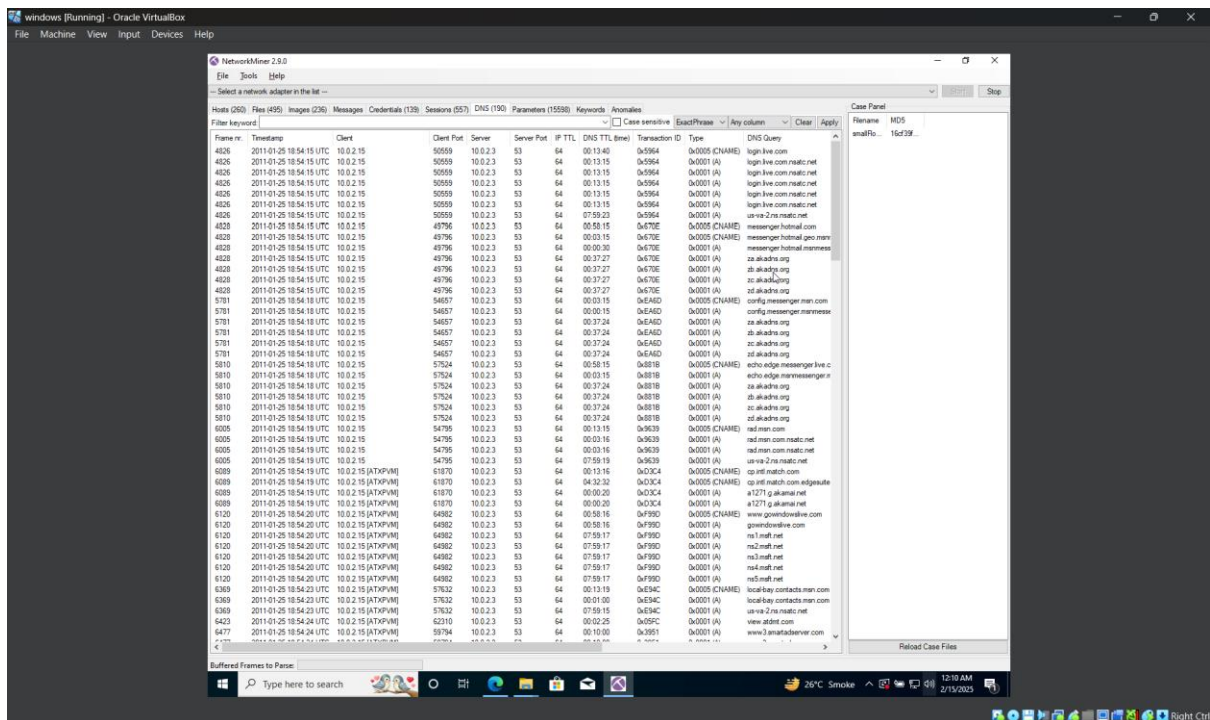


## Step 12: You can view all sessions below

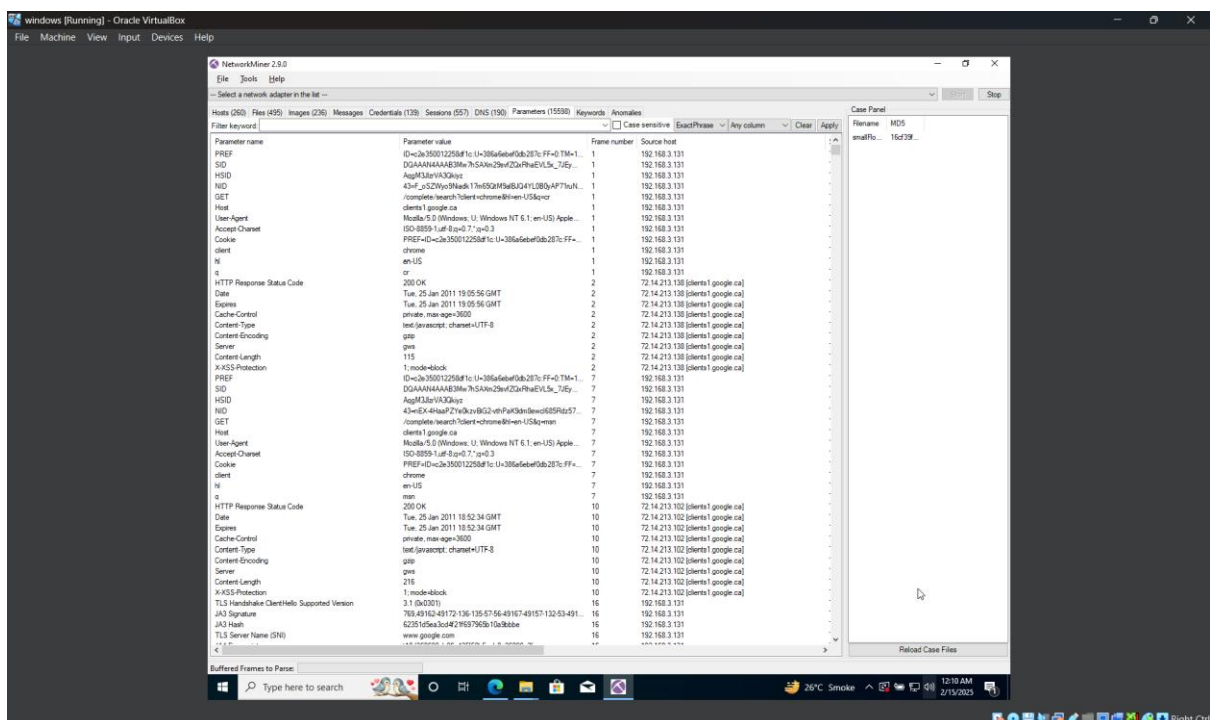




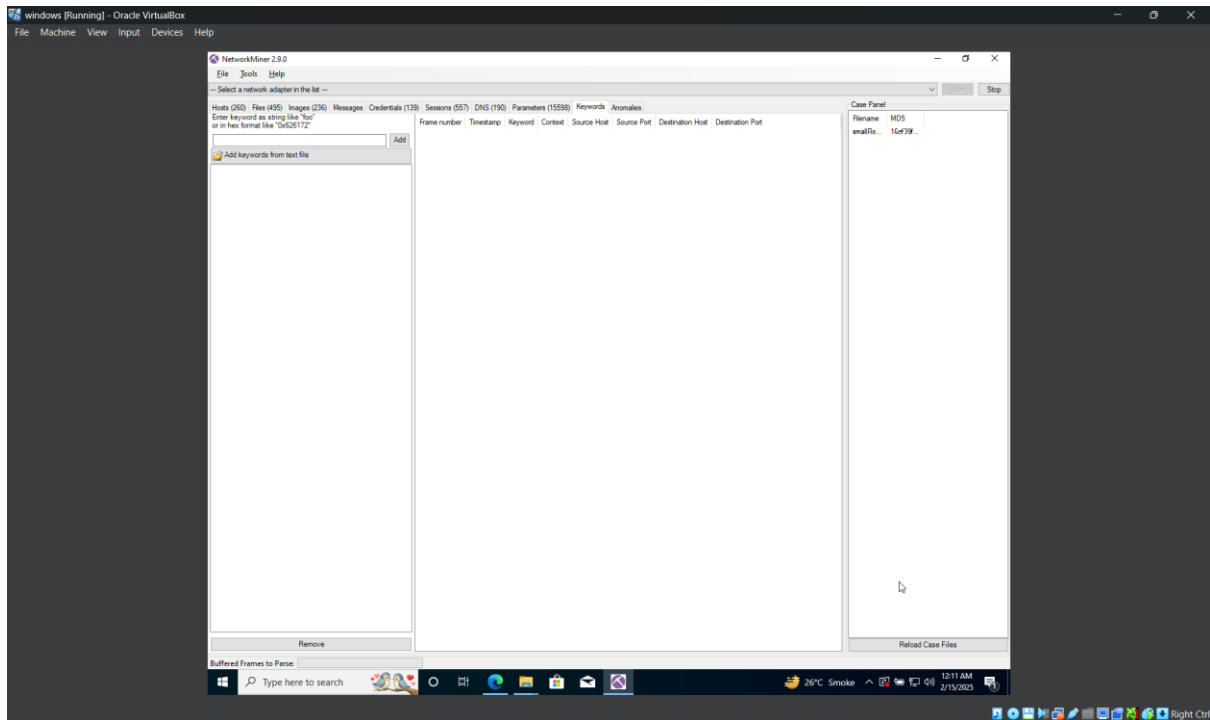
## Step 13: You can view all DNS below



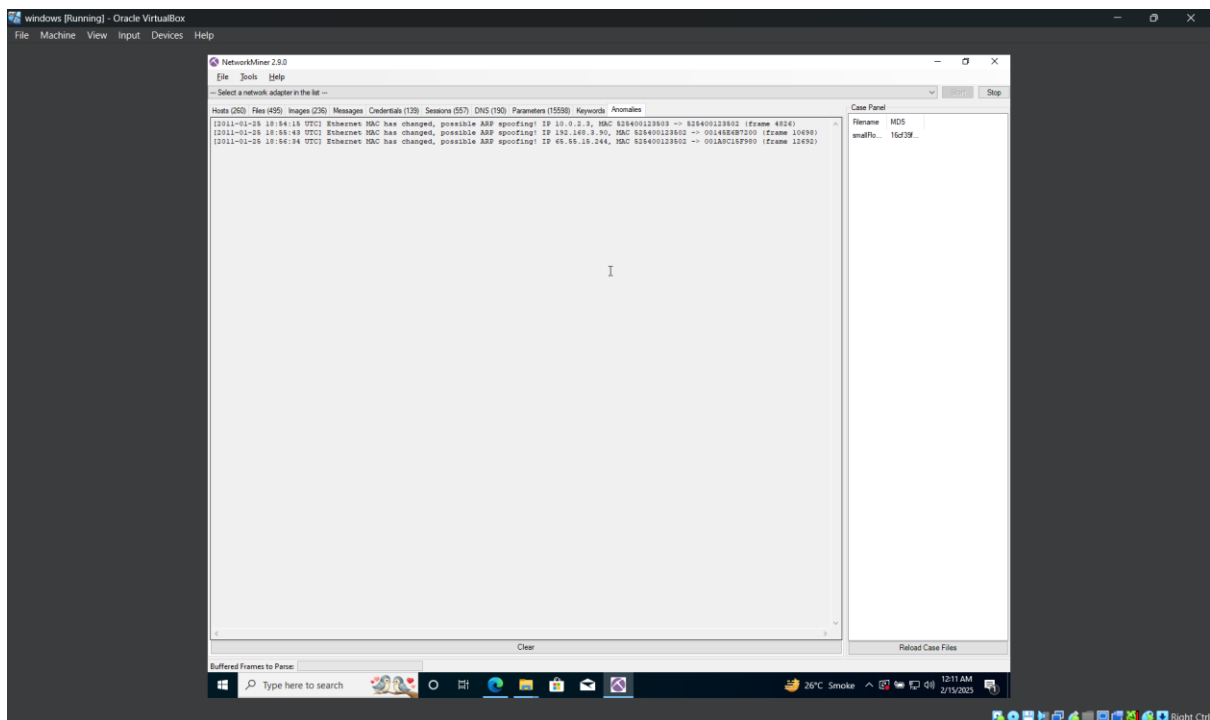
## Step 14: You can view all parameters below



Step 15: You can view all keywords below



Step 16: You can view any anomalies below



## Practical: 7

### Aim: Using Data Acquisition Tools [ProDiscover]

#### Theory

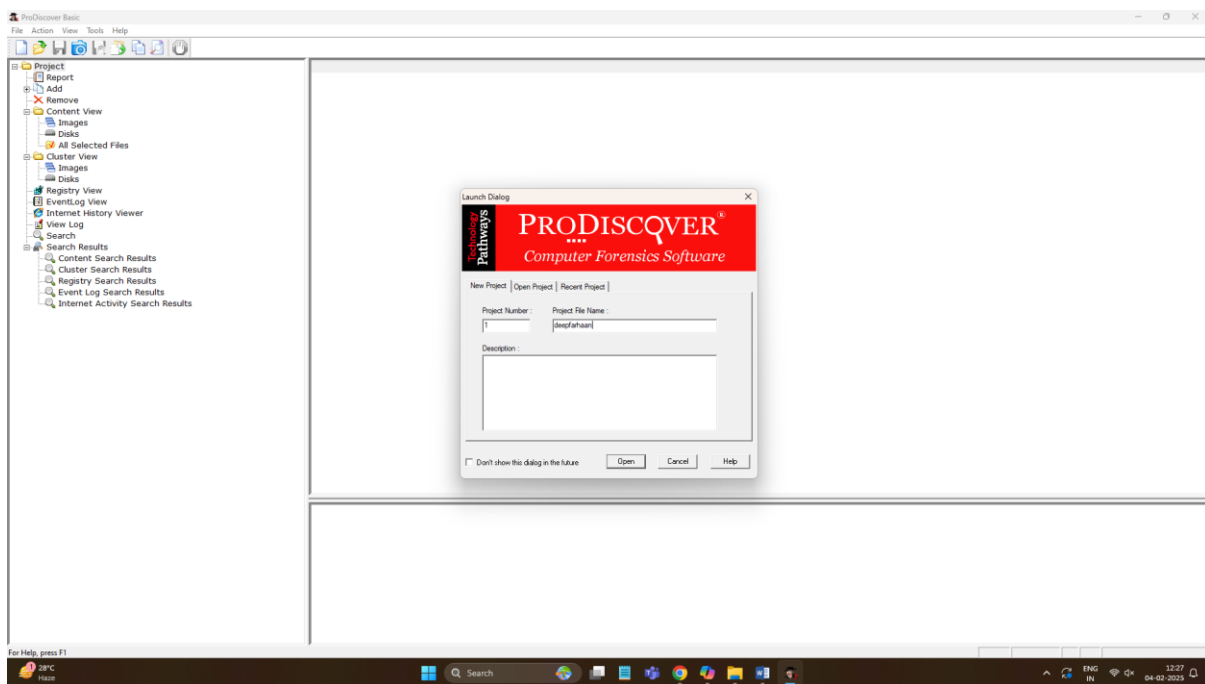
ProDiscover Forensics is a comprehensive digital forensics software that empowers investigators to capture key evidence from computer systems. ProDiscover has capabilities to handle all aspects of an in-depth forensic investigation to collect, preserve, filter, and analyze evidence.

#### Data Acquisition Tools:

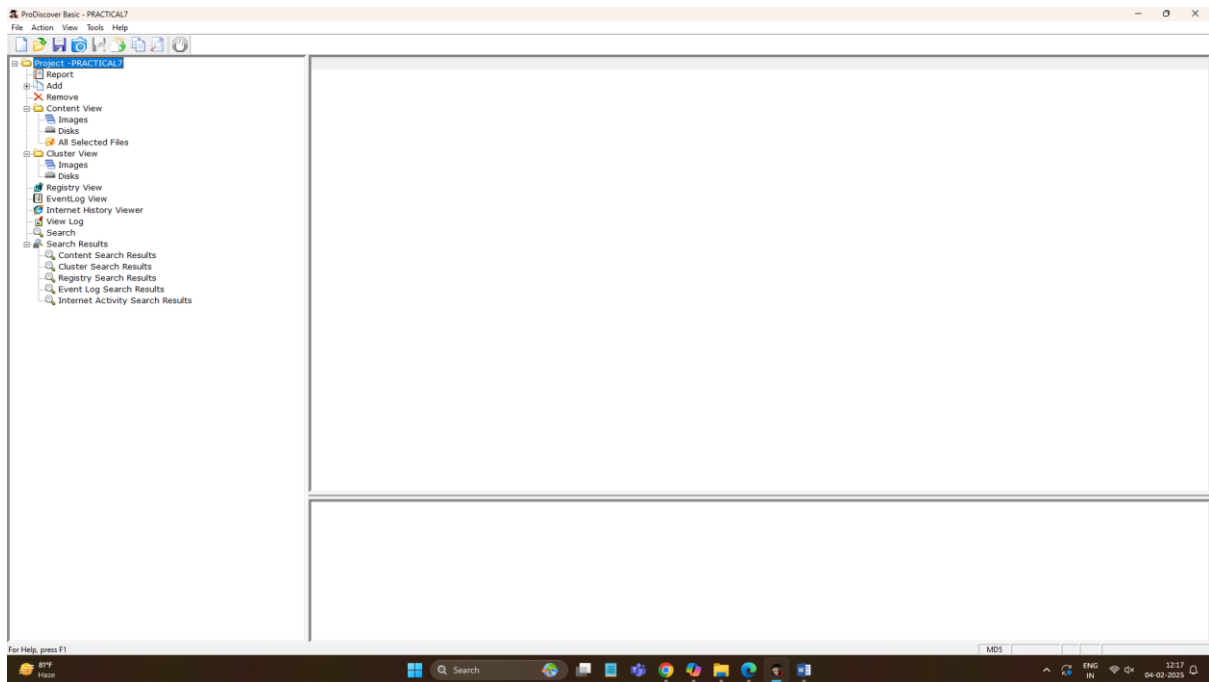
USB WRITE BLOCKER + FTK Imager

- USB Write Blocker enables you to activate or deactivate the write protection for your USB portable devices with just a few clicks. With USB Write Blocker, you will be able to restrict the access of others to your important documents and prevent unwanted, accidental modifications that might cause data loss.
- Forensic Toolkit, or FTK, is a computer forensics software made by Access Data. It scans a hard drive looking for various information.

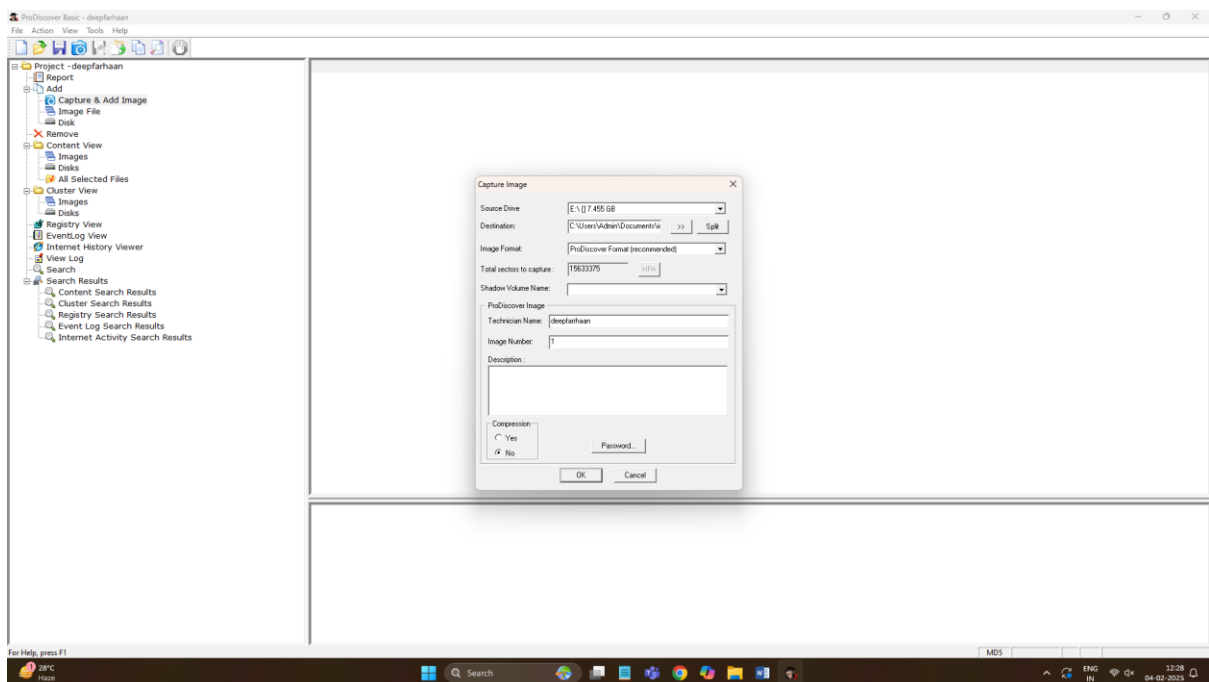
Step 1: Open ProDiscover Basic and start with new project.



Step 2: The new created project appears on the left side – Go on add and select Capture and Add Image

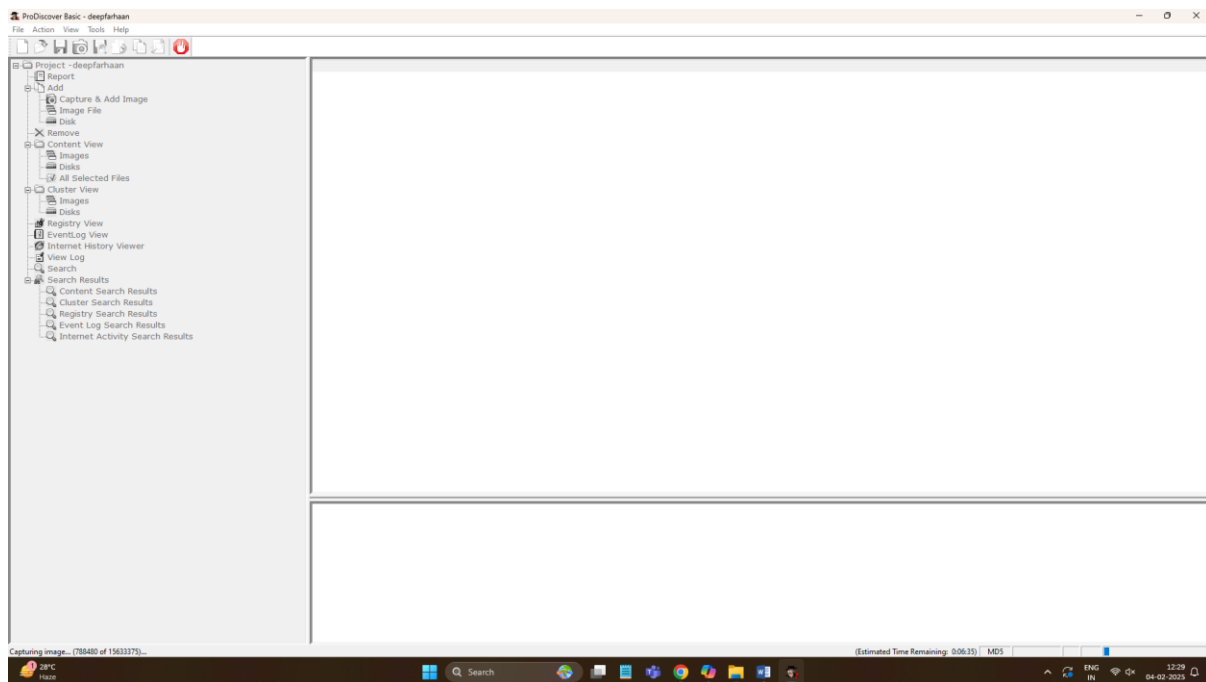


Step 3: Select your Source Drive file from the drop down, select Destination path and add Technician Name, Image Number and Description. Click on ok

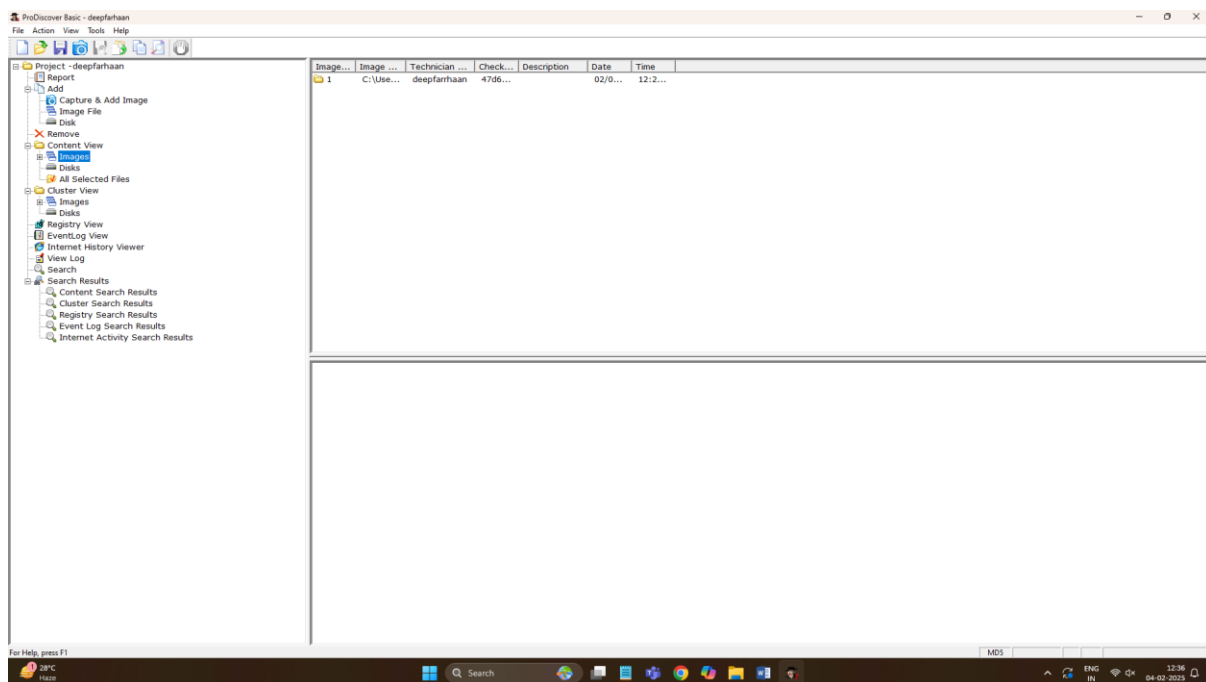


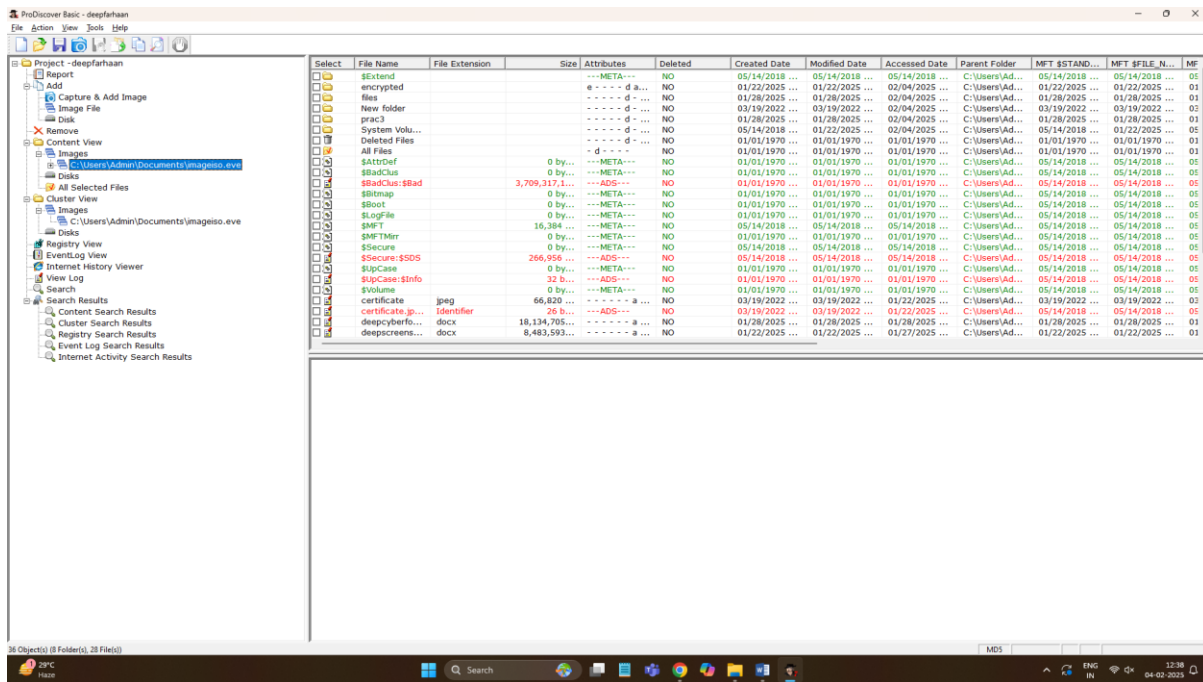
Step 4: The image capturing will start processing.



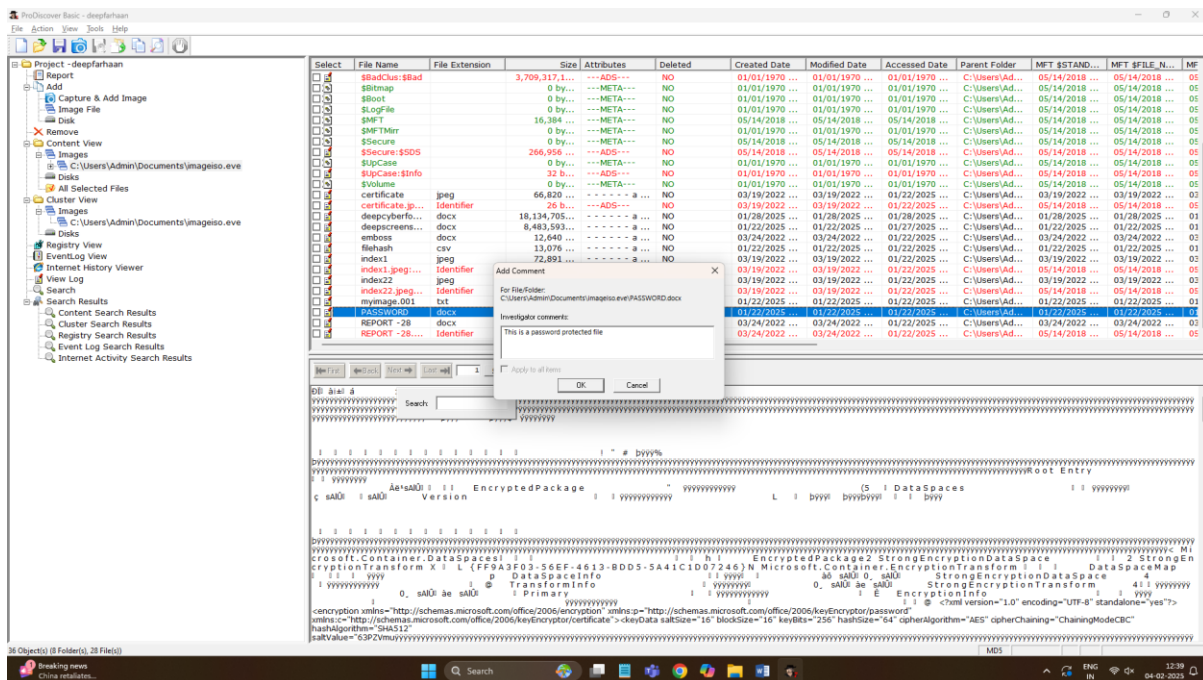


Step 5: The image has been captured. Click on Ok

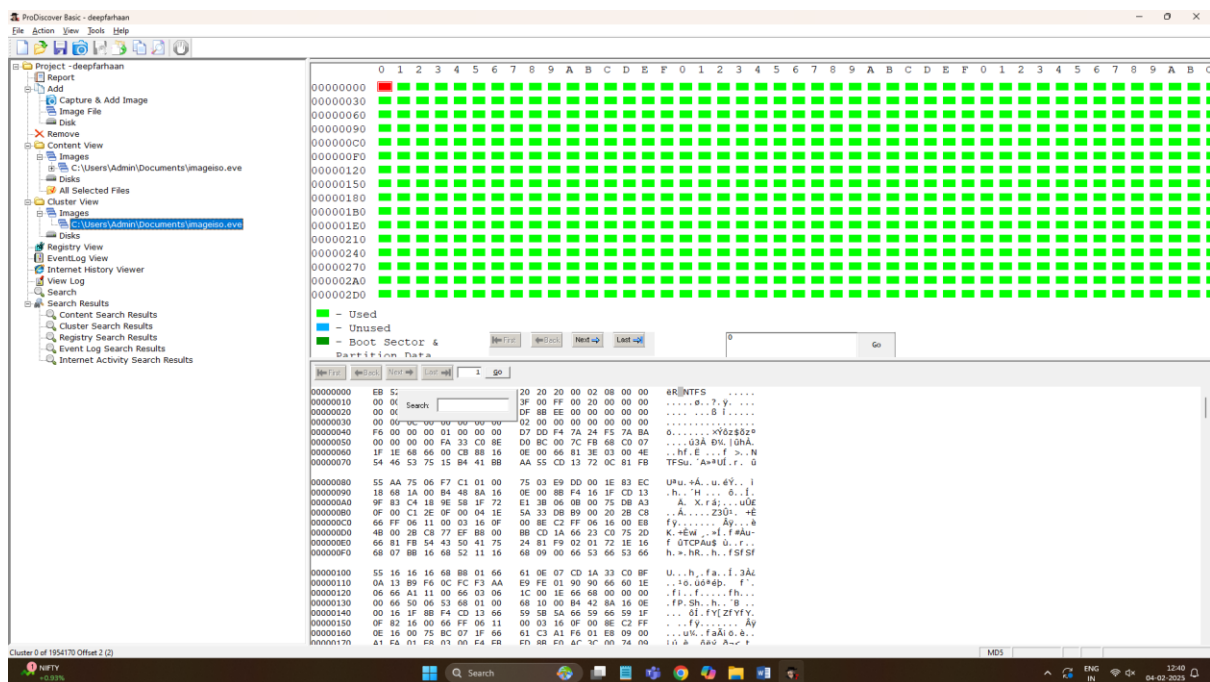




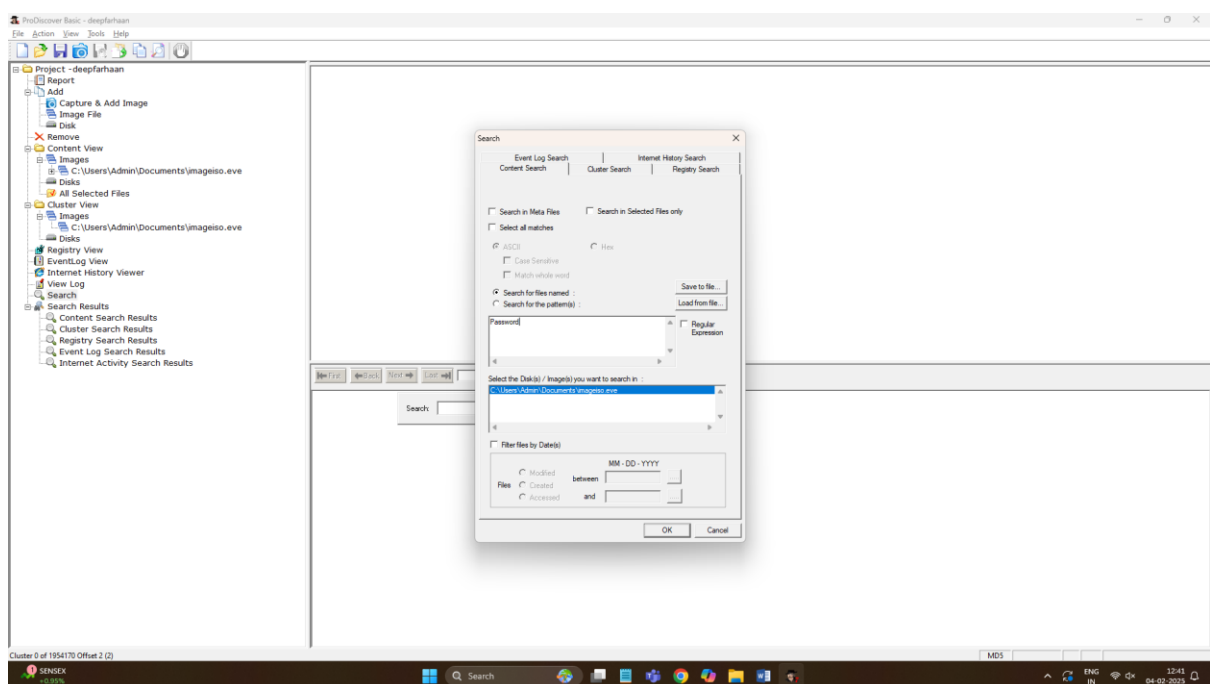
Step 7: Click on any of the file and type a comment.



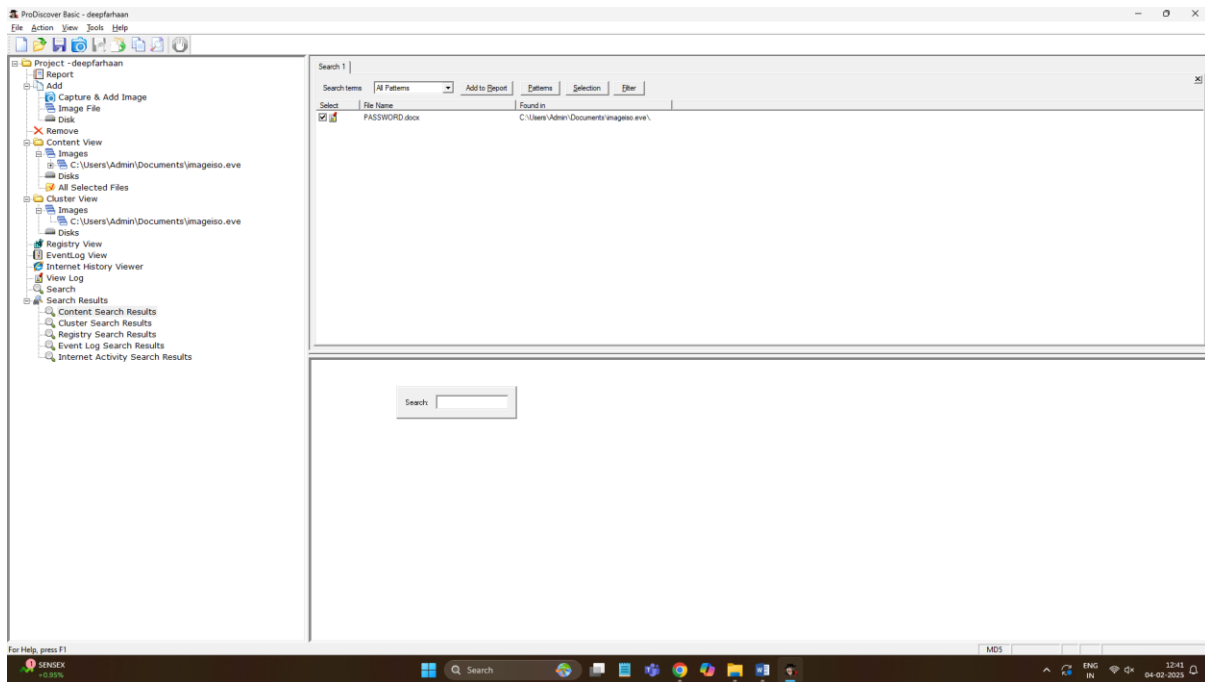
Step 8: To see the cluster view , click on cluster view tab on the left side panel.



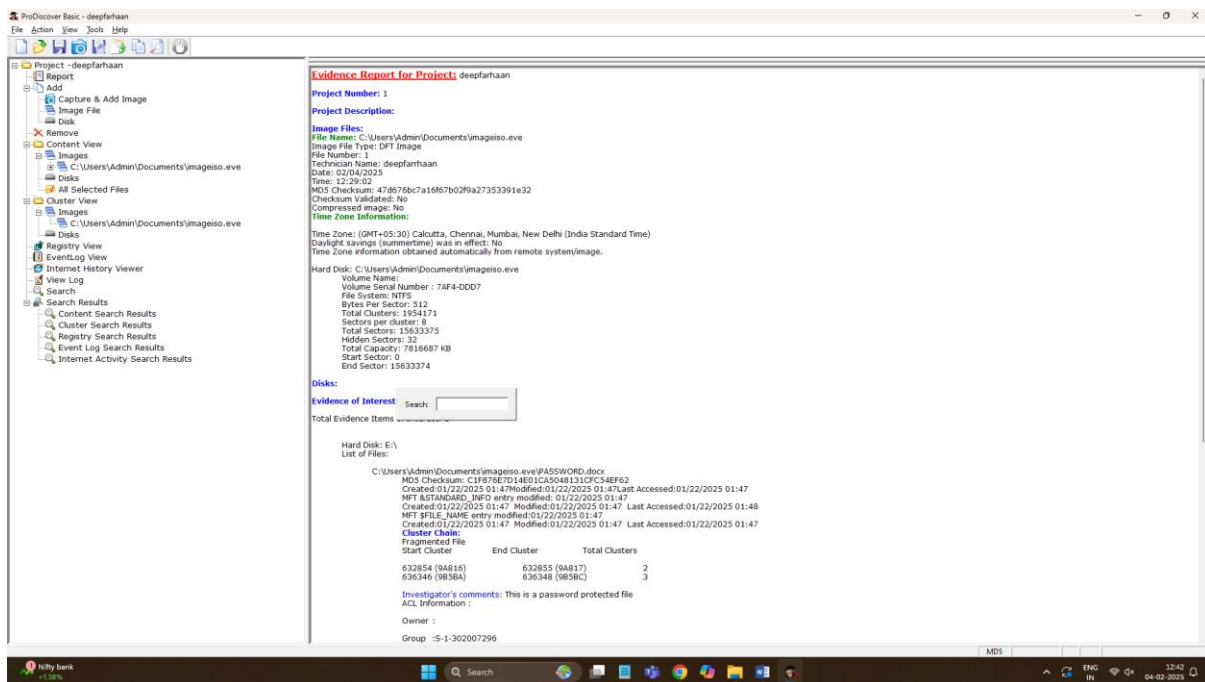
Step 9: For Keyword Search , Click on the Search option on left side of panel and Enter the filename to be searched in the image created.

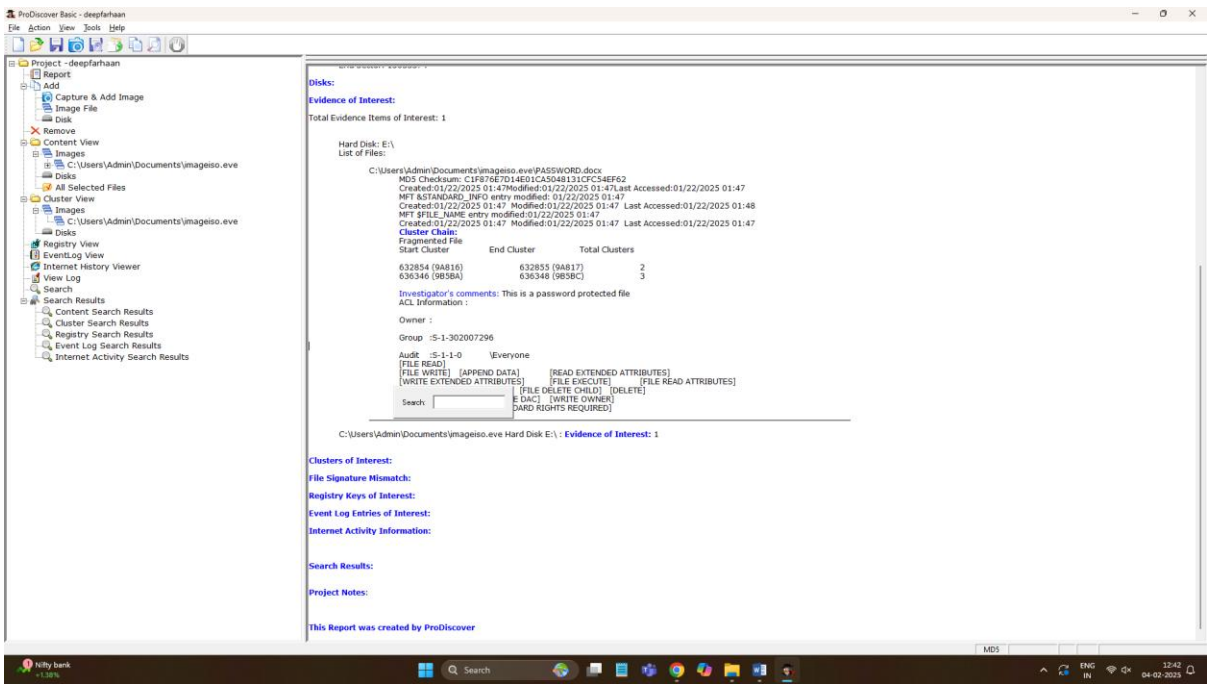


Step 10: The following is the Output of the keyword search.



Step 11: Click on View – Report This will show entire gist of report done in the project.





## Practical: 8

### Aim: Using Steganography tool (S-Tool)

#### Theory

Steganography is one of the oldest technique used to hide data in a image, hide image into image and hide data in a video/audio etc. You can easily hide any kind of video/audio/text/message/image into each other.

Best Tools to Perform Steganography:

There are many software available that offer steganography. Some offer normal steganography but a few offer encryption before hiding the data. These are the steganography tools which are available for free:

- Stegosuite is a free steganography tool which is written in Java. With Stegosuite you can easily hide confidential information in image files.
- Steg hide is an open source Steganography software that lets you hide a secret file in image or audio file.
- Xiao Steganography is a free software that can be used to hide data in BMP images or in WAV files.
- SSuit Pícel is another free portable application to hide text inside an image file but it takes a different approach when compared to other tools.
- OpenPuff is a professional steganographic tool where you can store files in image, audio, video or flash files

Install the latest version of this tool. Launch stegosuite using below command

In order to encrypt an image file with a confidential file, first, you need to load the image file through the File menu.

Select a file in BMP, GIF, JPG or PNG format from the file browser and then click the OK button. The image file will be loaded in the main Stegosuite window.

You can do the following three tasks through this window:

- Enter a secret message along with the file(s) you want to embed.
- Right-click and then add a confidential file in the “embedded files” area.
- Enter a password that will be used while extracting the embedded files and a secret message, later from the image.

In order to extract the original confidential file from the image file it was embedded into, you can right-click the image file from the file browser and select “Open With Other Application” from the menu and then select Stegosuite from the Select Application list as follows:

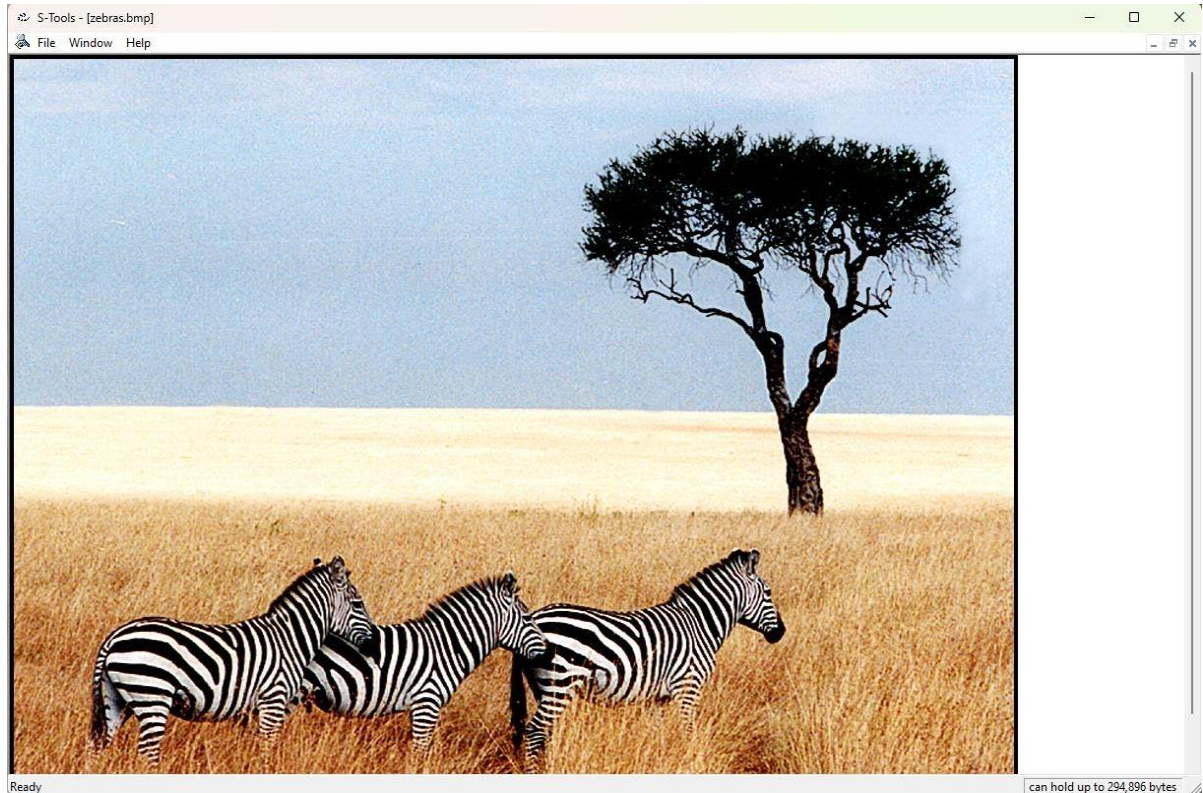
Or you can open the Stegosuite application and load an embedded image file from the



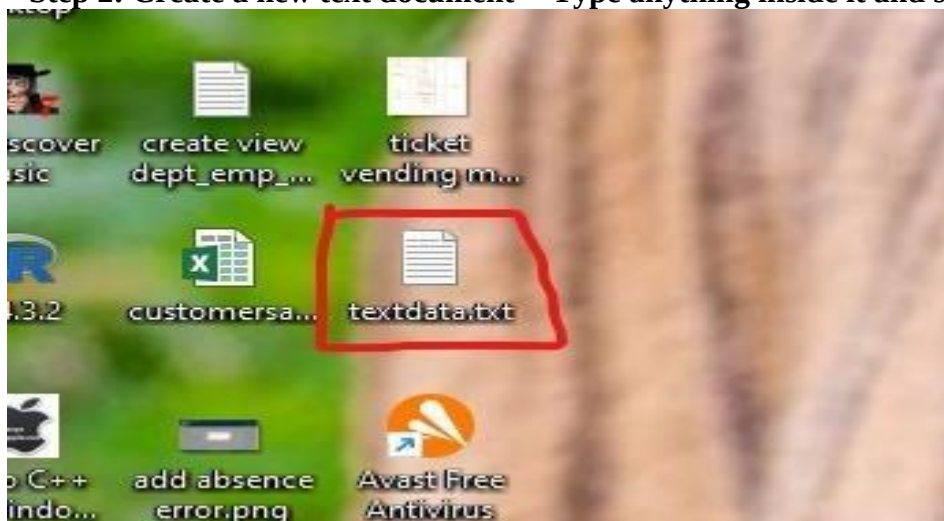
File menu. Once the file is loaded, simply provide the password and click the Extract button after which, the original confidential file will be extracted back to your system.

**Step 1: Download S-tool using giving S-link**

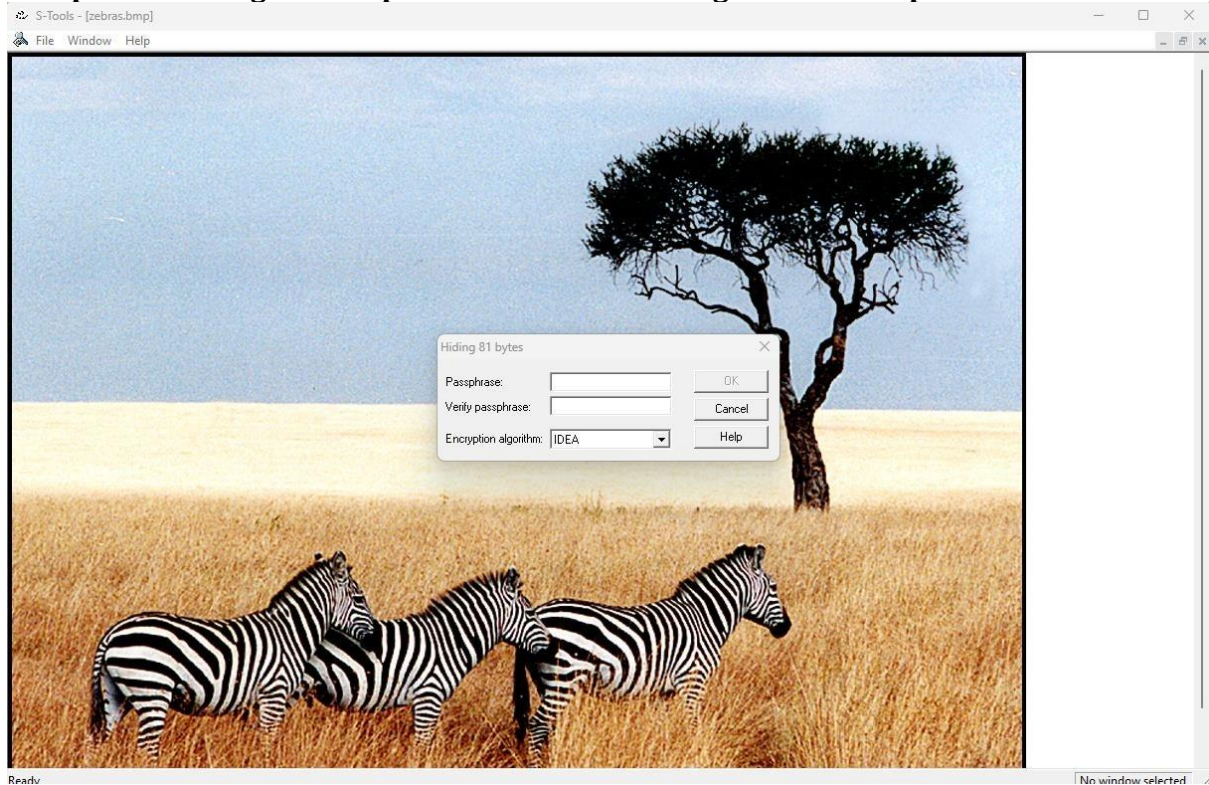
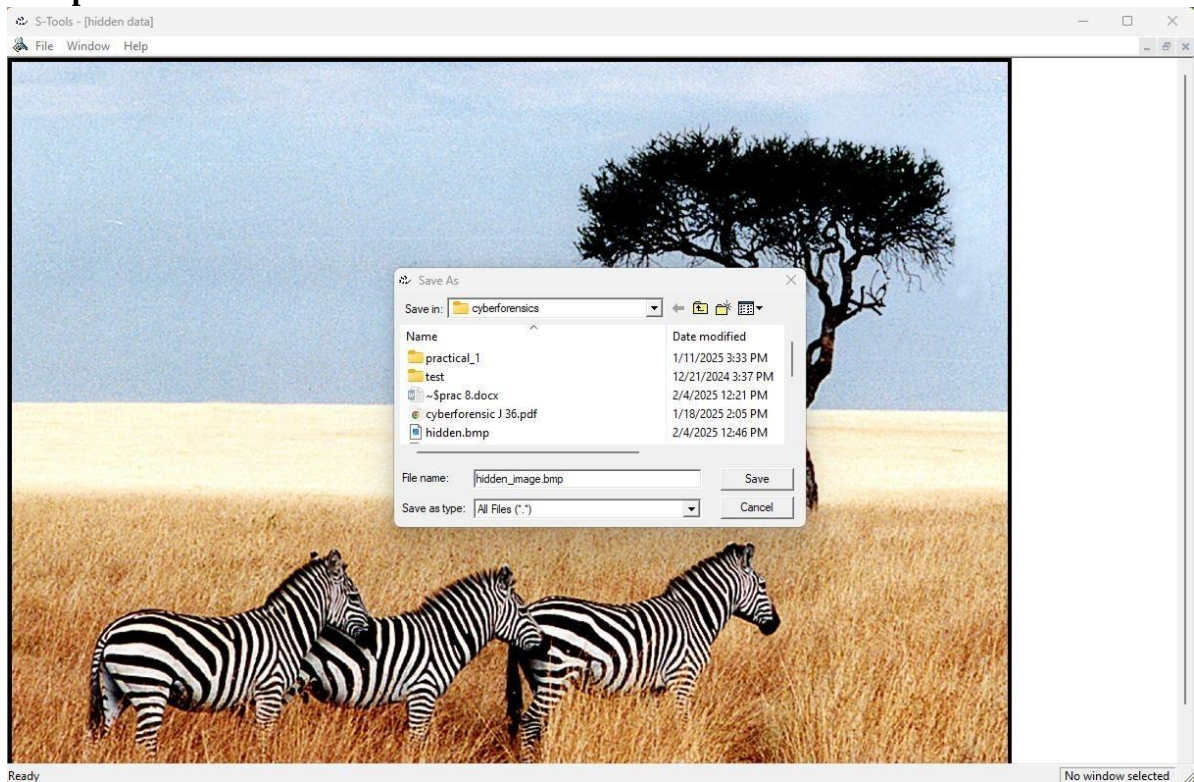
<https://www.cs.vu.nl/~ast/books/mos2/zebras.html>-> then drag and drop the Zebra.bmp image



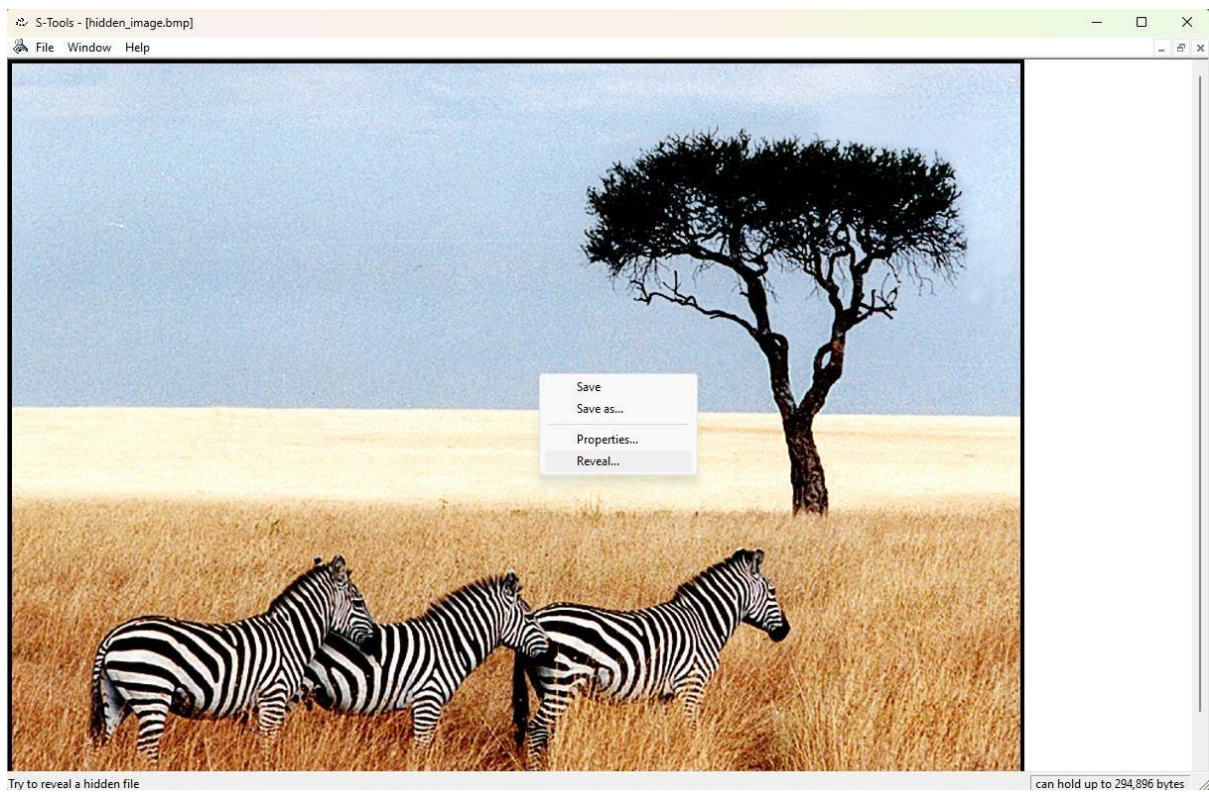
**Step 2: Create a new text document-> Type anything inside it and save it**



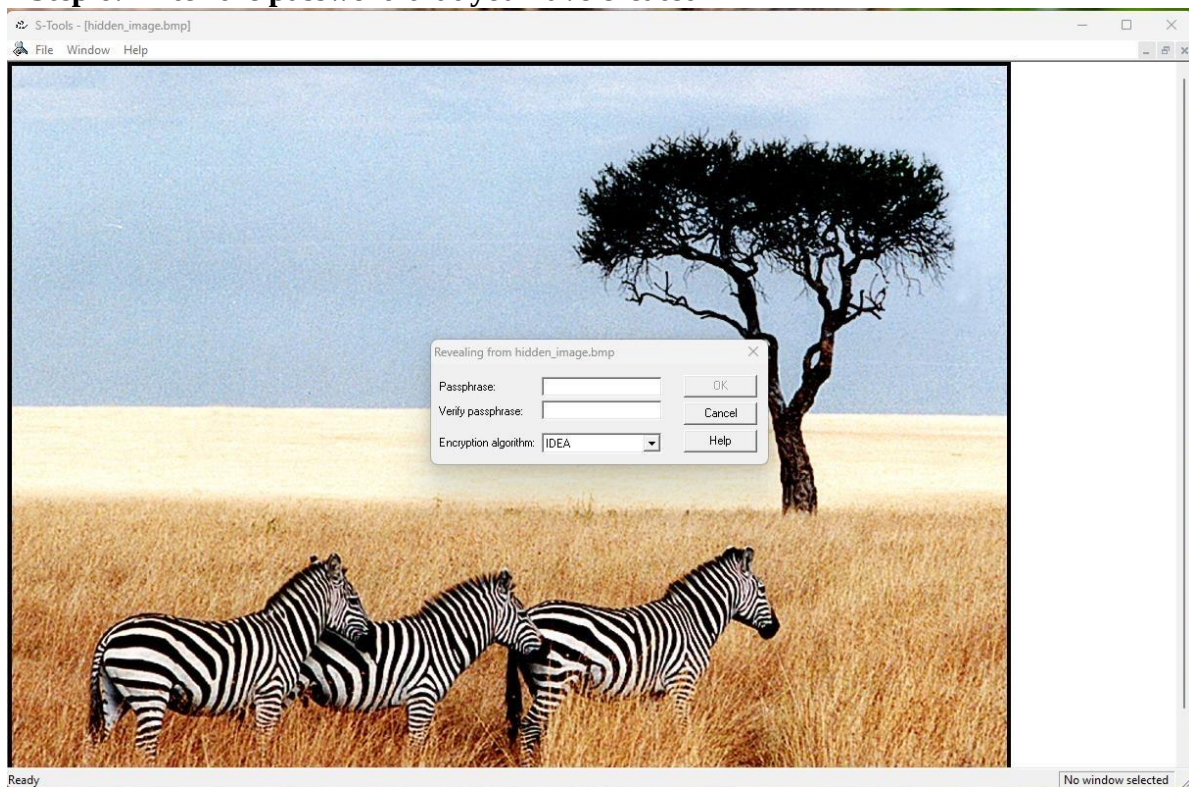


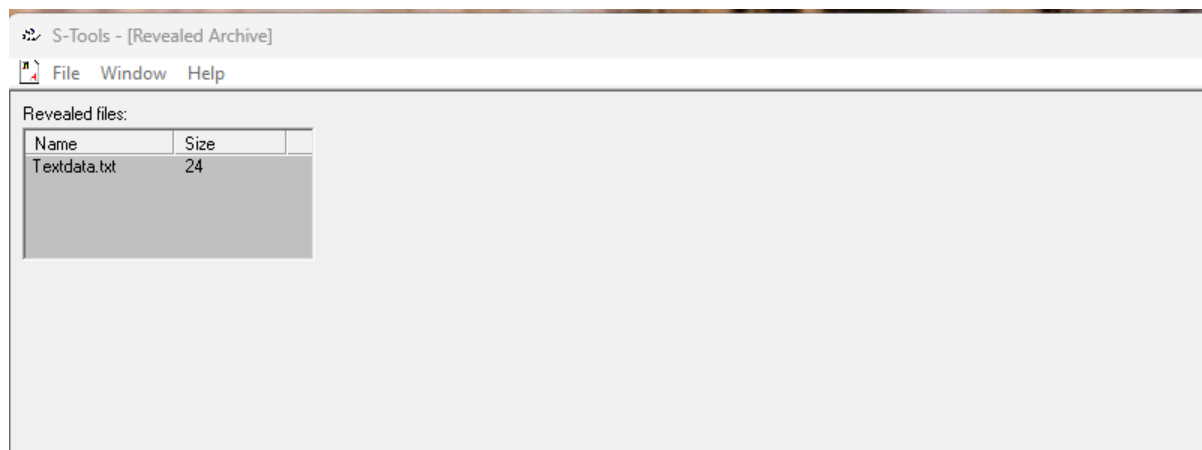
**Step 3: Now drag and drop the text file on the image-> enter the password****Step 4: Now save the file****Step 5: Open saved image -> right click on it-> Revel...**



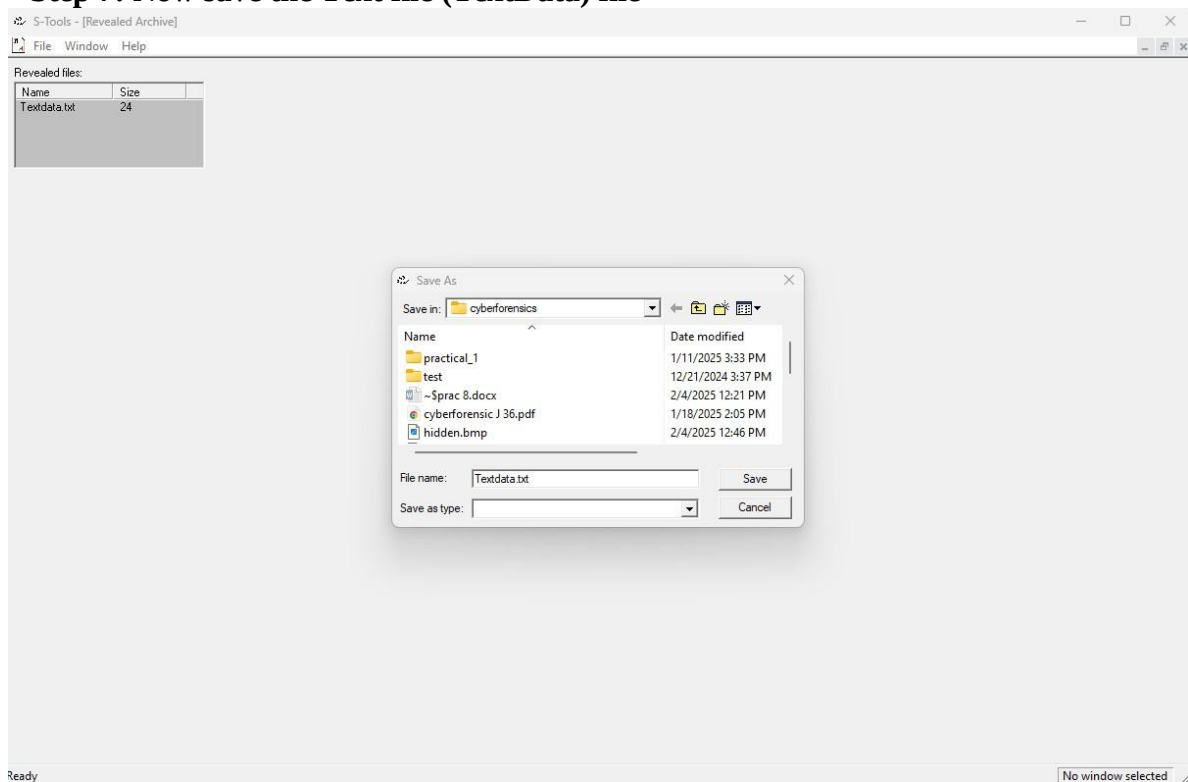


### Step 6: Enter the password that you have created





### Step 7: Now save the Text file (TextData) file



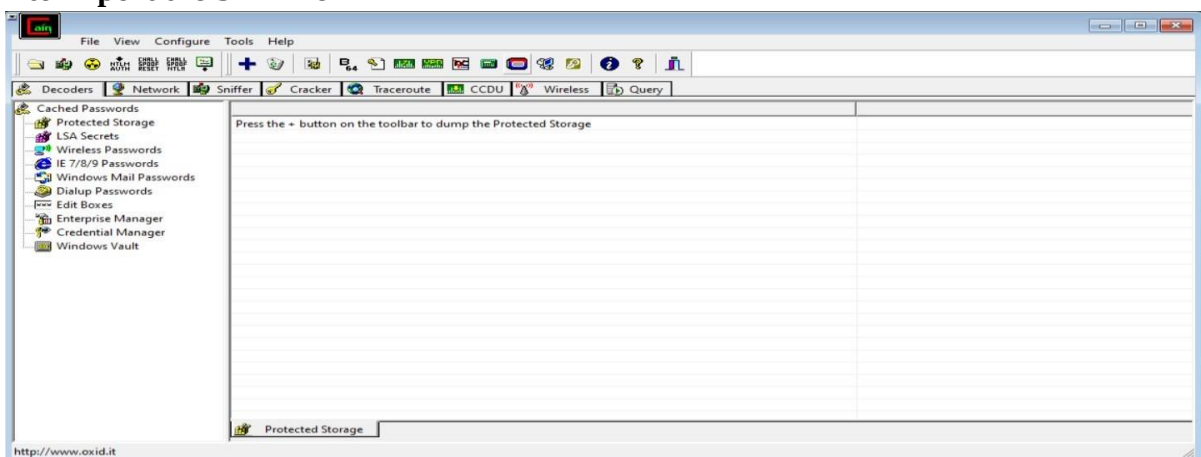
## Practical: 9

### Aim: (A) Performing Password Cracking [Cain & Abel].

#### Theory

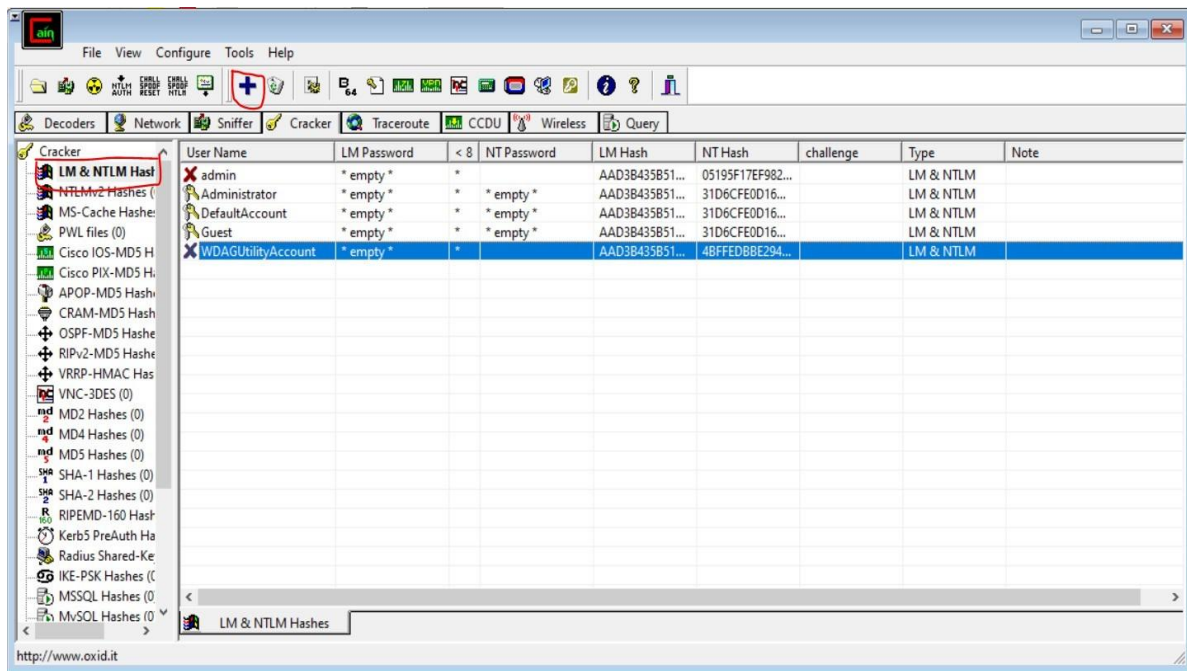
Cain & Abel is a tool that will be quite useful for network administrators, teachers, professional penetration testers, security consultants/professionals, forensic staff and security software vendors.

**Step1:Download Cain n able software using the given link [http://www.oxid.it/downloads/ca\\_setup.exe](http://www.oxid.it/downloads/ca_setup.exe) Now it's time to speak about the cracker tab,the most important feature of Cain.When Cain captures some LM and NTLM hashes or any kind of passwords for any supported protocols, Cain sends them automatically to the Cracker tab. We will import a local SAMfile just for demonstration purposes to illustrate this point. Here is how to import the SAMfile**

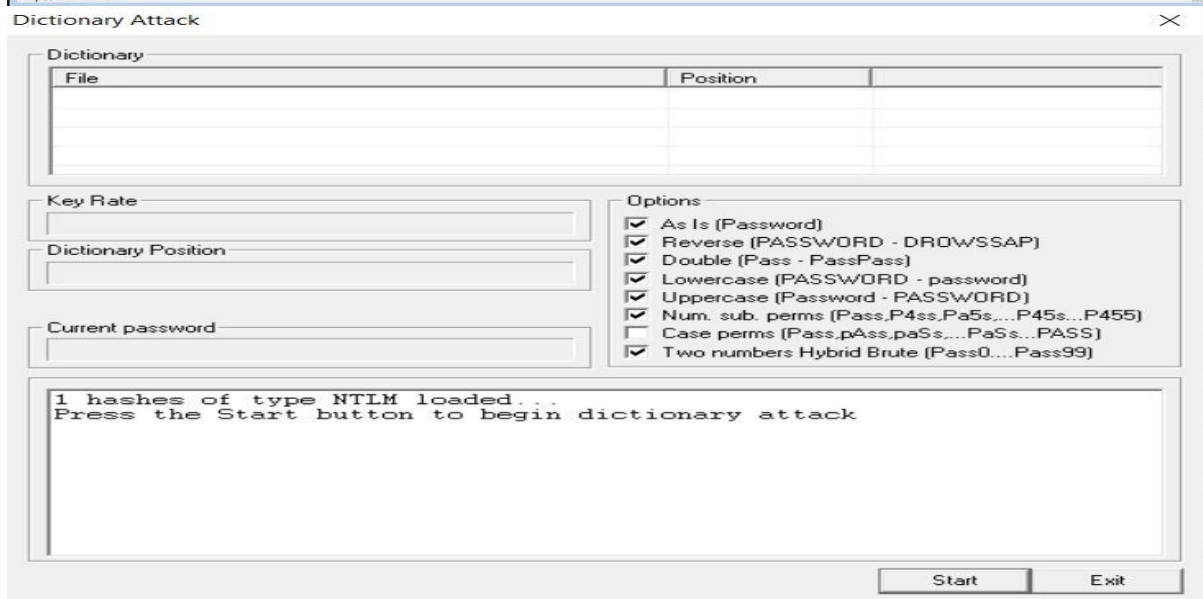
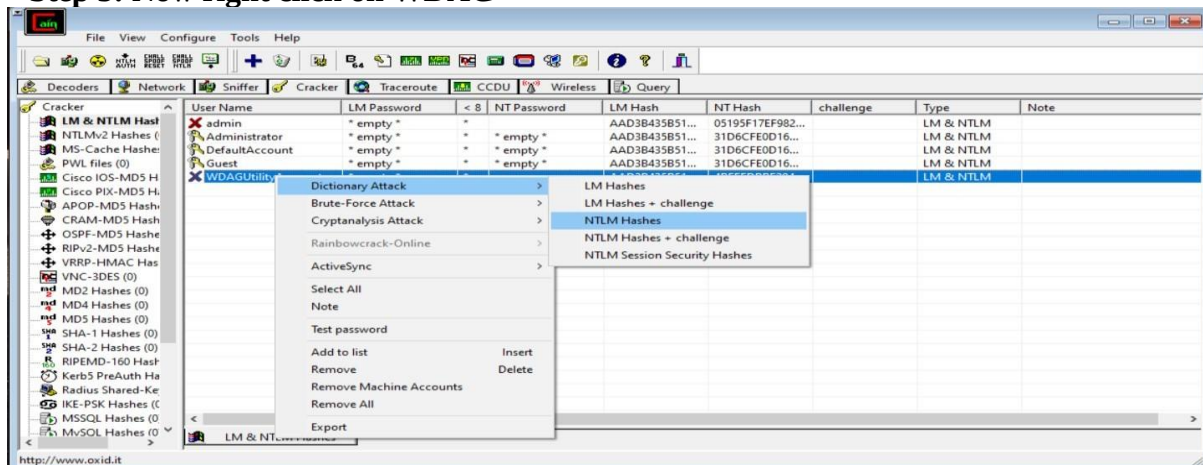


**Step 2: Click on Craker-> then plus icon-> then LM & NTLM head**

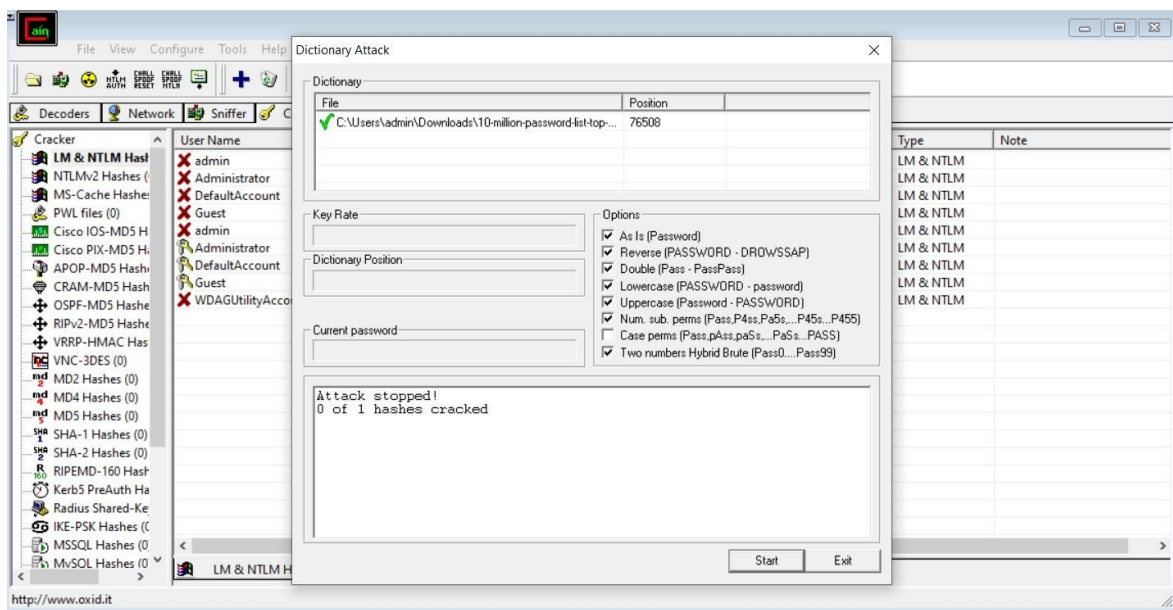
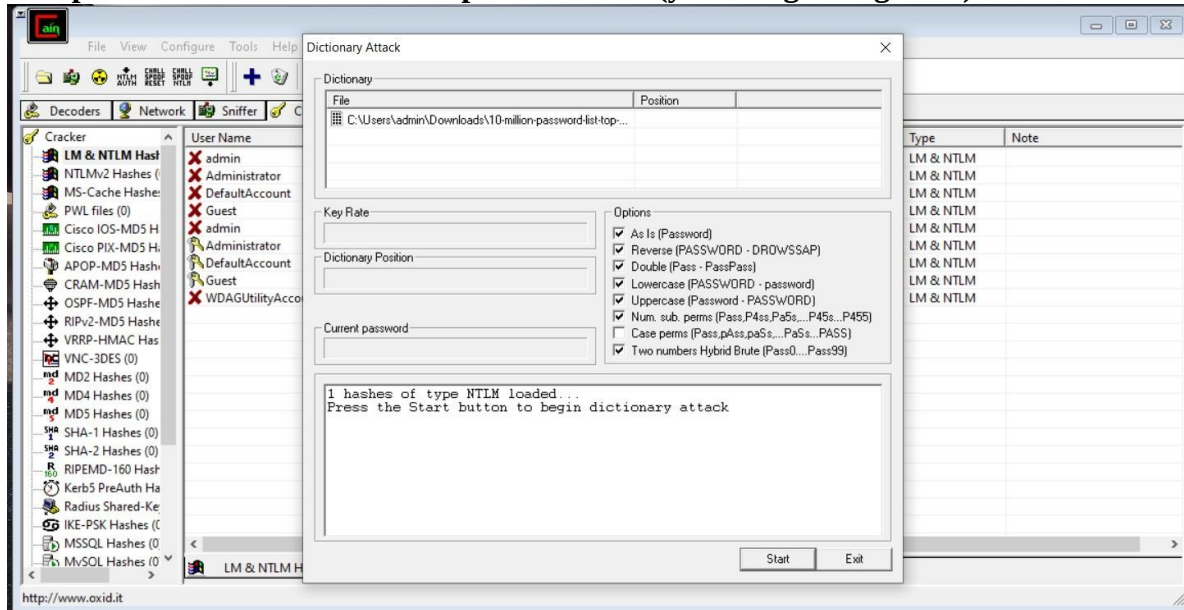




### Step 3: Now right click on WDAG



**Step 4: Choose a file 10-million passwords.txt (you will get on github)-> click on start**



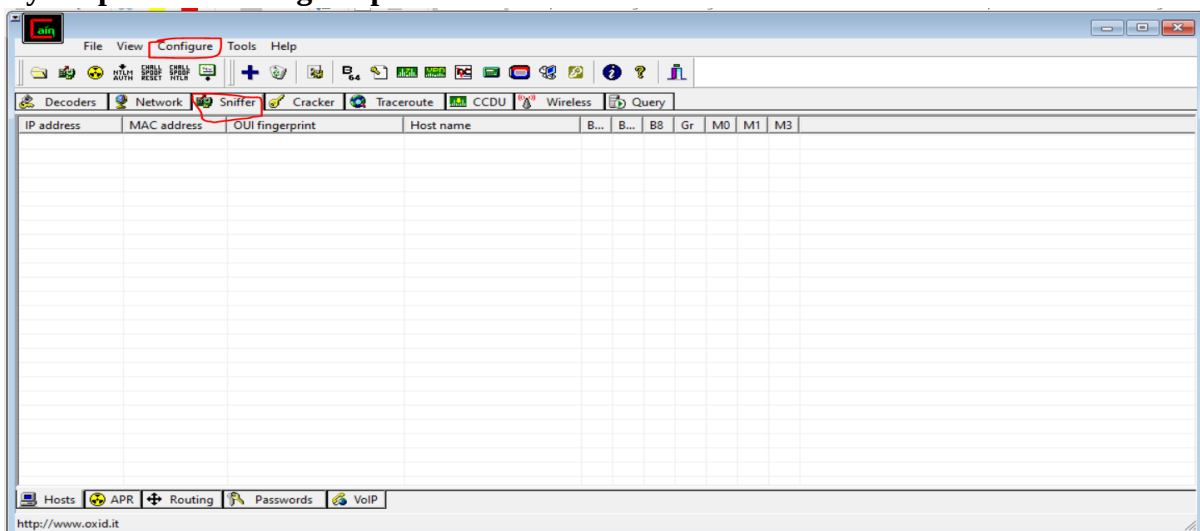
## B) Performing Sniffing [Cain & Abel].

### Theory

#### Sniffing:

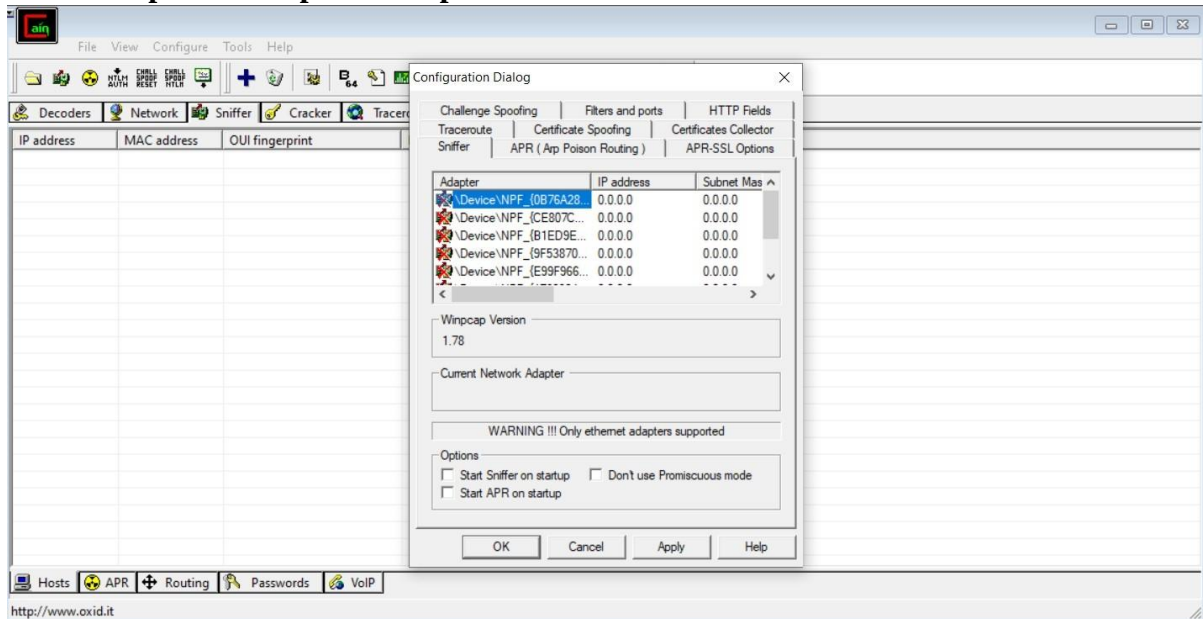
A packet analyser (also known as a network analyser, protocol analyser or packet sniffer, or for particular types of networks, an Ethernet sniffer or wireless sniffer) is a computer program or a piece of computer hardware that can intercept and log traffic passing over a digital network or part of a network. As data streams flow across the network, the sniffer captures each packet and, if needed, decodes the packet's raw data, showing the values of various fields in the packet, and analyses its content according to the appropriate RFC or other specifications.

**Step 1: Go to -> Sniffer Tab and click Configure in the main menu to configure your packet listening adapter.**

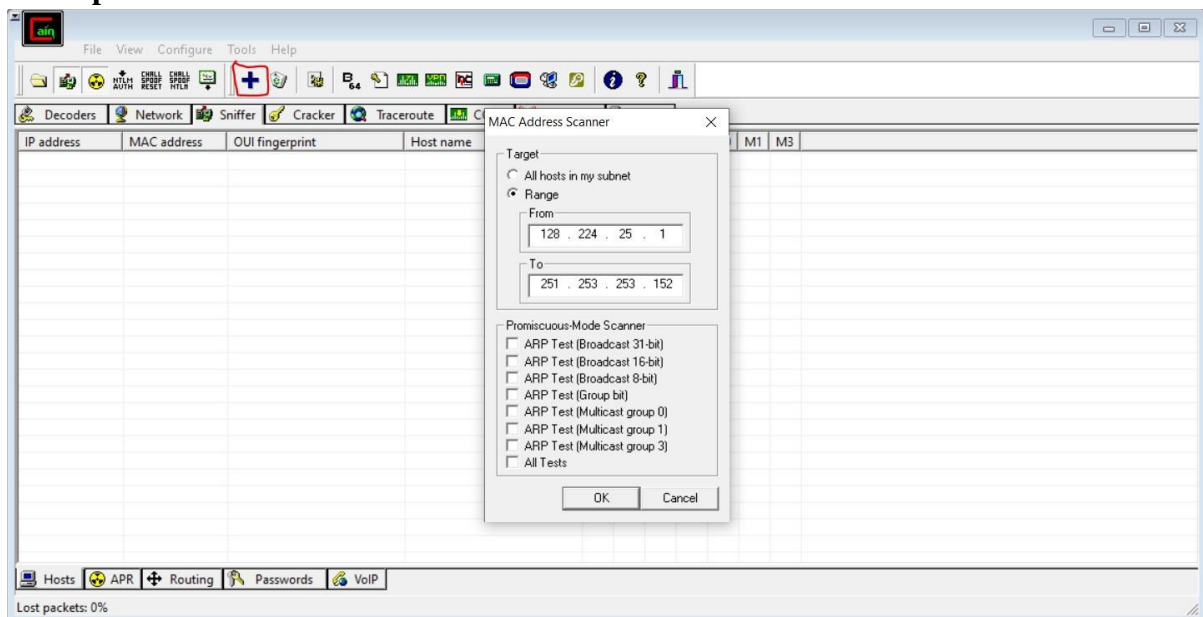


**Step 2: Select the appropriate network adapter for your network that you want to**

**sniff the packets for plain-text passwords. And Click Ok**

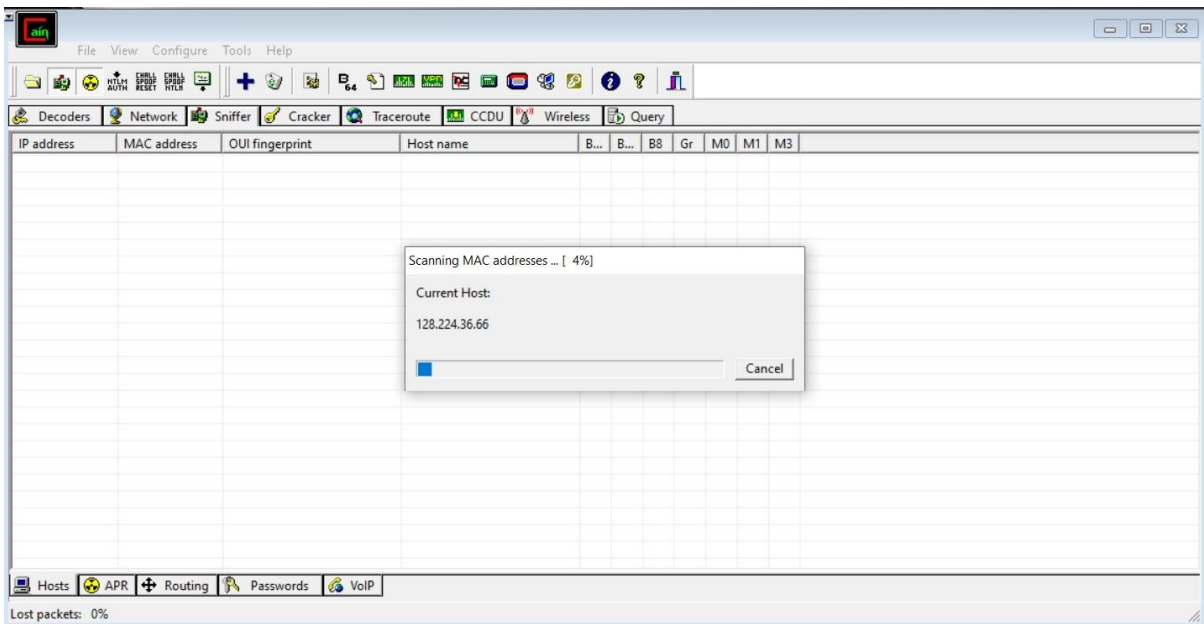


**Step 3: This is very interesting step. Now we need to select the router's IP address and click ok. This means that we want to listen to every packet that is sent to router. If we select any other IP address in our LAN network then we can listen to only that particular HOST on the network.**



**Step 4: After Scanning Mac Address. You will finally get the Output.**





## Practical: 10

### Aim: A) Scan Registry using RegScanner.

#### Theory

RegScanner is a small and lightweight system utility, which allows you to scan the Registry, find the desired Registry values that match the specified search criteria and then displays them in a list.

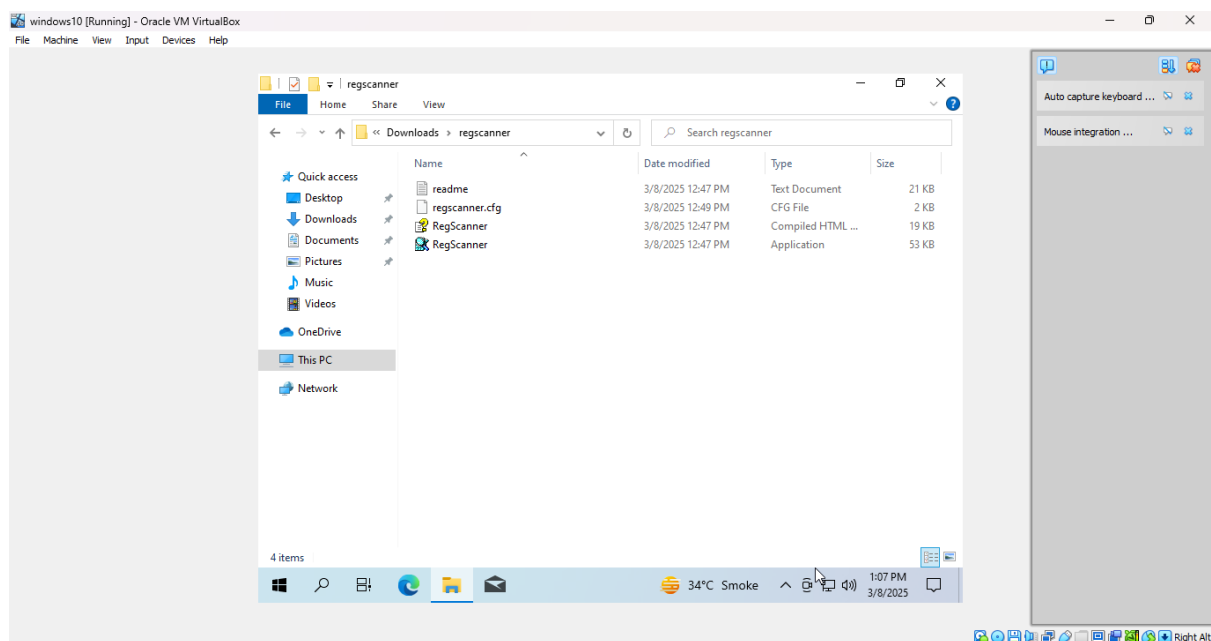
#### I. Problem Statement

A registry scanner or registry cleaner is third-party software for Microsoft Windows that searches for registry items that are redundant or no longer needed, then removes or resolves them. The objective is to help the system work more efficiently by "cleaning" it of these superfluous registry entries.

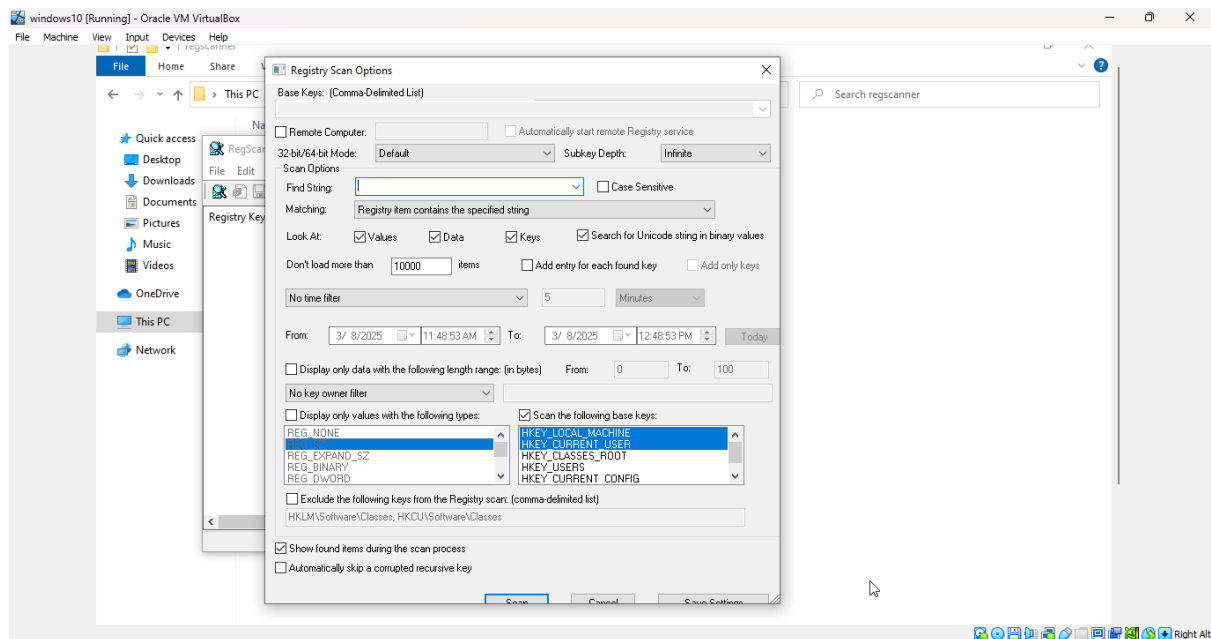
#### II. Literature/Theory

Some Registry cleaners make no distinction as to the severity of the errors, and many that do may erroneously categorize errors as "critical" with little basis to support it.[1] Removing or changing certain Registry data can prevent the system from starting, or cause application errors and crashes

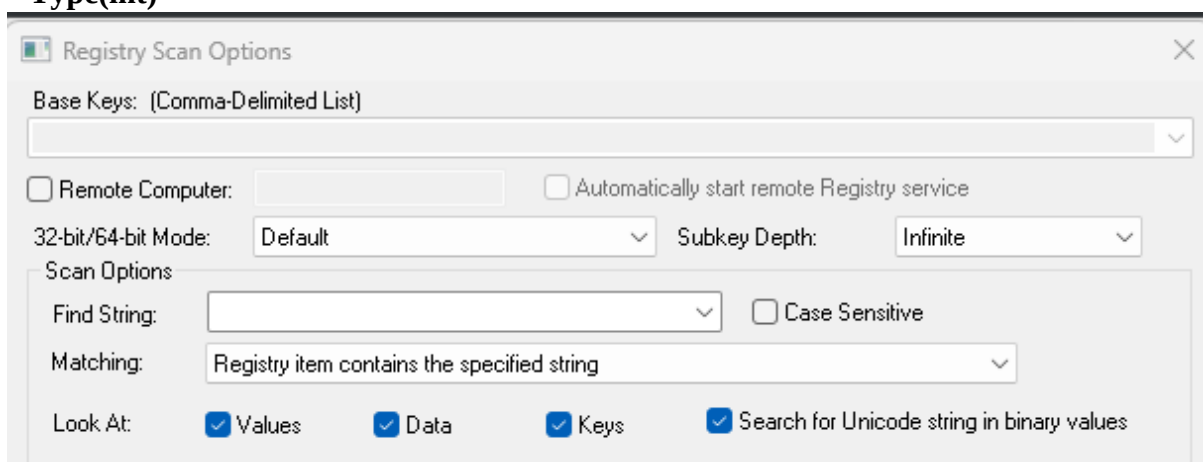
**Step 1: Install RegScanner > Double click on RegScanner .exe file to run the application.**



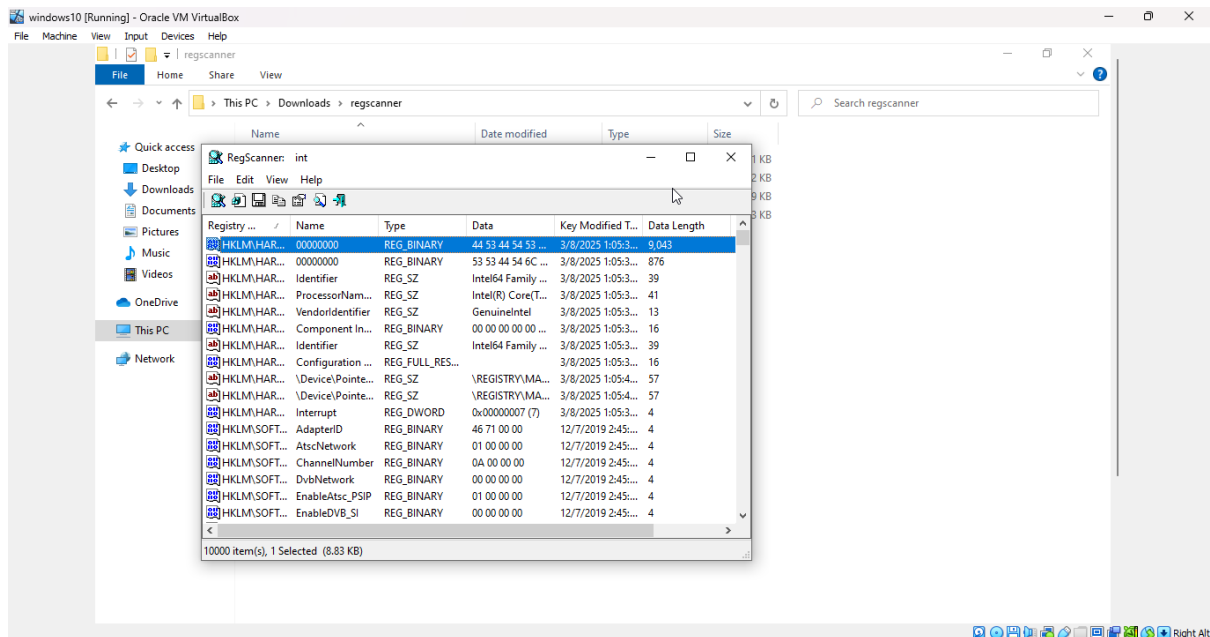
**Step 2: Quickly scan your registry and you can also search for the specific values with RegScanner.. You can select to scan the base keys**



**Step 3: In Matching drop-down > Select Registry item contains the specified string -> Type(int)**



**Step 4: Scan Result / Output:**



### B) Study Registry Viewer tool (Alien Registry

**Viewer). Theory**

## I. Title

Alien Registry Viewer allows you to explore registry files, search for specific key names and values, export registry data into a .REG or text file and bookmark registry keys as favorites. As well as the above mentioned files, Windows uses hidden files with the same names and extensions.

## II. Problem Statement

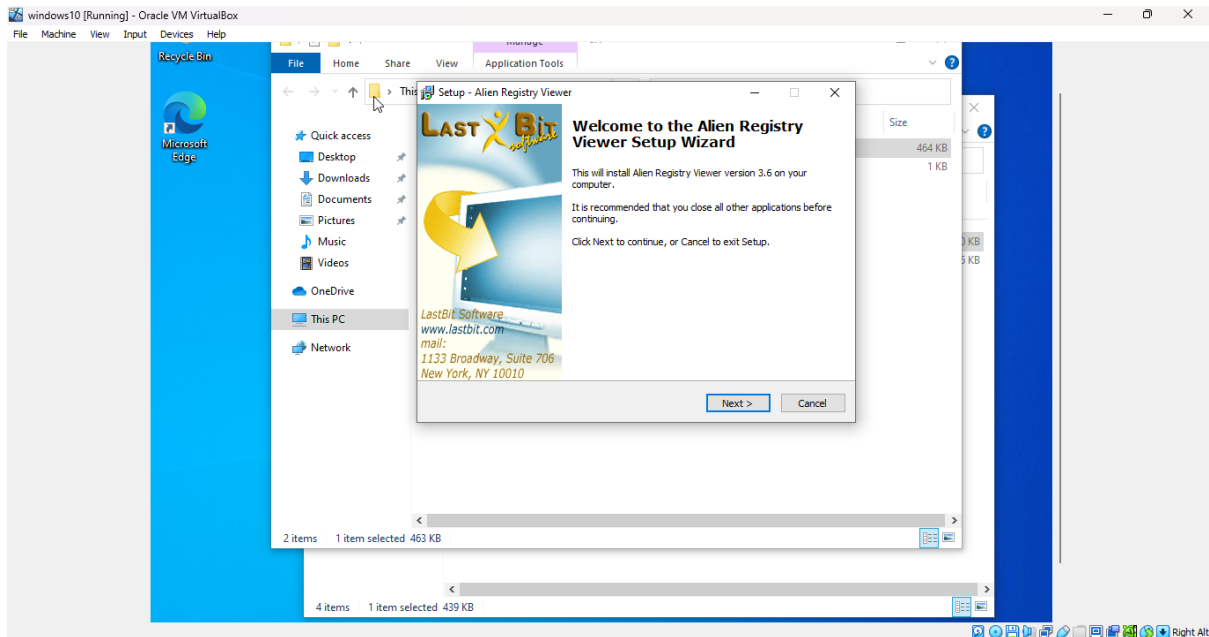
Registry Viewer allows you to view the protected storage of a registry. Passwords, usernames, and other information can be stored in the secured storage, which is not accessible using the Windows Registry Editor. Registry Viewer comes with a number of tools for retrieving and reporting vital registry data.

### III.Literature/Theory

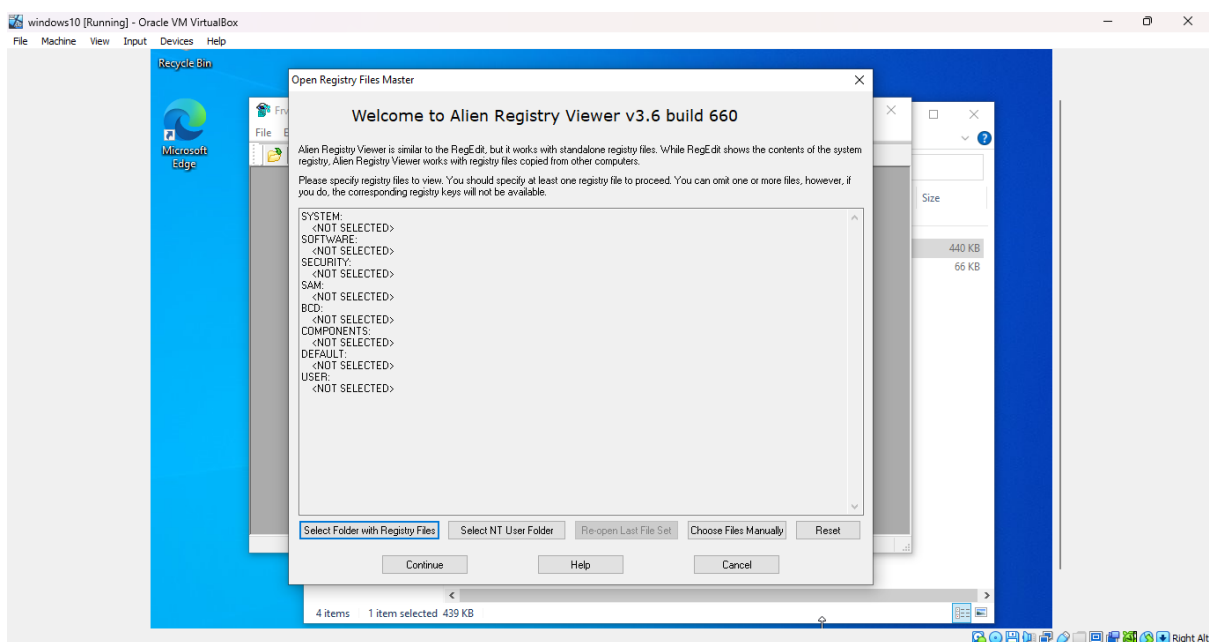
Alien Registry Viewer is similar to the RegEdit application included into Windows, but

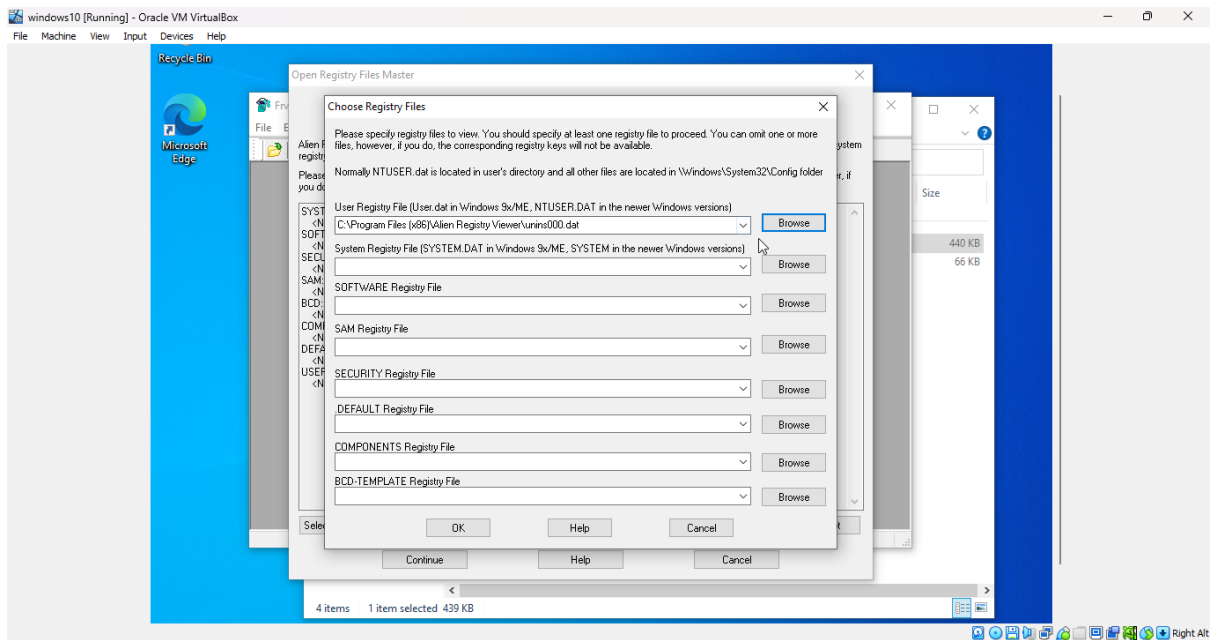
unlike RegEdit, it works with standalone registry files. The current version of Alien Registry Viewer works in the read-only mode, i.e. you can view but you cannot edit registry files.

**Step 1: Install the Alien registry viewer by double clicking on the .exe file.-> The following window would appear.Click on Next.**

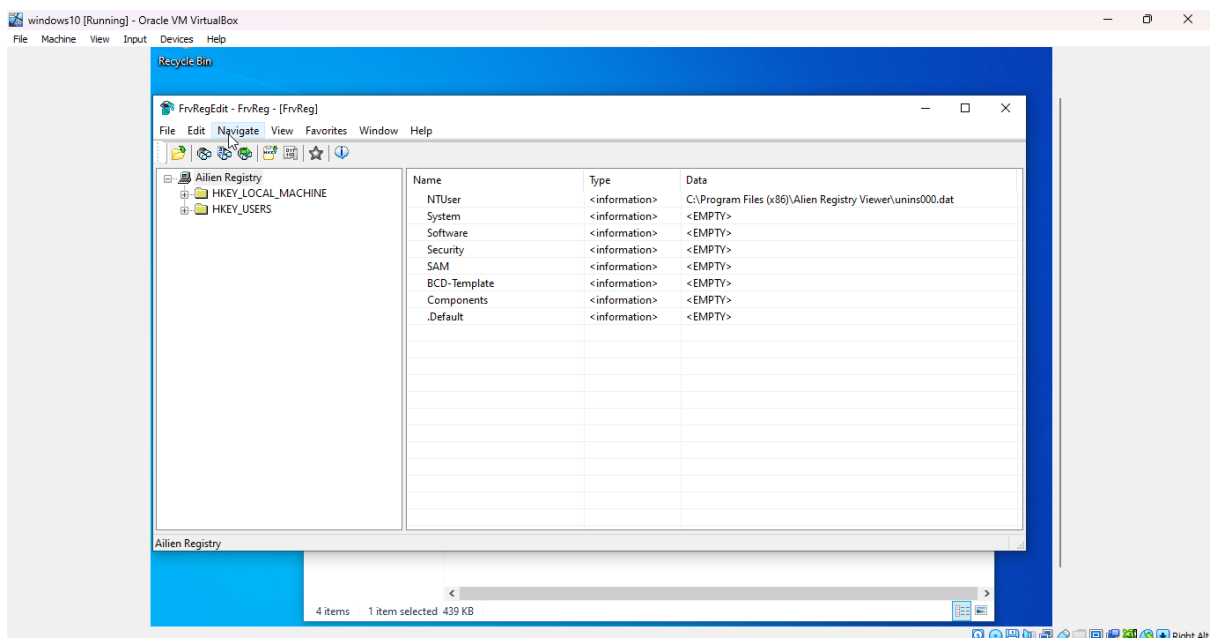


**Step 2: After installation the following window would appear.**





**Step 3: Click on HKEY\_CLASSES\_ROOT HKEY\_CURRENT\_USER.... Etc**



**Step 4: Press CTRL + F. The following dialog box will appear.in Find what textbox type .pdf > Click Find Next.**

